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1.1 General Linear Model (GLM): Extensions

R-Package 'galamm', 'gllvm', 'glmm', 'glm2'

GLM — GLiM **Little (2013)**: "The GLM is the concept that "all analytic methods are correlational ... and yield variance-accounted-for effect sizes analogous to r^2 (e.g., R^2, η^2, ω^2)" (Thompson, 2000, p. 263). All the GLM methods apply weights (e.g., regression β-weights, factor pattern coefficients) to the measured/observed variables to estimate scores on composite/latent/synthetic variables (e.g., regression $\hat{Y}$, factor scores). As Graham (2008) explained,

The vast majority of parametric statistical procedures in common use are part of [a single analytic family called] the General Linear Model (GLM), including the t test, analysis of variance (ANOVA), multiple regression, descriptive discriminant analysis (DDA), multivariate analysis of variance (MANOVA), canonical correlation analysis

(CCA), and structural equation modeling (SEM). Moreover, these procedures are hierarchical [italics added], in that some procedures are special cases of others. (p. 485)

In 1968, Cohen proved that multiple regression analysis subsumes all univariate parametric statistical analyses as special cases. For example, you can obtain ANOVA p-values and effect sizes by running a multiple regression computer program, but not vice versa.

Ten years later, Knapp (1978) showed that all commonly utilized univariate and multivariate analyses are special cases of canonical correlation analysis. Later, Bagozzi, Fornell and Larcker (1981) demonstrated that structural equation modeling (SEM) is an even more general case of the GLM.

All the statistical methods considered in the present chapter are part of a single analytic family in which members have more similarities than differences with each other (Zientek & Thompson, 2009). Thus, broad statements and principles can be stated that generalize across all the methods within the GLM. One fundamental principle is that traditional/classical statistical analyses partition the variances on outcome variables into subcomponents and also focus on computing ratios of the variance partitions to the total observed variances of the outcome variables. [...]

The starting point for generalized linear models is the familiar general linear model (GLM), the most widely taught and used method of data analysis in psychology and the behavioral sciences today. [...] The generalized linear model (GLiM), developed by Nelder & Wedderburn (1972) and expanded by McCullagh and Nelder (1983, 1989), extends linear regression to a broader family of outcome variables. The GLiM incorporates the basic structure of linear regression equations, but introduces two major additions to the framework. First, it accommodates typically nonlinear relationships of predictors to the criterion through transformation of the predicted score to a form that is linearly related to the predictors; a *link function* relates predicted to observed criterion scores. Second, the GLiM allows a variety of *error structures* (i.e., conditional distributions of outcome variance) beyond the normally distributed error structure of linear regression.

There are three components to the generalized linear model: the systematic portion, the random portion, and the link function. The systematic portion of the model parallels the model for the predicted value in OLS regression. It defines the relation between η, which is some function of the expected value of Y, and the independent variables in the model. This relationship is defined as linear in the variables,

$$\eta = b_0 + b_1X_1 + b_2X_2 + \cdots + b_pX_p. \tag{2}$$

Thus, the regression coefficients can be interpreted identically to those in linear regression: a 1 -unit change in X_1 results in a b_1-unit change in η, holding all other variables constant.

The random portion of the model defines the error distribution of the outcome variable. The error distribution of the outcome variable refers to the conditional distribution of the outcome given a set of specified values on the predictors. The GLiM allows any discrete or continuous probability distribution in the exponential family. Each of the members of this family have a probability density function (or probability mass function if the distribution is discrete) that can be written in a form that includes the natural logarithm e raised to a power that is a function of the parameters. The most familiar member of this family is the normal distribution,

$$f\left(Y \mid \mu, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(-\frac{(Y-\mu)^2}{2\sigma^2}\right)}, \tag{3}$$

in which the height of the normal curve is a function of two independent parameters, the population mean (μ) and variance (σ^2). Other distributions in the exponential family that are commonly used in GLiMs are the binomial, multinomial, Poisson, exponential, gamma, and beta distributions. Other distributions exist in the exponential family, but are less commonly used in GLiMs.

The link function relates the conditional mean of Y, also known as the expected value of Y, $\mathrm{E}(Y \mid X)$, or μ, to the linear combination of predictors η defined by Equation 2. The link function allows for nonlinear relations between the predictors and the predicted outcome; the link function transforms the predicted value of the dependent variable to a new form that has a linear relationship with the predictors. Several link functions are possible, but each error distribution has an associated special link function known as its canonical link. The canonical link satisfies special properties of the model, makes estimation simpler, and is the most commonly used link function. To cite three examples we will discuss in detail later in this chapter, the natural log(ln) link function is the canonical link for a conditional Poisson distribution. The logit or log-odds is the canonical link for a conditional binomial distribution, resulting in logistic regression. The canonical link for the normal error distribution is identity (i.e., a 1.0 or no transformation) resulting in linear regression.

In this framework, linear regression becomes just a special case of the GLiM: the error distribution for linear regression is a normal distribution and the link function is identity. A wide variety of generalized linear models are possible, depending on the proposed conditional distribution of the outcome variable. Table 3.1 shows the error distributions and their associated canonical links for several of the models within the GLiM framework. [...]"

GLMM Molenberghs and Verbeke (2005): "The generalized linear mixed model is the most frequently used randomeffects model in the context of discrete repeated measurements. Not only is it a rather straightforward extension of the generalized linear model for univariate data to the context of clustered measuerements [...]

As before, Y_{ij}, is the j th outcome measured for cluster (subject) $i, i = 1, \ldots, N, j = 1, \ldots, n_i$ and $\boldsymbol{Y}_i$ is the n_i-dimensional vector of all measurements available for cluster i. As introduced in Section 13.4.4, it is assumed that, conditionally on q-dimensional random effects $\boldsymbol{b}_i$, assumed to be drawn independently from the $N(\mathbf{0}, D)$, the outcomes Y_{ij} are independent with densities of the form

$$f_i\left(y_{ij} \mid \boldsymbol{b}_i, \boldsymbol{\beta}, \phi\right) = \exp\left\{\phi^{-1}\left[y_{ij}\theta_{ij} - \psi\left(\theta_{ij}\right)\right] + c\left(y_{ij}, \phi\right)\right\},$$

with $\eta\left(\mu_{ij}\right) = \eta\left[E\left(Y_{ij} \mid \boldsymbol{b}_i\right)\right] = \boldsymbol{x}_{ij}'\boldsymbol{\beta} + \boldsymbol{z}_{ij}'\boldsymbol{b}_i$ for a known link function $\eta(\cdot)$, with $\boldsymbol{x}_{ij}$ and $z_{ij} p$-dimensional and q-dimensional vectors of known covariate values, with $\boldsymbol{\beta}$ a

p-dimensional vector of unknown fixed regression coefficients, and with ϕ a scale parameter. Finally, let $f(\boldsymbol{b}_i \mid D)$ be the density of the $N(\mathbf{0}, D)$ distribution for the random effects $\boldsymbol{b}_i$. [...]"

GLLVM Bartholomew et al. (2011a): "The principal aim of this book is to give a unified treatment of latent variable models. This is to be achieved by deriving them as special cases of what we shall call the general linear latent variable model (GLLVM). [...]

We saw in Chapter 1 that a latent variable model consists of two parts. First there is the prior distribution of the latent variables represented in (1.10) by the density function $h(\boldsymbol{y})$. This was seen to be essentially arbitrary and, as we shall argue later, its choice is largely a matter of convention. In any event, most of the results of this chapter do not depend on the choice of $h(\boldsymbol{y})$ which, for most purposes, is therefore irrelevant. The second element in the model is the set of conditional distributions of the manifest variables $\{x_i\}$ given the latent variables. In (1.10) these were denoted by $g_i(x_i \mid \boldsymbol{y})\,(i = 1, 2, \ldots, p)$ where the subscript i on g reminds us that the form of the distribution can vary with i.

A convenient family of distributions which turns out to have many useful properties in other branches of statistics is the one-parameter exponential family. If we suppose that the latent variables combine to determine the value of the parameter then we may write

$$g_i(x_i \mid \theta_i) = F_i(x_i)\, G_i(\theta_i) \exp(\theta_i u_i(x_i)) \quad (i = 1, 2, \ldots, p), \tag{2.1}$$

where θ_i is some function of $\boldsymbol{y}$. The simplest assumption about the form of this function is to suppose that it is a linear function, in which case we have

$$\theta_i = \alpha_{i0} + \alpha_{i1} y_1 + \alpha_{i2} y_2 + \cdots + \alpha_{iq} y_q \quad (i = 1, 2, \ldots, p). \tag{2.2}$$

This is the general linear latent variable model. The term linear refers to its linearity in the α s for reasons which will appear shortly.

The similarity between this model and the so-called generalised linear model (GLIM) used in statistics will be immediately apparent. The chief difference is that here we have a set of x s rather than a single dependent variable and here the ys are unobservable. In view of this connection it may seem somewhat perverse of us to have used x to denote the dependent variable and y the independent (or regressor) variables when the usual convention is to reverse their roles. The reason for this will become apparent when we come to discuss the posterior analysis of the model. In that context we shall be predicting the y s given the x s, in which case our notation conforms to the standard usage.

The exponential family of (2.1) includes the normal, gamma and Bernoulli distributions as special cases. If we allow x_i and θ to be vector-valued it also includes the multinomial distribution which we shall need for our treatment of categorical variables. [...]"

GLLAMM Rabe-Hesketh et al. (2004): "A unifying framework for generalized multilevel structural equation modeling is introduced. The models in the framework, called generalized linear latent and mixed models (GLLAMM), combine features of generalized linear mixed models (GLMM) and structural equation models (SEM) and consist of a response model and a structural model for the latent variables. The response model generalizes GLMMs to incorporate factor structures in addition to random intercepts and coefficients. As in GLMMs, the data can have an arbitrary number of levels and can be highly unbalanced with different numbers of lower-level units in the higher-level units and missing data. A wide range of response processes can be modeled including ordered and unordered categorical responses, counts, and responses of mixed types. The structural model is similar to the structural part of a SEM except that it may include latent and observed variables varying at different levels. [...]"

1.2 GLM — NLFM — GGM — EGA

Bartholomew et al. (2011b): "We have already introduced the normal linear factor (NLFM) model in Chapter 1 as an example of our general approach. Its main properties arise as special cases of the General Linear Latent Variable Model (GLLVM) given in Chapter 2."

H. F. Golino and Epskamp (2017): "The relationship between latent variables on the one hand and network clusters on the other goes deeper than mere philosophical speculation and empirical findings. It can directly be seen that if a latent variable model is the true underlying causal model, we would expect indicators in a network model to feature strongly connected clusters for each latent variable. Since edges correspond to partial correlation coefficients between two variables after conditioning on all other variables in the network, and two indicators cannot become independent after conditioning on observed variables given that they are both caused by a latent variable, the edge strength between two indicators should not be zero. In fact, in a mathematical point of view, network models can be shown to be equivalent under certain conditions to latent variable models in both binary [42, 49] and Gaussian datasets [50], in which case each latent variable is represented by a rank-1 cluster. Thus, when defining a cluster as a group of connected nodes regardless of edge weight, we can state the following relationship as a fundamental rule of network psychometrics: Clusters in network = latent variables.

It should be noted that when multiple correlated latent variables underlie distinct sets of indicators, none of the edges should be missing as we technically cannot condition on any observed variable to make two indicators independent. However, we would expect the partial correlation between two indicators of the same latent variable to be much stronger than the partial correlation between two indicators of different latent variables. Furthermore, when using LASSO estimation, we would expect these already small edge weights to be pushed more easily to zero simply due to the penalization. As such, we expect an algorithm to detect weighted network clusters to indicate indicators of the same latent variable.

In a more mathematical point of view, let y represent a centered random vector of K responses, which we assume to be multivariate normally distributed with some variancecovariance matrix $\boldsymbol{\Sigma}$:

$$\boldsymbol{y} \sim N_p(\boldsymbol{0}, \boldsymbol{\Sigma}).$$

In factor analysis, we typically assume that the response of subject p is caused through a linear factor model by a set of M latent variables $\boldsymbol{\eta}$ plus random error $\boldsymbol{\varepsilon}$:

$$\boldsymbol{y}_p = \boldsymbol{\Lambda}\boldsymbol{\eta}_P + \boldsymbol{\varepsilon}_P,$$

in which $\boldsymbol{\Lambda}$ is a $K \times M$ factor loadings matrix. This leads to the well known factor analysis model:

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Psi}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta},$$

in which $\boldsymbol{\Theta} = \mathrm{Var}(\boldsymbol{\varepsilon})$ and $\boldsymbol{\Psi} = \mathrm{Var}(\boldsymbol{\eta})$. Often it is assumed that each item only loads on one factor (simple structure), and thus that $\boldsymbol{\Lambda}$ can be reordered to be block diagonal:

$$\boldsymbol{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_M \end{bmatrix},$$

Furthermore, we assume $\boldsymbol{\Theta}$ to be diagonal (local independence):

$$\boldsymbol{\Theta} = \begin{bmatrix} \Theta_1 & 0 & \cdots & 0 \\ 0 & \Theta_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Theta_K \end{bmatrix}.$$

In network modeling, the Gaussian graphical model [13] is used in which the inverse variance-covariance matrix is modeled [51]:

$$\boldsymbol{\Sigma} = \boldsymbol{K}^{-1}.$$

A zero element in $\boldsymbol{K}$ indicates conditional independence:

$$y_i \perp\!\!\!\perp y_j \mid \boldsymbol{y}^{-(i,j)} \Longleftrightarrow k_{i,j} = k_{j,i} = 0,$$

in which $\boldsymbol{y}^{-(i,j)}$ indicates $\boldsymbol{y}$ without elements i and j, and negative elements of $\boldsymbol{K}$ can be standardized to equal partial correlation coefficients:

$$\rho_{ij} = -\frac{k_{ij}}{\sqrt{k_{ij}k_{jj}}}.$$

These partial correlation coefficients are thus proportional to the inverse variance-covariance matrix, and can be used to form a partial correlation network. As such, off-diagonal elements of $\boldsymbol{K}$ encode a network structure.

Relating the above expressions and applying the Woodbury matrix identity we obtain:

$$\boldsymbol{K} = \left(\boldsymbol{\Lambda}\boldsymbol{\Psi}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta}\right)^{-1} = \boldsymbol{\Theta}^{-1} - \boldsymbol{\Theta}^{-1}\,\Lambda\left(\,\Psi^{-1} + \boldsymbol{\Lambda}^\top\boldsymbol{\Theta}^{-1}\boldsymbol{\Lambda}\right)^{-1}\boldsymbol{\Lambda}^\top\boldsymbol{\Theta}^{-1}$$

Since $\mathbf{\Theta}$ is diagonal, so is $\mathbf{\Theta}^{-1}$, leading to $\mathbf{\Theta}^{-1}\mathbf{\Lambda}$ to be block diagonal and $\mathbf{\Lambda}^{\mathrm{T}}\mathbf{\Theta}^{-1}\mathbf{\Lambda}$ to be diagonal. Let $\boldsymbol{X} = \left(\mathbf{\Psi}^{-1} + \mathbf{\Lambda}^{\mathrm{T}}\mathbf{\Theta}^{-1}\mathbf{\Lambda}\right)^{-1}$. Then, $\boldsymbol{K}$ becomes a block matrix in which every block is constructed of the inner product of factor loadings and inverse residual variances, every diagonal block is scaled by diagonal elements of $\boldsymbol{X}$ and every off-diagonal block is scaled by off-diagonal values of $\boldsymbol{X}$.

Since ψ must be positive definite it follows that $\boldsymbol{X}$ must be positive definite as well. Typically in factor analysis the first factor loadings or the latent variance-covariances are fixed to 1 to identify the model. We can, however, without loss of information, also constrain the diagonal of $\boldsymbol{X}$ to equal 1. It then follows that every absolute off-diagonal value of $\boldsymbol{X}$ must be smaller than 1. From the formation of $\boldsymbol{X}$ follows that off-diagonal values of $\boldsymbol{X}$ equal zero if the latent factors are orthogonal. Hence, the above decomposition shows that:

1. If the latent factors are orthogonal, the resulting GGM consists of unconnected clusters.
2. Assuming factor loadings and residual variances are reasonably on the same scale for every item, the off-diagonal blocks of $\boldsymbol{K}$ will be scaled closer to zero than the diagonal blocks of $\boldsymbol{K}$. Hence, the resulting GGM will contain weighted clusters for each factor.

This line of reasoning leads us to develop Exploratory Graph Analysis (EGA), in which firstly we estimate the correlation matrix of the observable variables, then the graphical LASSO estimation is used to obtain the sparse inverse covariance matrix, with the regularization parameter defined via EBIC over 100 different values. Finally, the walktrap algorithm [52] is used to find the number of dense subgraphs (communities or clusters) of the partial correlation matrix computed in the previous step. The walktrap algorithm provides a measure of similarities between vertices based on random walks which can capture the community/cluster structure in a graph [52]. The number of clusters identified equals the number of latent factors in a given dataset."

1.2.1 EGA — EGM – DynEGA

R-Package: 'EGAnet'

Christensen and Golino (2024): "EGM represents a novel class of clustered network models. A core assumption of EGM is that the $\mathbb{P}(\text{in}) \geq \mathbb{P}(\text{out})$ meaning that within-community connections are expected to be as dense or more dense than between community connections for a given structure. In practice, EGM can be applied by setting both parameters based on some a priori expectations of $\mathbb{P}(\text{out})$ and $\mathbb{P}(\text{out})$ for each community in the network. Setting both parameters, however, is unlikely to be known in advance. A search procedure using $\mathbb{P}(\text{in})$ can be used such that values greater than or equal to some probability can be search over in combination with values less than or equal to the same probability for $\mathbb{P}(\text{out})$. Over the multiple models estimated, some fit criterion such as SRMR, log-likelihood, AIC, or BIC on the model-implied zero-order correlation matrix can be used to establish the best fitting combination of values for $\mathbb{P}(\text{in})$ and $\mathbb{P}(\text{out})$.

One consequence of this more general approach for the EGM is that the original EGA, on which the model is based, becomes a special case of EGM. Specifically, the graphical least aboslute shrinkage and selection operator (GLASSO; Friedman, Hastie, & Tibshirani, 2008) with extended BIC (EBIC; Chen & Chen, 2008; Epskamp & Fried, 2018; Foygel & Drton, 2010) model selection (EBICglasso) defines the EGM parameters $\mathbb{P}$(in) and $\mathbb{P}$ (out) indirectly when combined with a community detection algorithm such as the Walktrap (Pons & Latapy, 2006) or other community detection algorithm (Christensen, Garrido, et al., 2024). The network estimation method defines the overall sparsity of the network while the community structure defines the communities and subsequently the parameters $\mathbb{P}$(in) and $\mathbb{P}$(out). The network and the number of communities or item assignments of the communities could otherwise be based on theory or researcher expectation."

H. Golino et al. (2022): "The DynEGA approach uses time delay embedding to preprocess each variable (e.g., time series of words counts) and estimates the derivatives from each variable using generalized local linear approximation (GLLA; Boker et al., 2010). Finally, a network psychometrics approach for dimensionality assessment termed exploratory graph analysis (EGA; Golino & Epskamp, 2017; Golino et al., 2020b) is used to identify clusters of variables that are changing together (i.e., dynamic latent topics). The DynEGA approach automatically estimates the number of latent topics and their short-term temporal dynamics with the results displayed as a network plot, which facilitates their interpretation. Importantly, the DynEGA approach does not assume that the topics are uncorrelated (e.g., LDA and other commonly applied topic modeling techniques), it can accommodate different time scales, and it can estimate the latent structure at different levels of analysis (i.e., population, groups, and individuals)."

1.3 GLM — SEM

Graham (2008):"[...] all GLM procedures (both univariate and multivariate) are special cases of SEM (Bagozzi, Fornell, & Larcker, 1981; Fan, 1997)"

Little (2013): "Knapp (1978) showed that all commonly utilized univariate and multivariate analyses are special cases of canonical correlation analysis. Later, Bagozzi, Fornell and Larcker (1981) demonstrated that structural equation modeling (SEM) is an even more general case of the GLM. All the statistical methods considered in the present chapter are part of a single analytic family in which members have more similarities than differences with each other (Zientek & Thompson, 2009). Thus, broad statements and principles can be stated that generalize across all the methods within the GLM.One fundamental principle is that traditional/classical statistical analyses partition the variances on outcome variables into subcomponents and also focus on computing ratios of the variance partitions to the total observed variances of the outcome variables."

1.4 SEM — Path Analysis

R-Package: 'sem', 'lavaan'

Epskamp et al. (2017): "In Confirmatory Factor Analysis (CFA), $\boldsymbol{Y}$ is typically assumed to be a causal linear effect of a set of M centered latent variables, $\boldsymbol{\eta}$, and independent residuals or error, $\boldsymbol{\varepsilon}$:

$$\boldsymbol{y} = \Lambda\boldsymbol{\eta} + \boldsymbol{\varepsilon}.$$

Here, $\boldsymbol{\Lambda}$ represents a $P \times M$ matrix of factor loadings. This model implies the following model for $\hat{\boldsymbol{\Sigma}}$:

$$\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Lambda}\boldsymbol{\Psi}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Theta} \tag{2}$$

in which $\boldsymbol{\Psi} = \mathrm{Var}(\boldsymbol{\eta})$ and $\boldsymbol{\Theta} = \mathrm{Var}(\boldsymbol{\varepsilon})$. In Structural Equation Modeling (SEM), $\mathrm{Var}(\boldsymbol{\eta})$ can further be modeled by adding structural linear relations between the latent variables [2] :

$$\boldsymbol{\eta} = \boldsymbol{B}\boldsymbol{\eta} + \zeta,$$

in which ζ is a vector of residuals and $\boldsymbol{B}$ is an $M \times M$ matrix of regression coefficients. Now, $\hat{\boldsymbol{\Sigma}}$ can be more extensively modeled as:

$$\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Lambda}(\boldsymbol{I} - \boldsymbol{B})^{-1}\boldsymbol{\Psi}(\boldsymbol{I} - \boldsymbol{B})^{-1\top}\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Theta}, \tag{3}$$

in which now $\boldsymbol{\Psi} = \mathrm{Var}(\boldsymbol{\zeta})$. This framework can be used to model direct causal effects between observed variables by setting $\boldsymbol{\Lambda} = \boldsymbol{I}$ and $\boldsymbol{\Theta} = \boldsymbol{O}$, which is often called path analysis (Wright, 1934).

The $\boldsymbol{\Theta}$ matrix is, like $\hat{\boldsymbol{\Sigma}}$ and $\boldsymbol{S}$, a $P \times P$ matrix; if $\boldsymbol{\Theta}$ is fully estimated-contains no restricted elements - then $\boldsymbol{\Theta}$ alone constitutes a saturated model. Therefore, to make either (2) or (3) identifiable, $\boldsymbol{\Theta}$ must be strongly restricted. Typically, $\boldsymbol{\Theta}$ is set to be diagonal, a restriction often termed local independence (Lord et al., 1968; Holland and Rosenbaum, 1986) because indicators are independent of each other after conditioning on the set of latent variables. To improve fit, select off-diagonal elements of $\boldsymbol{\Theta}$ can be estimated, but systematic violations of local independence - many nonzero elements in $\boldsymbol{\Theta}$ - are not possible as that will quickly make (2) and (3) saturated or even over-identified. More precisely, $\boldsymbol{\Theta}$ can not be fully-populated -some elements of $\boldsymbol{\Theta}$ must be set to equal zero-when latent variables are used. An element of $\boldsymbol{\Theta}$ being fixed to zero indicates that two variables are locally independent after conditioning on the set of latent variables. As such, local independence is a critical assumption in both CFA and SEM; if local independence is systematically violated, CFA and SEM will never result in correct models."

1.5 SEM — LNM/RNM — GGM

R-Package 'lvnet'

Epskamp et al. (2017): "We propose two generalizations of both SEM and the GGM that both allow the modeling of network structures in SEM. In the first generalization, we adopt the CFA decomposition in (2) and model the variance-covariance matrix of latent variables as a GGM:

$$\boldsymbol{\Psi} = \Delta_{\boldsymbol{\Psi}} \left(\boldsymbol{I} - \boldsymbol{\Omega}_{\boldsymbol{\Psi}}\right)^{-1} \Delta_{\boldsymbol{\Psi}}.$$

This framework can be seen as modeling conditional independencies between latent variables not by directed effects (as in SEM) but as an undirected network. As such, we term this framework latent network modeling (LNM).

In the second generalization, we adopt the SEM decomposition of the variance-covariance matrix in (3) and allow the residual variance-covariance matrix $\boldsymbol{\Theta}$ to be modeled as a GGM:

$$\boldsymbol{\Theta} = \Delta_{\Theta} \left(I - \Omega_{\Theta}\right)^{-1} \Delta_{\Theta}.$$

Because this framework conceptualizes associations between residuals as pairwise interactions, rather than correlations, we term this framework Residual Network Modeling (RNM). Using this framework allows - as will be described below-for a powerful way of fitting a confirmatory factor structure even though local independence is systematically violated and all residuals are correlated."

1.6 SEM — MSEM-RS — Curie-Weiss

Chow et al. (2015): "Catastrophe theory (Thom, 1972, 1993) is the study of the many ways in which continuous changes in a system's parameters can result in discontinuous changes in 1 or several outcome variables of interest. Catastrophe theory—–inspired models have been used to represent a variety of change phenomena in the realm of social and behavioral sciences. Despite their promise, widespread applications of catastrophe models have been impeded, in part, by difficulties in performing model fitting and model comparison procedures. We propose a new modeling framework for testing 1 kind of catastrophe model—–the cusp catastrophe model——as a mixture structural equation model (MSEM) when cross-sectional data are available; or alternatively, as an MSEM with regime-switching (MSEM-RS) when longitudinal panel data are available. [...]

Cusp Catastrophe In catastrophe systems, changes in a system over time are governed by a deterministic potential function, $V(\boldsymbol{y}; \boldsymbol{\theta})$, written as

$$\frac{d\boldsymbol{y}}{dt} = -\frac{\partial V(\boldsymbol{y}; \boldsymbol{\theta})}{\partial \boldsymbol{y}}, \tag{1}$$

where y is a vector of dependent variables of interest, usually referred to as the *behavioral variables* in the catastrophe literature, and $\boldsymbol{\theta}$ is a vector of control parameters. Equation 1 indicates that the instantaneous changes in the behavioral variables with each unit of increase in time (as the increments in time get infinitely small), $\frac{dy}{dt}$, are negatively related to changes in the potential function with respect to each of the behavioral variables contained in the vector, $\frac{\partial V(j;\boldsymbol{\theta})}{\partial y}$. If the change in the potential function with respect to the pth behavioral variable, $\frac{\partial V(j;\boldsymbol{\theta})}{\partial y_0}$, is positive, the potential function increases with increase in that behavioral variable; if this value is negative, the potential function decreases with increase in the behavioral variable. If $\frac{\partial V(\boldsymbol{y};\boldsymbol{\theta})}{\partial \boldsymbol{y}} = \mathbf{0}$, then all the dependent variables are "static" or do not manifest any changes over time. Thus, the equilibrium points of the cusp catastrophe systems are values of $\boldsymbol{y}$ for which $\frac{\partial V(v;\boldsymbol{\theta})}{\partial y} = \mathbf{0}$. Finding the equilibrium points of a catastrophe system and understanding the dynamics of the system around each of the equilibrium points is a key step toward deducing the long-term behaviors of the system.

One of the simplest forms of catastrophe is the cusp catastrophe (shown in Figure 1) in which

$$-V(y;\boldsymbol{\theta}) = \frac{1}{2}\beta y^2 + \alpha y - \frac{1}{4}y^4, \tag{2}$$

where there is only one behavioral variable and $\boldsymbol{\theta} = [\alpha\beta]'$ consists of the two control parameters in the system (Gilmore, 1981; Stewart & Peregoy, 1983). In practice, these parameters are often expressed as linear, fixed functions of one or more covariates (e.g., Cobb & Zacks, 1985; Grasman, Van der Maas, & Wagenmakers, 2009; Guastello, 1984; van der Maas & Molenaar, 1992). Thus, the control parameters are sometimes referred to as control variables. The parameter α is referred to as the normal variable or asymmetry variable, whereas β is often referred to as the splitting or bifurcation variable. They are responsible for driving hysteresis and bifurcation, respectively, in the system (Gilmore, 1981; Grasman et al., 2009 ; Poston & Stewart, 1978). The equilibrium points of the cusp catastrophe system can be obtained by differentiating Equation 2 with respect to y and setting it to zero as

$$\beta y + \alpha - y^3 = 0. \tag{3}$$

That is, the roots of (3) depict the equilibrium points of the system under different values of α and β. A stable equilibrium is defined as a point into which a system settles in the long run. The different combinations of α and β values, together with their associated equilibrium points, constitute a three-dimensional landscape—referred to as the cusp landscape—as depicted in Figure 1. The nature of the equilibrium points that arise under different values of α and β can be deduced from the Cardan's discriminant of Equation 3 (Grasman et al., 2009; Poston & Stewart, 1978), defined as

$$D = 27\alpha^2 - 4\beta^3. \tag{4}$$

The Cardan's discriminant, D, conveys information about the nature of the roots of Equation 3 and, thus, helps classify the behaviors of the system under different α and β values into four broad categories or sets ($\mathrm{A}, \mathrm{B}, \mathrm{C}_1$, and C_2), which are shown in subplot (A) in Figure 1. [...]

MSEM — MSEM-RS Structural equation modeling (Jöreskog, 1973) is a statistical technique for representing multivariate relationships involving both observed and latent variables. A mixture SEM (MSEM; Dolan & van der Maas, 1998; Jedidi, Jagpal, & DeSarbo, 1997; Vermunt & Magidson, 2005) is an extension of SEM that allows for heterogeneities in the mean and covariance structures of a SEM model conditional on individual i's unobserved group membership, $C_{i0}, (i = 1 \ldots, n)$, typically referred to as the individual's latent class.

In a MSEM, the measurement and structural models for an individual i in latent class h, namely, $C_{i0} = h$, can be expressed, respectively, as

$$(z_i \mid C_{i0} = h) = \nu_h + \mathbf{\Lambda}_h \eta_i + \boldsymbol{\epsilon}_i, (\boldsymbol{\epsilon}_i \mid C_{i0} = h) \sim \mathcal{N}\left(\mathbf{0}, \mathbf{\Psi}_{\epsilon,h}\right), \tag{5}$$

$$(\boldsymbol{\eta}_i \mid C_{i0} = h) = \boldsymbol{\tau}_h + \mathbf{\Gamma}_h \boldsymbol{\eta}_i + \zeta_i, (\zeta_i \mid C_{i0} = h) \sim \mathcal{N}\left(\mathbf{0}, \mathbf{\Psi}_{\zeta,h}\right), \tag{6}$$

where z_i is a $p \times 1$ vector of continuous observed variables for individual i, $\boldsymbol{\eta}_i$ is a $w \times 1$ vector of latent variables, ν_h and $\boldsymbol{\tau}_h$ are, respectively, a $p \times 1$ and a $w \times 1$ vector of intercepts, $\mathbf{\Lambda}_h$ is a $p \times w$ matrix of factor loadings, $\boldsymbol{\epsilon}_i$ is a $p \times 1$ vector of measurement errors, $\mathbf{\Gamma}_h$ is a $w \times w$ matrix of class-dependent regression effects among the latent variables and ζ_i is a $w \times 1$ vector of disturbances. The subscript h highlights the class-specific nature of the parameters and the index of "0" in C_{i0} is used to highlight the timeinvariant nature of the latent class membership in MSEM in contrast to the MSEM-RS to be described later. In cases involving discrete or categorical indicators, the corresponding elements in z_i are unobserved and appropriate link functions are used to relate these unobserved continuous variables to their observed indicators (see, e.g., Jöreskog & Moustaki, 2001). For instance, with ordinal observed variables, the l th unobserved continuous variable, z_{il}, is linked to the corresponding manifest ordinal response, z^*_{il}, as (Jöreskog & Moustaki, 2001; Muthén, 1984)

$$z^*_{il} = s \Leftrightarrow \kappa_{l,s-1} < z_{il} \le \kappa_{l,s}, s = 1, \ldots, S, \tag{7}$$

where $\kappa_{l,h}$, is a set of threshold values for variable l that is held invariant across individuals.

A multinomial logistic regression model is typically used to represent the class probabilities as

$$\Pr\left(C_{i0} = h \mid \boldsymbol{x}_{i0}\right) \triangleq \pi_{h,i0} = \frac{\exp\left(a_{h_0} + \boldsymbol{b}'_{h_0} \boldsymbol{x}_{i0}\right)}{\sum_{s_0=1}^{K_0} \exp\left(a_{s_0} + \boldsymbol{b}'_{s_0} \boldsymbol{x}_{i0}\right)}, \tag{8}$$

where a_{h_0} is the logit intercept for class h; $\boldsymbol{x}_{i0}$ is a vector of covariates used to predict class membership, $\boldsymbol{b}_{h_0}$ is the associated vector of logit slopes and K_0 is the number of latent classes. For identification purposes, one of the classes has to be designated to be the reference class, with a_{K_0} and $\boldsymbol{b}_{K_0}$ set to zeros.

Mixture structural equation models with regime switching (MSEMs-RS) are longitudinal extensions of MSEMs in which individuals are allowed to transition among different latent classes over time (Chow, Grimm, Guillaume, Dolan, & McArdle, 2013; Kaplan, 2008; Muthén & Asparouhov, 2011; Nylund-Gibson, Muthén, Nishina, Bellmore, & Graham, 2013). These different latent classes can be conceived as distinct phases of a process over time and are commonly referred to as regimes in the time series and econometric literature (Hamilton, 1994; Kim & Nelson, 1999). Here, we use the terms latent class and regime interchangeably, and refer to within-person switches between latent classes as regime switching.

To deal with repeated measures in SEM, z_i and η_i are expanded to include $z_i = \left[z'_{i1} \ldots z'_{iT}\right]'$ and $\boldsymbol{\eta}_i = \left[\boldsymbol{\eta}'_{i1} \ldots \boldsymbol{\eta}'_{iT}\right]'$. The sizes of all other components in Equations 5 and 6 are also expanded accordingly to accommodate the presence of multiple time points (e.g., $\boldsymbol{\nu}_h$ is now of size $pT \times 1$, as opposed to $p \times 1$; $\boldsymbol{\Lambda}_h$ is of size $pT \times wT$, etc.). The C_{i0} in Equation 8 may then be conceptualized as an indicator of membership in an initial (baseline) class. A separate multinomial logistic regression model is used to describe each individual i 's class membership at time t, denoted as C_{it}, conditional on membership at time $t-1$. In the general MSEM-RS framework, the transition in class membership can depend on all previous class membership information (e.g., including C_{i0}; Asparouhov & Muthén, 2011). One of the simpler special cases is a first-order Markov specification, which allows the class membership at time t to only depend on the class membership at time $t-1$ as

$$\Pr\left(C_{it} = k \mid C_{i,t-1} = j, \boldsymbol{x}_{it}\right) \triangleq \pi_{jk,it} = \frac{\exp\left(a_{k_t} + \boldsymbol{b}'_{jk'_t}\boldsymbol{x}_{it}\right)}{\sum_{s_t=1}^{K_t} \exp\left(a_{s_t} + \boldsymbol{b}'_{js_t}\boldsymbol{x}_{it}\right)} \tag{9}$$

where $\pi_{jk,it}$ is individual i 's transition probability of moving from class j at time $t-1$ to class k at time t; K_t denotes the number of classes at time t, highlighting the possibility that latent classes may emerge or diminish over time. The parameter a_{k_i} is the logit intercept for the k th class at time $t, \boldsymbol{x}_{it}$ is a vector of fixed covariates for predicting the transition probabilities, with an associated vector of logit slopes, $\boldsymbol{b}_{jk_i}$. Included in $\boldsymbol{x}_{it}$ are $K_{t-1} - 1$ binary constants reflecting the deviation in log-odds of switching into latent class k at time t from latent class j (i.e., $C_{i,t-1} = j, C_{it} = k$), compared with switching into $C_{it} = k$ from the reference class, that is, $C_{i,t-1} = K_{t-1}$. Note that no binary constants are included in $\boldsymbol{x}_{i0}$ in Equation 8 because at Time 1, there are no transition probabilities to be predicted, only class probabilities. In addition, time index is included in elements such as K_v, a_{k_t} and $\boldsymbol{b}_{jk_t}$ to allow the possibility for them to vary over time. Similar to the need to select a reference class among the initial latent classes for model identification purposes, a reference class also has to be selected for each time point and all the logit-related parameters for the reference class have to be set to zeros. This has the effect of setting each row of person i's $K_{t-1} \times K_t$ transition probability matrix at time t to sum to 1 .

Maximum likelihood estimates of the parameters in the MSEM and MSEM-RS can be obtained via the expectation-maximization (EM; Dempster, Laird, & Rubin, 1977; Everitt & Hand, 1981; Titterington, Smith, & Makov, 1985) algorithm. The estimation details have been documented elsewhere (see, e.g., Asparouhov & Muthén, 2011; Muthén & Shedden, 1999) and are not reiterated here. All model fitting presented herein was performed using a commercial structural equation modeling software, Mplus (L. K. Muthén & B. O. Muthén, 2001).

MSEM(-RS) — Cusp Catastrophe In this section, we describe one possible way of circumventing the methodological difficulties behind fitting the nonlinear cusp catastrophe model by approximating it using variations of MSEM/ MSEM-RS composed of multiple regimes within which the system's dynamics are all linear, but differ in meaningful and measurable ways. In this way, we are approximating the multimodal density of y by combining a series of conditional distributions of y that are assumed to be normal within a regime. This approach is similar in rationale to those adopted by others to approximate nonnormal data using mixture of normal distributions (e.g., Dolan & van der Maas, 1998), or to approximate nonlinear functions using mixture of linear functions (e.g., Bauer, 2005). Thus, conditional on (i.e., within) a particular regime, a linear structural equation model can be used, and the presence of selected cusprelated features can be tested by means of model fit indices and diagnostic tools designed for use with MSEM and MSEM-RS. [...]"

Cusp — Curie-Weiss van der Maas (2024): "Interestingly, in the case of a fully connected Ising network (also called the Curie—Weiss model), the emergent behavior—— what is called the mean field behavior——can be described by the cusp (Abe et al. 2017; Poston and Stewart 2014)."

2 Gaussian Graphical Model (GGM)

R-Package: 'ggm', 'GGMridge'

Epskamp, Waldorp, et al. (2018): Let $\boldsymbol{y}_C^\top = \begin{bmatrix} Y_{C1} & Y_{C2} & \dots & Y_{Cm} \end{bmatrix}$ denote a random vector with $\boldsymbol{y}_c$ as its realization. We assume $\boldsymbol{y}_C$ is centered [4] and normally distributed with some variance-covariance matrix $\boldsymbol{\Sigma}$:

$$\boldsymbol{y}_C \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}). \tag{1}$$

The subscript C denotes a case (a row in the spreadsheet). We currently do not define the nature of the observed variables. Thus, $\boldsymbol{y}_C$ can consist of variables relating to one or more subjects, could contain repeated measures on one or more variables, could contain variables of a single subject that do not vary within-subject, and so forth. Consider three examples: (1) Y_1 could represent the level of anxiety of subject p on day 1 and Y_2 the level of anxiety of subject p on day 2, (2) Y_1 could represent the length of subject p and Y_2 the number of times subject p bumps his or her head, and (3) Y_1 could represent the number of cigarettes subject p smokes per day and Y_2 the number of cigarettes another subject $p+1$ smokes per day (case C then represents a dyadic pair).

Partial correlation networks. Assuming multivariate normality, $\boldsymbol{\Sigma}$ encodes all the information necessary to determine how the observed measures relate to one another. However, we will not focus on $\boldsymbol{\Sigma}$ in this paper but rather on its inverse-the precision matrix $\boldsymbol{K}$:

$$\boldsymbol{K} = \boldsymbol{\Sigma}^{-1}.$$

Of particular importance is that the precision matrix can be standardized to encode partial correlation coefficients of two variables, given all other variables (dropping subscript C for notational clarity; Lauritzen 1996): [5]

$$\mathrm{Cor}\left(Y_i, Y_j \mid \boldsymbol{y}_{-(i,j)}\right) = -\frac{\kappa_{ij}}{\sqrt{\kappa_{ii}}\sqrt{\kappa_{jj}}}, \tag{2}$$

in which κ_{ij} denotes an element of $\boldsymbol{K}$, and $\boldsymbol{y}_{-(i,j)}$ denotes the set of variables without i and j. These partial correlations can be graphically displayed in a weighted network, in which each variable Y_i is represented as a node, and connections (edges) between these nodes represent the partial correlation between two variables. When the partial correlation (thus the corresponding element in $\boldsymbol{K}$) equals zero, no edge is drawn. Thus, modeling the inverse variance-covariance matrix, such that every nonzero element is treated as a freely estimated parameter, allows for a sparse model for $\boldsymbol{\Sigma}$ (i.e., every element in $\boldsymbol{\Sigma}$ may be nonzero while some elements in $\boldsymbol{K}$ are zero; Epskamp, Rhemtulla, & Borsboom 2017d). Such a model is termed a GGM (Lauritzen, 1996). Of note, when the sample-variance-covariance matrix is inverted and standardized, no partial correlation will be exactly equal to zero and the GGM will therefore be saturated. To obtain a sparse model with testable implications, in this paper partial correlations are forced to zero either by using thresholding rules or regularization techniques.

When drawing a GGM as a network (often termed a partial correlation network), positive partial correlations are typically visualized with blue or green edges and negative partial correlations with red edges, [6] and the absolute strength of a partial correlation is represented by the width and saturation of an edge (Epskamp et al., 2012). When a partial correlation is zero, we draw no edge between two nodes. As such, the GGM can be seen as a network model of conditional associations; no edge indicates that two variables are independent after conditioning on all other variables in the data set. This allows us to model conditional associations, which we might expect to be zero, rather than marginal associations, which we rarely expect to be zero (Meehl, 1990)."

Epskamp et al. (2017): "In the case of multivariate Gaussian data, this model is termed the Gaussian Graphical Model (GGM; Lauritzen 1996). Here, the partial correlation coefficient is sufficient to test the degree of conditional independence of two variables after conditioning on all other variables; if the partial correlation coefficient is zero, there is conditional independence and hence no edge in the network. As such, partial correlation coefficients can directly be used in the network as edge weights; the strength of connection between two nodes [3]. Such a network is typically encoded in a symmetrical and real valued $p \times p$ weight matrix, $\boldsymbol{\Omega}$, in which element ω_{jk} represents the edge weight between node j and node k:

$$\mathrm{Cor}\left(y_j, y_k \mid \boldsymbol{y}^{-(j,k)}\right) = \omega_{jk} = \omega_{kj}$$

The partial correlation coefficients can be directly obtained from the inverse of variancecovariance matrix $\hat{\boldsymbol{\Sigma}}$, also termed the precision matrix $\hat{\boldsymbol{K}}$ (Lauritzen, 1996):

$$\mathrm{Cor}\left(y_j, y_k \mid \boldsymbol{y}^{-(j,k)}\right) = -\frac{\kappa_{jk}}{\sqrt{\kappa_{kk}}\sqrt{\kappa_{jj}}}$$

Thus, element κ_{jk} of the precision matrix is proportional to to the partial correlation coefficient of variables y_j and y_k after conditioning on all other variables. Since this process simply involves standardizing the precision matrix, we propose the following model:

$$\hat{\boldsymbol{\Sigma}} = \hat{K}^{-1} = \boldsymbol{\Delta}(\boldsymbol{I} - \boldsymbol{\Omega})^{-1}\boldsymbol{\Delta} \tag{4}$$

in which $\boldsymbol{\Delta}$ is a diagonal matrix with $\delta_{jj} = \kappa_{jj}^{-\frac{1}{2}}$ and $\boldsymbol{\Omega}$ has zeroes on the diagonal. This model allows for confirmative testing of the GGM structures on psychometric data. Furthermore, the model can be compared to a saturated model (fully populated off-diagonal values of $\boldsymbol{\Omega}$) and the independence model ($\boldsymbol{\Omega} = \boldsymbol{O}$), allowing one to obtain χ^2 fit statistics as well as fit indices such as the RMSEA (Browne and Cudeck, 1992) and CFI (Bentler, 1990). Such methods of assessing model fit have not yet been used in network psychometrics.

Similar to CFA and SEM, the GGM relies on a critical assumption; namely, that covariances between observed variables are not caused by any latent or unobserved variable. If we estimate a GGM in a case where, in fact, a latent factor model was the true data generating structure, then generally we would expect the GGM to be saturated - i.e., there would be no missing edges in the GGM (Chandrasekaran et al., 2012). A missing edge in the GGM indicates the presence of conditional independence between two indicators given all other indicators; we do not expect indicators to become independent given subsets of other indicators (see also Ellis and Junker 1997; Holland and Rosenbaum 1986). Again, this critical assumption might not be plausible. While variables such as ”Am indifferent to the feelings of others” and ”Inquire about others’ well-being” quite probably interact with each other, it might be far-fetched to assume that no unobserved variable, such as a personality trait, in part also causes some of the variance in responses on these items.”

2.1 Cross-Sectional Data (N=Large, P=Large, T=1)

- Latent Variable GGM (LVGGM; Chandrasekaran et al., 2012; R. Li et al., 2023; K. Wang et al., 2023)
- Gaussian Graphical Interaction Model (GGIM; Fitch, 2019)
- latent GGM (Zheng et al., 2022)
- covariate-dependent GGM (cdexGGM; J. Wang and Gao, 2025)
- Random GGM Cai, 2017

2.1.1 LVGGM

Chandrasekaran et al. (2012): ”Suppose we observe samples of a subset of a collection of random variables. No additional information is provided about the number of latent variables, nor of the relationship between the latent and observed variables. Is it possible to discover the number of latent components, and to learn a statistical model over the entire collection of variables? We address this question in the setting in which the latent and observed variables are jointly Gaussian, with the conditional statistics of the observed variables conditioned on the latent variables being specified by a graphical model. [...]

[...] More concretely, X is a Gaussian random vector in $\mathbb{R}^{p+h}$, O and H are disjoint subsets of indices in $\{1, \ldots, p+h\}$ of cardinalities $|O| = p$ and $|H| = h$, and the corresponding subvectors of X are denoted by X_O and X_H, respectively. Let the covariance matrix underlying X be denoted by $\Sigma^*_{(OH)}$. The marginal statistics corresponding to the observed variables X_O are given by the marginal covariance matrix Σ^*_O, which is simply a submatrix of the full covariance matrix $\Sigma^*_{(OH)}$. However, suppose that we parameterize our model by the concentration matrix $K^*_{(OH)} = \left(\Sigma^*_{(OH)}\right)^{-1}$, which as discussed above reveals the connection to graphical models. Here the submatrices $K^*_O, K^*_{O,H}, K^*_H$ specify (in the full model) the dependencies among the observed variables, between the observed and latent variables, and among the latent variables, respectively. In such a parameterization, the marginal concentration matrix $\left(\Sigma^*_O\right)^{-1}$ corresponding to the observed variables X_O is given by the Schur complement [16] with respect to the block K^*_H :

$$\tilde{K}^*_O = (\Sigma^*_O)^{-1} = K^*_O - K^*_{O,H}\,(K^*_H)^{-1}\,K^*_{H,O}. \tag{1.1}$$

Thus if we only observe the variables X_O, we only have access to Σ^*_O (or $\tilde{K}^*_O$). The two terms that compose $\tilde{K}^*_O$ above have interesting properties. The matrix K^*_O specifies the concentration matrix of the conditional statistics of the observed variables given the latent variables. If these conditional statistics are given by a sparse graphical model, then K^*_O is sparse. On the other hand, the matrix $K^*_{O,H}\left(K^*_H\right)^{-1} K^*_{H,O}$ serves as a summary of the effect of marginalization over the latent variables X_H. This matrix has small rank if the number of latent, unobserved variables X_H is small relative to the number of observed variables X_O. Therefore the marginal concentration matrix $\tilde{K}^*_O$ is generally not sparse due to the additional low-rank term $K^*_{O,H}\left(K^*_H\right)^{-1} K^*_{H,O}$. Hence standard graphical model selection techniques applied directly to the observed variables X_O are not useful.

A modeling paradigm that infers the effect of the latent variables X_H would be more suitable in order to provide a concise explanation of the underlying statistical structure. Hence we approximate the sample covariance by a model in which the concentration matrix decomposes into the sum of a sparse matrix and a low-rank matrix, which reveals the conditional graphical model structure in the observed variables as well as the number of and effect due to the unobserved latent variables. Such a method can be viewed as a blend of principal component analysis and graphical modeling. In standard graphical modeling one would directly approximate a concentration matrix by a sparse matrix to learn a sparse graphical model, while in principal component analysis the goal is to explain the statistical structure underlying a set of observations using a small number of latent variables (i.e., approximate a covariance matrix as a low-rank matrix). In our framework we learn a sparse graphical model among the observed variables conditioned on a few (additional) latent variables. These latent variables are not principal components, as the conditional statistics (conditioned on these latent variables) are given by a graphical model. Therefore we refer to these latent variables informally as latent components."

K. Wang et al. (2023): "We focus on estimating $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$, the precision matrix encoding the graph structure of interest [8, 20, 41, 52]. When latent confounders are present, the covariance matrix for the observed data, $\boldsymbol{\Sigma}_{\text{obs}}$ can be expressed as

$$\Sigma_{obs} = \Sigma + L_{\Sigma} \tag{1}$$

where the positive semidefinite matrix $\boldsymbol{L}_{\boldsymbol{\Sigma}}$ reflects the effect of latent confounders. One approach, which we will call PCA+GGM, is motivated by confounders that affect the marginal correlation between observed variables [41] and uses principal component analysis (PCA) as a preprocessing step to remove the effect of these confounders [4,28]. PCA removes the leading eigencomponents from $\boldsymbol{\Sigma}_{obs}$ which are assumed to be $\boldsymbol{L}_{\boldsymbol{\Sigma}}$, then a second stage of standard GGM inference follows. PCA+GGM has shown to be useful in estimating gene co-expression networks, where correlated measurement noise and batch effects induce large extraneous marginal correlations between observed variables [19,21,22,27,34,48].

Alternatively, Eq. (1) can be reparametrized as the observed precision matrix by applying the Sherman-Morrison identity [26] as,

$$\Omega_{obs} = \Sigma_{obs}^{-1} = \Omega - L_{\Omega}, \tag{2}$$

where $\boldsymbol{L}_{\Omega}$ again reflects the effect of unobserved confounding, i.e., unobserved nodes in a graph [12]. One such approach, known as latent variable Gaussian Graphical Models (LVGGM), uses parameterization (2) and involves joint inference for $\boldsymbol{\Omega}$ and $\boldsymbol{L}_{\Omega}$. The motivation behind LVGGM is to address the effect of unobserved variables in the complete data graph, which affect the partial correlations of the variables in the observed precision matrix $\boldsymbol{\Omega}$. This perspective can be particularly useful when the unobserved variables would have been included in the graph, had they been observed.

In previous work, either parameterization (1), e.g., PCA+GGM, or (2), e.g., LVGGM, has been used, depending on the source of confounding and motivations as described earlier. Typically, LVGGM is appropriate when confounding is induced by unobserved nodes in a complete data graph of interest, whereas PCA+GGM is more appropriate when confounding corresponds to nuisance variables, e.g., from batch effects.

In practice, the selection between these two methods will depend on user's belief about the type of confounding present in the observed data. In this paper, our goal is to explore a way to address the effect of confounders in order to obtain the graph structure encoded in Ω without making such selection.

To achieve this goal, we generalize two seemingly different methods, PCA+GGM and LVGGM, into a common framework for addressing the effect of $\boldsymbol{L}_{\boldsymbol{\Sigma}}$ in order to obtain the graph structure encoded in $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$. Based on the generalization, We propose a new method, PCA+LVGGM, to address two different sources of confounding. The combined approach is more general, since PCA+LVGGM contains both LVGGM and PCA+GGM as special cases. To our knowledge, the two methods of addressing confounding have not been discussed together in the literature. [...]

LVGGM One method for controlling the effects of confounders is the Latent Variable Gaussian Graphical Model (LVGGM) approach first proposed by [11]. They assume that the number of latent factors is small compared to the number of observed variables, and that the conditional dependencies among the observed variables conditional on the latent factors is sparse. Consider a $(p + r)$ dimensional mean-zero normal random variable $\boldsymbol{X} = [\boldsymbol{X}_O, \boldsymbol{X}_H]^{\mathrm{T}}$, where $\boldsymbol{X}_O \in \mathbb{R}^p$ is observed and $\boldsymbol{X}_H \in \mathbb{R}^r$ is latent. Let $\boldsymbol{X}$ have precision matrix $\boldsymbol{\Omega} \in \mathbb{R}^{(p+r)\times(p+r)}$, and the submatrices $\boldsymbol{\Omega}_O \in \mathbb{R}^{p\times p}$, $\boldsymbol{\Omega}_H \in \mathbb{R}^{r\times r}$ and $\Omega_{O,H} \in \mathbb{R}^{p\times r}$ specify the dependencies between observed variables, between latent variables and between the observed and latent variables respectively. By Schur complement, the inverse of the observed covariance matrix satisfies:

$$\boldsymbol{\Omega}_{obs} = \boldsymbol{\Sigma}_{obs}^{-1} = \boldsymbol{\Omega}_O - \boldsymbol{\Omega}_{O,H}\boldsymbol{\Omega}_H^{-1}\boldsymbol{\Omega}_{O,H}^{\mathrm{T}} = \boldsymbol{\Omega} - \boldsymbol{L}_{\boldsymbol{\Omega}}. \tag{4}$$

where $\boldsymbol{\Omega} = \boldsymbol{\Omega}_0$ encodes the conditional independence relations of interest and is sparse by assumption. $\boldsymbol{L}_\Omega = \boldsymbol{\Omega}_{O,H}\boldsymbol{\Omega}_H^{-1}\boldsymbol{\Omega}_{O,H}^{\mathrm{T}}$ reflects the low-rank effect of latent variables $\boldsymbol{X}_H$. Based on this sparse plus low-rank decomposition, [11] proposed the following problem:

$$\begin{array}{ll} \underset{\Omega, \boldsymbol{L}_\Omega}{\text{minimize}} & -\ell\left(\boldsymbol{\Omega} - \boldsymbol{L}_\Omega; \boldsymbol{\Sigma}_n\right) + \lambda\|\boldsymbol{\Omega}\|_1 + \gamma\left\|\boldsymbol{L}_\Omega\right\|_* \\ \text{subject to} & \boldsymbol{L}_\Omega \succeq \boldsymbol{0}, \boldsymbol{\Omega} - \boldsymbol{L}_\Omega \succ \boldsymbol{0} \end{array} \tag{5}$$

where $\boldsymbol{\Sigma}_n$ is the observed sample covariance matrix and $\ell(\boldsymbol{\Omega}, \boldsymbol{\Sigma}) = \ln(\det(\boldsymbol{\Omega})) - \mathrm{Tr}(\boldsymbol{\Omega}\boldsymbol{\Sigma})$ is the Gaussian log-likelihood function. The ℓ_1-norm encourages sparsity on Ω and the nuclear norm encourages low-rank structure on $\boldsymbol{L}_\Omega$. [...]

PCA+GGM Unlike LVGGM, which involves a decomposition of the observed data precision matrix, PCA+GGM involves a decomposition of the observed data covariance matrix:

$$\boldsymbol{\Sigma}_{obs} = \boldsymbol{\Omega}_{obs}^{-1} = (\boldsymbol{\Omega} - \boldsymbol{L}_\Omega)^{-1} = \boldsymbol{\Omega}^{-1} + \boldsymbol{L}_{\boldsymbol{\Sigma}}. \tag{6}$$

Motivated by confounding from measurement error and batch effects, [41] proposed the principal components correction (PC-correction) for removing $\boldsymbol{L}_{\boldsymbol{\Sigma}}$. Consider observed data $\boldsymbol{X}_{\mathrm{obs}}$, such that

$$\boldsymbol{X}_{obs} = \boldsymbol{X} + \boldsymbol{A}\boldsymbol{Z}, \tag{7}$$

where $\boldsymbol{X} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma})$ and $\boldsymbol{Z} \sim N(\boldsymbol{0}, \boldsymbol{I}_r)$. Matrix $\boldsymbol{A} \in \mathbb{R}^{p\times r}$ is non-random so that $\boldsymbol{L}_{\boldsymbol{\Sigma}} = \boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}$. In general, additional structural assumptions are needed to distinguish $\boldsymbol{L}_{\boldsymbol{\Sigma}}$ from $\boldsymbol{\Sigma}$. As we will discuss in Section 3, one of our contributions is to show that if the spectral norm of $\boldsymbol{L}_{\boldsymbol{\Sigma}}$ is large relative to that of $\boldsymbol{\Sigma}$, then under mild conditions, $\boldsymbol{L}_{\boldsymbol{\Sigma}}$ is close to the sum of the first few eigencomponents of $\boldsymbol{\Sigma}_{\mathrm{obs}}$. Therefore, one can remove the first r eigencomponents from $\boldsymbol{\Sigma}_{\mathrm{obs}}$ [41]. This PCA+GGM method is described in Procedure 1. Note that the number of principal components needs to be determined a priori, which we discuss in subsequent sections.

PCA+LVGGM As previously mentioned, while LVGGM and PCA+GGM solve the same problem, they are motivated by different sources of confounding. In applications, the observed data may be corrupted by multiple sources of confounding, and thus elements from both methods are needed. For example, in the biological application discussed in Section 5.1, both batch effects and unmeasured biological variables likely confound estimates of graph structure between observed variables. This motivates us to propose the PCA+LVGGM strategy described below.

As (4) illustrated, the observed precision matrix Ω' may have been corrupted by a latent factor $\boldsymbol{L}_\Omega$:

$$\boldsymbol{\Omega}' = \boldsymbol{\Omega} - \boldsymbol{L}_\Omega. \tag{8}$$

Now, rewriting (8) in terms of $\boldsymbol{\Sigma} = \boldsymbol{\Omega}^{-1}$ and $\boldsymbol{\Sigma}' = \boldsymbol{\Omega}'^{-1}$, applying the Sherman-Morrison identity on $\boldsymbol{\Omega}'$ gives,

$$\boldsymbol{\Sigma}' = \boldsymbol{\Sigma} + \boldsymbol{L}'_\Omega \tag{9}$$

where $\boldsymbol{L}'_\Omega$ is still a low-rank matrix. If $\boldsymbol{\Sigma}'$ is further corrupted by an additive latent factor represented by $\boldsymbol{L}_\Sigma$, the following equation described the observed matrix $\boldsymbol{\Sigma}_{\mathrm{obs}}$:

$$\boldsymbol{\Sigma}_{obs} = \boldsymbol{\Sigma}' + \boldsymbol{L}_\Sigma = \boldsymbol{\Sigma} + \boldsymbol{L}'_\Omega + \boldsymbol{L}_\Sigma \tag{10}$$

In the above example, following our theoretical analysis in Section 3, if the spectral norm of $\boldsymbol{L}_\Sigma$ is much larger than that of $\boldsymbol{\Sigma}$ and $\boldsymbol{L}'_\Omega$, then removing $\boldsymbol{L}_\Sigma$ using the PC-correction is likely to be effective. If the spectral norm of $\boldsymbol{L}'_\Omega$ is not much larger than that of $\boldsymbol{\Sigma}$, then PC-correction is not a good choice to remove $\boldsymbol{L}'_\Omega$. If $\boldsymbol{L}'_\Omega$ is dense, then $\boldsymbol{\Omega}$ and $\boldsymbol{L}_\Omega$ can be well estimated by LVGGM. In (10), the overall confounding $\boldsymbol{L}'_\Omega + \boldsymbol{L}_\Sigma$ is the sum of two low-rank components with different norms, we can consider using both methods: first remove $\boldsymbol{L}_\Sigma$ via eigendecomposition, then apply LVGGM to estimate $\boldsymbol{\Omega}$ and $\boldsymbol{L}_\Omega$. We call this procedure PCA+LVGGM and it is shown in Procedure 2. We discuss methods for setting the ranks for $\boldsymbol{L}_\Sigma$ (defined as r_p) and $\boldsymbol{L}'_{\boldsymbol{\Omega}}$ (defined as r_L) in Section 3.5. [...]"

2.1.2 GGIM

Fitch (2019): "In this paper, we introduce a new directed graphical model from Gaussian data: the Gaussian graphical interaction model (GGIM). The development of this model comes from considering stationary Gaussian processes on graphs, and leveraging the equations between the resulting steady-state covariance matrix and the Laplacian matrix representing the interaction graph. Through the presentation of conceptually straightforward theory, we develop the new model and provide interpretations of the edges in the graphical model in terms of statistical measures. We show that when restricted to undirected graphs, the Laplacian matrix representing a GGIM is equivalent to the standard inverse covariance matrix that encodes conditional dependence relationships. Furthermore, our approach leads to a natural definition of directed conditional independence of two elements in a stationary Gaussian process. [...]

While GGMs and sparse inverse covariance estimation provide a straightforward and generalizable framework for approximating conditional independence relationships between variables from a sample set of observations, the question remains of how to graphically model directed interactions. For example, consider a random vector with three variables, a, b, and c, where a influences b, b influences c, and c influences a. The conditional independence graph would contain undirected edges between all three nodes, however, the true sparsest model of interaction would be a directed cycle from a to b to c to a. This is to say that conditional dependence between any two variables does not imply that the two variables influence each other equally, therefore, inverse covariance estimation provides limited information on the interactions between variables.

Learned directed graphical models such as Bayesian networks (Barber, 2012), and linear Gaussian structure equation models (SEMs) (Bollen, 1989) overcome some of the limitations of inverse covariance estimation by representing casual relationships between variables as edges in directed acyclic graphs (DAGs). While these models have been successfully and broadly implemented for a wide variety of applications, they still have non-trivial disadvantages. Most notably, due to their hierarchical nature, neither Bayesian networks nor SEMs can directly model cyclic interactions. As a result, the simple directed cycle from the previous paragraph cannot be accurately reproduced through these methods. This is an issue because the ability to learn and model cyclic interactions is critical to our understanding of complex systems. In a biological system, for example, cyclic interactions can be indicative of regulatory feedback loops. Thus, when attempting to learn graphical structure from data, the failure to identify feedback loops results in an incomplete understanding of the fundamental mechanisms of the biological system. There is, therefore, a need for learned graphical models of Gaussian data that represent directed relationships between variables without topological restrictions. [...]

It is noted that GGIMs inherently differ from Bayesian networks and linear Gaussian SEMs. The models in this paper arise from the steady-state characteristics of a set of equations where the rate of change of a variable is a linear combination of the state of the remaining variables. SEMs arise from a set of equations where the state of a variable is a linear combination of the state of parent variables. Bayesian networks and SEMs model strictly hierarchical relationships through directed edges that represent causality, whereas GGIMs model the interactions between variables. The hierarchical nature of Bayesian networks and SEMs mandates that they are represented by DAGs, while no restriction is placed on GGIMs, which can have any general graph structure. Due to these conceptual differences we do not seek to apply notions such as Markov equivalence (Chickering, 2002), morality (Lauritzen and Spiegelhalter, 1988), or faithfulness (Zhang and Spirtes, 2003) to GGIMs. [...]

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$ be a connected, directed graph representing interactions between variables, where $\mathcal{V} = \{1, 2, \ldots, p\}$ is the set of variables (also refered to as nodes), and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of m edges. $A \in \mathbb{R}^{p \times p}$ is the adjacency matrix where element $a_{i,j}$ is the weight on edge (i, j). If $(i, j) \in \mathcal{E}$ then $a_{i,j} > 0$; otherwise $a_{i,j} = 0$. The out-degree of node i is calculated as $d_i = \sum_{j=1}^{p} a_{i,j}$. The out-degree matrix is a diagonal matrix of node out-degrees, $D = \operatorname{diag}\left\{d_1, d_2, \ldots, d_p\right\}$. The associated directed Laplacian matrix is defined as $L = D - A$. This corresponds to a 'sensing' interpretation of node interactions where an edge from i to j indicates that node i senses the state of node j. We primarily consider this interpretation of edge orientation to maintain consistency with previous research on Gaussian processes on directed graphs, e.g. (Fitch, 2019; Young et al., 2016). The transposed Laplacian corresponds to the 'sending' interpretation where an edge from i to j indicates that the state of node i is available to node j.

Throughout the paper, partitioning of a given matrix, G, will follow

$$G = \begin{bmatrix} G_{a,a} & G_{a,b} \\ G_{a,b}^T & G_{b,b} \end{bmatrix}, \tag{1}$$

where $G_{a,a} \in \mathbb{R}^{2 \times 2}$. [...]

GGM — stationary Gaussian Process We begin this section by reviewing the relationship between conditional independence and the inverse covariance matrix for a random vector $\boldsymbol{X}$ with Gaussian distribution $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$. Then, we provide a lemma relating the inverse covariance matrix of $\boldsymbol{X}$ to the undirected interaction graph for a stationary Gaussian process $\boldsymbol{x}$ with distribution $\mathcal{N}(0, \Sigma)$. Next, we discuss how the Gaussian process viewpoint allows us to consider directed interactions between variables. The section concludes with the definition for a new directed graphical model, the Gaussian graphical interaction model.

There are multiple approaches for demonstrating that the sparsity pattern of the precision matrix is representative of the underlying undirected independence graph structure. For example, by using Schur complements it is possible to show that zeros in the precision matrix correspond to conditional independence relationships (Anderson, 2003). Consider a covariance matrix partitioned as in (1), where $\Sigma_{a,a} \in \mathbb{R}^{2\times 2}$ contains elements indexed by 1, 2, and let $\boldsymbol{P} = \boldsymbol{\Sigma}^{-1}$. Then, the submatrix $\boldsymbol{P}_{a,a}$ can be written as

$$\boldsymbol{P}_{a,a} = \left(\boldsymbol{\Sigma}_{a,a} - \boldsymbol{\Sigma}_{a,b} \left(\boldsymbol{\Sigma}_{b,b}\right)^{-1} \boldsymbol{\Sigma}_{a,b}^{T}\right)^{-1} = \left(\boldsymbol{\Sigma}_{a|b}\right)^{-1}.$$

Therefore $\boldsymbol{P}_{a,a}$ is the inverse of the conditional covariance, $\boldsymbol{\Sigma}_{a|b}$, and two-by-two matrix inversion, $\boldsymbol{P}_{1,2} = 0$ implies that $\boldsymbol{\Sigma}_{1,2|b} = 0$. In other words, variables indexed by i and j are conditionally independent given the remaining variables if and only if $\boldsymbol{P}_{i,j} = \boldsymbol{\Sigma}_{i,j}^{-1} = 0$.

We now demonstrate that the same relationship between the precision matrix and the underlying graph can be established from dynamic systems perspective. Rather than think of the n i.i.d. samples as drawn from a multivariate normal distribution, they can be thought of as belonging to a stationary Gaussian process. Recall, for a stochastic process to be stationary its unconditional joint probability distribution is unchanged under time shifts. This further implies that the mean and covariance do not change over time. Before stating the lemma regarding the relationship between inverse covariance matrix of a random vector and the interaction graphs of stationary Gaussian processes, we first provide a theorem from (Arnold, 1992). In the following, $d\boldsymbol{W}$ represents increments drawn from independent Wiener processes.

Theorem 2 (From (Arnold, 1992)) The solution of the equation

$$d\boldsymbol{x} = (M(t) + a(t))\boldsymbol{x}dt + B(t)d\boldsymbol{W}, \quad \boldsymbol{x}_{t_0} = c, \tag{2}$$

is a stationary Gaussian process if $M(t) \equiv M$, $a(t) \equiv 0$, $B(t) \equiv B$, the eigenvalues of M have negative real parts, and c is $\mathcal{R}(0, K)$-distributed, where Σ, the steady-state covariance matrix, is

$$\Sigma = \int_0^\infty e^{Mt} BB^T e^{M^T t} dt$$

or equivalently the solution of the Lyapunov equation

$$M\Sigma + \Sigma M^T = -BB^T.$$

Lemma 3 Consider a stationary Gaussian process $\boldsymbol{x} \in \mathbb{R}^p$ with multivariate Gaussian distribution $\mathcal{N}(0, \Sigma)$ and a random vector $X \in \mathbb{R}^p$ with multivariate Gaussian distribution $\mathcal{N}(0, \Sigma)$. Then, up to a scaling, the unique symmetric Laplacian matrix corresponding to the steady-state undirected interaction graph associated with $\boldsymbol{x}$ is equal to the inverse covariance matrix associated with the random vector X.

Proof Consider equation (2) in the context of a stochastic process on a graph by replacing M with the negative graph Laplacian $-L$ and taking $B = \sigma I_p$. The resulting equation,

$$d\boldsymbol{x} = -L\boldsymbol{x}dt + \sigma d\boldsymbol{W} \tag{3}$$

represents a noisy diffusion process where each node i in the graph averages its state, $\boldsymbol{x}_i$, with the state of its neighbors, subject to noise in the state of each node with standard deviation σ. Without loss of generality, let $\sigma^2 = 2$. As the eigenvalues of $-L$ have negative real parts, the solution $\boldsymbol{x}$ is a stationary process. The corresponding steady-state covariance matrix is the unique solution to

$$L\Sigma + \Sigma L^T = 2I_p \tag{4}$$

When restricting to only undirected graphs (L symmetric), the unique solution $\Sigma = L^{-1}$ satisfies (4).

Therefore, we have demonstrated that for undirected graphical models, the stationary Gaussian diffusion process represented by the steady-state dynamics of $\boldsymbol{x}$ recovers the relationship between the precision matrix and graph structure, up to a scaling. More specifically, by the relationship $L = \Sigma^{-1}$, the non-zero elements of the inverse covariance matrix indicate both conditional dependencies and edges in the underlying interaction graph.

Returning to the stationary Gaussian process (3), we can ask what happens when L represents a directed graph and is no longer symmetric. In this case, the solution Σ to (4) still exists, is unique, and is symmetric, but is no longer a scale of the inverse Laplacian matrix. In the reverse direction, given a covariance matrix, Σ, a corresponding L is not unique. We formalize the relationship between a L and Σ in the following proposition, which follows from considering the relationships provided for rank deficient matrices in (Fitch, 2019) in the context of full-rank L and Σ.

Proposition 4 Consider a stationary Gaussian process $\boldsymbol{x} \in \mathbb{R}^p$ with multivariate Gaussian distribution $\mathcal{N}(0, \Sigma)$. Then the Laplacian matrices corresponding to the family of steady-state equivalent interaction graphs between the p variables of $\boldsymbol{x}$ are given as a function of $\kappa \in \mathbb{R}^{p \times p}$ skew-symmetric matrices, by

$$L(\kappa) = \left(I_p + \kappa\right)\Sigma^{-1}. \tag{5}$$

Proof Direct substitution yields

$$L\Sigma + \Sigma L^T = \left(I_p + \kappa\right)\Sigma^{-1}\Sigma + \Sigma\Sigma^{-1}\left(I_p - \kappa\right) = I_p + \kappa + I_p - \kappa = 2I_p$$

It can be shown (i.e. (Barnett and Storey, 1967)) that there do not exist any Laplacian matrices corresponding steady-state equivalent interaction graphs outside of those that can be decomposed as (5).

This leads to the definition of the Gaussian Graphical Interaction Model:

GGIM **Definition 5** Consider a stationary Gaussian process with multivariate distribution $\mathcal{N}(0, \Sigma)$. Then a Gaussian Graphical Interaction Model for the stationary process is the graph induced by a Laplacian matrix $\hat{L}$ where

$$\hat{L} = \left(I_p + \kappa\right)\Sigma^{-1}, \quad \kappa \in so(p). \tag{6}$$

Remark 6 Similarly to most structure learning problems, learning a GGIM is inherently indeterminate, as there are an infinite number of matrices $\hat{L}$ that can satisfy the definition of a GGIM for a given stationary Gaussian process. In practice, we will add a regularization term penalizing the l_1-norm of $\hat{L}$. While constructing a directed interaction model based on a symmetric covariance matrix may seem at first counter intuitive, we note that for the l_1-norm, finding the sparsest directed matrix L that satisfies (4) inherently provides a generalization of a very natural assumption on edge directions. To illustrate, consider a two variable example consistent with the sending interpretation of variable interaction (corresponding to L^T). If $\Sigma_{1,1} > \Sigma_{2,2}$ then the resulting l_1-norm sparsest GGIM will contain an edge from 2 to 1 . In a sense, this assumes the variable with less uncertainty passes information to the variable with higher uncertainty. The directed edge in this example represents the sparsest interaction that could lead to the covariance matrix Σ.

[...] the submatrix equation from (4) pertaining to the conditional covariance matrix of two nodes allows us to write the off-diagonal elements of P in terms of two directed components. This then naturally leads to a notion of directed conditional dependence. [...]"

2.1.3 latent GGM

Zheng et al. (2022): "In this article, we focus on the latent Gaussian graphical model for both binary data and mixed types of data with continuous and binary variables. Our model assumes that each binary variable is generated by discretizing a latent Gaussian variable. We propose to perform maximum likelihood estimation using a modified expectation-maximization (EM) algorithm. [...]

for binary data **Definition 1.** Let $\boldsymbol{X} = \left(X_1, \ldots, X_p\right)^T \in \{0,1\}^p$ be a p-dimensional binary vector. The binary random vector $\boldsymbol{X}$ satisfies the latent Gaussian graphical model if there exists a p-dimensional random vector $\boldsymbol{Z} = \left(Z_1, \ldots, Z_p\right)^T \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\operatorname{diag}(\boldsymbol{\Sigma}) = \boldsymbol{I}_p$, such that

$$X_j = \mathrm{I}\left(Z_j > 0\right) \quad \text{for all } j = 1, \ldots, p,$$

where $\boldsymbol{I}(\cdot)$ is the indicator function, $\operatorname{diag}(\boldsymbol{\Sigma})$ denotes the diagonal matrix with the same diagonal entries as $\boldsymbol{\Sigma}$, and $\boldsymbol{I}_p$ is a $p \times p$ identity matrix. Then we denote $\boldsymbol{X} \sim \mathrm{LN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Here, the restriction on the covariance matrix $\boldsymbol{\Sigma}$, $\operatorname{diag}(\boldsymbol{\Sigma}) = \boldsymbol{I}_p$, will ensure model identifiability. This model is equivalent to the latent Gaussian copula model for binary data (Fan et al., 2017). With the Gaussian assumption, the conditional dependence structure among the elements of the latent variable $\boldsymbol{Z}$ is encoded in the sparsity pattern of the precision matrix $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$. Specifically, Z_j and Z_k are conditionally independent given the remaining variables $\boldsymbol{Z}_{-(j,k)}$ if and only if $\Omega_{jk} = 0$. Thus, it is crucial to obtain a good estimate of $\boldsymbol{\Omega}$ in order to infer the graph structure under the latent Gaussian model.

Let $\Theta = \left\{(\boldsymbol{\mu}, \boldsymbol{\Sigma}) : \boldsymbol{\mu} \in \mathbb{R}^p, \boldsymbol{\Sigma} \in \mathrm{SP}_+^p, \operatorname{diag}(\boldsymbol{\Sigma}) = \boldsymbol{I}_p\right\}$ denote the parameter space of the latent Gaussian model for binary data, where SP_+^p denotes the space of $p \times p$ symmetric positive definite matrices. Let $\boldsymbol{\mu}^\star$ and $\boldsymbol{\Sigma}^\star$ denote the true parameters. Given n i.i.d. vector-valued observations $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)} \sim \mathrm{LN}(\boldsymbol{\mu}^\star, \boldsymbol{\Sigma}^\star)$, the marginal likelihood is

$$L\left(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}; \boldsymbol{\theta}\right) = \prod_{i=1}^{n} \int_{\boldsymbol{I}(z>\mathbf{0})=\boldsymbol{x}^{(i)}} \phi_\theta(z) dz, \tag{1}$$

where $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma}) \in \Theta$ denotes all the parameters in the latent Gaussian model, $\boldsymbol{I}(z > \mathbf{0}) = \left(\mathrm{I}(z_1 > 0), \ldots, \mathrm{I}\left(z_p > 0\right)\right)^T$, and where $\phi_\theta(z) = \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(z - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(z - \boldsymbol{\mu})\right\}$ denotes the probability density function (pdf) of a multivariate Gaussian distribution with parameters θ.

With no known closed-form solution, it is challenging to obtain the MLE of $\boldsymbol{\theta}$ by directly maximizing the marginal likelihood $L\left(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}; \boldsymbol{\theta}\right)$. In the next subsection, we propose a modified EM algorithm to estimate the parameters $\boldsymbol{\theta}$ in the latent Gaussian model for binary data. [...]

for mixed data **Definition 2** Let $\boldsymbol{X} = \left(\boldsymbol{X}_1^T, \boldsymbol{X}_2^T\right)^T$, where $\boldsymbol{X}_1 \in \mathbb{R}^{p_1}$ denotes a p_1-dimensional continuous random vector, and $\boldsymbol{X}_2 \in \{0,1\}^{p_2}$ denotes a p_2-dimensional binary random vector. We say that the random vector $\boldsymbol{X}$ follows a latent Gaussian model for mixed data if there exists a p_2-dimensional random vector $\boldsymbol{Z}$ such that [...]"

2.1.4 cdexGGM

J. Wang and Gao (2025): "In this article, we propose covariate-dependent GGM (cdexGGM) via a novel parameterization that guarantees the positive definiteness of $\boldsymbol{\Sigma}^{-1}$ under modest assumptions. We further relax the restriction on the diagonal parameters and allow both off-diagonal and diagonal parameters of the precision matrix to change with covariates. We propose to employ the likelihood approach to jointly estimate all the edge and vertex parameters. [...]

Let $\boldsymbol{Y} = (\boldsymbol{Y}_1, \boldsymbol{Y}_2, \ldots, \boldsymbol{Y}_n)^T$ be a collection of independent random vectors, with each $\boldsymbol{Y}_m \sim N(\boldsymbol{\mu}_m(x_m), \boldsymbol{\Sigma}_m(x_m))$, $m = 1, \ldots, n$, where x_m denotes the observed covariate and $Y_m = \left(Y_{m1}, Y_{m2}, \ldots, Y_{mp}\right)^T$. For the sake of simplicity, we assume that the observations are centered, namely $\boldsymbol{\mu}_m(x_m) = \mathbf{0}$. First consider the simple case that x_m is a univariate covariate, which is min-max scaled so that $0 \le x_m \le 1$. We assume that $\Sigma_m^{-1}(x_m)$ can be formulated as

$$\begin{aligned} \boldsymbol{\Sigma}_m^{-1}(x_m) = \boldsymbol{K}_m &= x_m \boldsymbol{Q}_1 + (1 - x_m)\boldsymbol{Q}_0 \\ &= \boldsymbol{Q}_0 + x_m(\boldsymbol{Q}_1 - \boldsymbol{Q}_0) = \boldsymbol{Q}_0 + x_m \boldsymbol{P}_1 \end{aligned} \tag{1}$$

The matrix $\boldsymbol{Q}_0$ is the precision matrix when $x_m = 0$, and $\boldsymbol{Q}_1$ is the precision matrix when $x_m = 1$. For any arbitrary value of $x_m \in [0,1]$, the precision matrix is a weighted average of these two matrices. If $\boldsymbol{Q}_0$ and $\boldsymbol{Q}_1$ are assumed to be positive definite, $\boldsymbol{K}_m$ is guaranteed to be positive definite. After the reformulation, we have $\boldsymbol{K}_m$ expressed in a regression format where $\boldsymbol{Q}_0$ is the intercept matrix, and $\boldsymbol{P}_1$ is the slope matrix associated with the observed covariate.

We can further extend the model (1) to the general setting of multiple covariates

$$\boldsymbol{\Sigma}_m^{-1}(\boldsymbol{x_m}) = \boldsymbol{K}_m = \sum_{h=1}^{H} x_m^{(h)} \boldsymbol{Q}_h + \left(1 - \sum_{h=1}^{H} \frac{x_m^{(h)}}{H}\right) \boldsymbol{Q}_0 \tag{2}$$

$$= \boldsymbol{Q}_0 + \sum_{h=1}^{H} x_m^{(h)} (\boldsymbol{Q}_h - \boldsymbol{Q}_0/H)$$

$$= \boldsymbol{Q}_0 + \sum_{h=1}^{H} x_m^{(h)} \boldsymbol{P}_h \tag{3}$$

where $\boldsymbol{P}_h = \boldsymbol{Q}_h - \boldsymbol{Q}_0/H$, H is the number of covariates, $\boldsymbol{x_m} = \left(x_m^{(1)}, \ldots, x_m^{(H)}\right)^T$, and $x_m^{(h)} \in [0, 1]$ represents the hth covariate of the m th observation. In (2), $\boldsymbol{\Sigma}_m^{-1}(\boldsymbol{x_m})$ is a linear combination of $Q_0, \ldots, Q_H$ with nonnegative coefficients. Under the assumption that all these matrices are positive definite, $\boldsymbol{\Sigma}_m^{-1}(\boldsymbol{x_m})$ is guaranteed to be positive definite for any combinations of possible values of covariates. The $\boldsymbol{Q}_0$ is the baseline precision matrix when all covariates are set to zero. The matrices $\boldsymbol{P}_1, \ldots, \boldsymbol{P}_H$ can be treated as the slopes of the covariates to describe how the current network structure deviates from its baseline state resulting from $x_m^{(h)}$. This general model of covariate-dependent GGMs (cdexGGM) encompasses standard GGMs, group-specific GGMs, and time-varying GGMs as special cases, as summarized in Table 1 (Standard GGMs: $\forall h = 1, \ldots, H, x_m^{(h)} = 0$; Group-specific GGMs: $\forall h = 1, \ldots, H, x_m^{(h)}$ is categorical; Time-varying GGMs: $h = 1, x_m^{(h)}$ is continuous; General covariate-dependent $\forall h = 1, \ldots, H, x_m^{(h)}$ is continuous or categorical).

We use the following notations $(\boldsymbol{Q}_0)_{ij} = \alpha_{ij}, (\boldsymbol{P}_h)_{ij} = \left[\theta_{ij}\right]_h$ representing the (i, j)th element of $\boldsymbol{Q}_0$ and $\boldsymbol{P}_h$ respectively. The set of unknown parameters can be formulated as $\boldsymbol{\beta} = (\boldsymbol{\alpha}, \boldsymbol{\theta})^T$, where $\boldsymbol{\alpha}$ is the vector of all $\alpha_{ij}, i \le j$ and $\boldsymbol{\theta}$ is the vector of all $\left[\theta_{ij}\right]_h, i \le j$. Furthermore, we define an index set $E = \{(i, j) : i \le j; i, j = 1, , \ldots, p\}$. For $\forall u \in E$, we define a indicator matrix T^u as

$$(T^u)_{ij} = \begin{cases} 1 & \text{when } (i, j) = u \text{ or } (j, i) = u, \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

Therefore, each matrix can be written as

$$\boldsymbol{Q}_0 = \sum_{s:s\in E} \alpha_s \boldsymbol{T}^s, \quad \boldsymbol{P}_h = \sum_{u:u\in E} [\theta_u]_h \boldsymbol{T}^u. \tag{5}$$

Using the notations in (4) and (5), the precision matrix can be rewritten as

$$\boldsymbol{K}_m = \sum_{s:s\in E} \alpha_s \boldsymbol{T}^s + \sum_{h=1}^{H} x_m^{(h)} \left\{ \sum_{u:u\in E} [\theta_u]_h \boldsymbol{T}^u \right\}." \tag{6}$$

2.1.5 Random GGM

Cai (2017): "We are interested in modeling networks in which the connectivity among the nodes and node attributes are random variables and interact with each other. We propose a probabilistic model that allows one to formulate jointly a probability distribution for these variables. This model can be described as a combination of a latent space model and a Gaussian graphical model: given the node variables, the edges will follow independent logistic distributions, with the node variables as covariates; given edges, the node variables will be distributed jointly as multivariate Gaussian, with their conditional covariance matrix depending on the graph induced by the edges. [...]

A reason that motivates us to propose such a model is that it provides a sensible framework for modeling network dynamics in which the edge status and the node variables change their values over time, as we will explain in the end of Section 2. We will see that the dynamical system updates edge status and node variables alternatively according to the conditional distributions between edges and nodes. The equilibrium (stable) distribution of the dynamical system is then exactly the joint distribution we propose here. In other words, our model can be viewed as the stable probability law of a dynamical network process.

We will call our model the random Gaussian graphical model and formulate it in this section. Let $G = (V, E)$ be a finite simple graph (undirected, unweighted, no loops, no multiple edges) with E being a subset of $V \times V$ which is fixed. Suppose $|V| = m$ and $|E| = n$. For convenience, we identify V as the integer set $V = \{1, \ldots, m\}$. Suppose associated with each node $i \in V$ there is a random variable X_i, representing an attribute of node i. Let $X = \{X_1, \ldots, X_m\}$. We will use $x \in \mathcal{R}^m$ to denote a generic value of X. We write (i, j) for the edge in E which is incident with the nodes $i, j \in V$.

We will consider random sub-graphs of G in which V remains the same and E is reduced randomly to some subset of itself. Such a random graph can be represented by a random adjacency matrix $A = \left\{A_{ij}, i, j \in V\right\}$ in which all the diagonal elements $A_{ii} = 0$, and for each edge $(i, j) \in E$, $A_{ij} = A_{ji} = 1$ if the edge is present in the random graph, and $A_{ij} = A_{ji} = 0$ if otherwise. It is always understood that $A_{ij} \equiv 0$ for all $(i, j) \notin E$. We will call A_{ij} an edge variable. With a slight abuse of notation, we let $\mathcal{A} = \{0, 1\}^E$ be the set of all possible values of A. We will use $a = a^T \in \mathscr{A}$ to denote a generic value of the adjacency matrix A.

By "random Gaussian graphical model" we mean the following joint probability density for variables A and X, defined on the space $\mathcal{A} \times \mathcal{R}^m$,

$$\mu(a, x) \equiv \frac{1}{Z} \exp\left\{-\frac{1}{2} H(a, x)\right\}, \quad (a, x) \in \mathcal{A} \times \mathcal{R}^m \tag{1}$$

where

$$H(a, x) = \alpha \sum_i x_i^2 + \beta \sum_{(i,j)\in E} a_{ij} \left(x_i - x_j\right)^2 \tag{2}$$

for some parameters $\alpha > 0$ and $\beta \geq 0$, and Z is the normalizing constant. It is clear that this Z is always finite.

We note that in this model, if all $a_{ij} = 1$ it becomes an usual Gaussian graphical model (as we will see below). Therefore, we can consider the Gaussian graphical model as a "full model" relative to the given edge set E while model (1) as a model that allows us to "turn off" some edges in E at random according to the values of a_{ij}'s. In particular, if all $a_{ij} = 0$ and therefore there are no connections among the nodes in the graph, X_i's are independent $N(0, 1/\alpha)$ random variables. The joint distribution of a_{ij}'s in turn depends on X_i's. In general, the likelihood of connectivity among the nodes is determined by the magnitudes of the differences between the corresponding node variables and the value of β. On the other hand, the connectivity of the nodes will, in turn, affect the distribution of the node variables. [...]"

2.2 Time-Series Data (N=Large, P=Large, T=Large)

- Dynamic GGM → Group Fused Graphical Lasso (GFGL; Gibberd and Nelson, 2017)
- Time-Varying Graphical Lasso (TVGL; Hallac et al., 2017; Masuda and Lambiotte, 2016)
- Latent variable Time-varying Graphical Lasso (LTGL; Tomasi et al., 2018)
- Latent variable Time-varying Graphical Lasso with temporal kernel / pattern inference (LTGL$_{\kappa/p}$; Tomasi et al., 2021)
- Time Adaptive GGM (TAGM; Tozzo et al., 2021)

2.2.1 Dynamic GGM

Gibberd and Nelson (2017): "The time-evolving precision matrix of a piecewise-constant Gaussian graphical model encodes the dynamic conditional dependency structure of a multivariate time-series. Traditionally, graphical models are estimated under the assumption that data are drawn identically from a generating distribution. Introducing sparsity and sparse-difference inducing priors, we relax these assumptions and propose a novel regularized M-estimator to jointly estimate both the graph and change point structure. The resulting estimator possesses the ability to therefore favor sparse dependency structures and/or smoothly evolving graph structures, as required. Moreover, our approach extends current methods to allow estimation of change points that are grouped across multiple dependencies in a system. [...]

An alternative approach as followed by Ahmed and Xing (2009); Kolar and Xing (2012); and others, is to formulate a convex optimization problem with suitable constraints that encourage desired dynamical properties. We refer to such approaches as fused graphical models.

Our contribution investigates and extends this class of models to enable the estimation of change points that are grouped over multiple edges in a graphical model. Such grouped change points indicate changes in dependency across a system that influence many variables at once. In many situations there may be a priori knowledge of potential groups, for example, grouping over different gene function in genetic data, or asset classes in finance. Grouped changes often indicate some sort of regime or phase change in the system dynamics, which may be of interest to the practitioner. To this end, we propose the *group-fused graphical lasso (GFGL)* method for joint change point and graph estimation. [...]

Dynamic GGM Given a P-variate time-series $\boldsymbol{y}^t \sim \mathcal{N}(\boldsymbol{\mu}^t, \boldsymbol{\Sigma}^t)$ for $t = 1, \ldots, T$, if the precision matrices $\boldsymbol{\Theta}^t := (\boldsymbol{\Sigma}^t)^{-1}$ are well defined then the dependency structure of the series can be captured by a dynamic Gaussian graphical model (GGM). This comprises a collection of graphs $G^t = (V^t, E^t)$, where the vertices $V^t = \{1, \ldots, P\}$ represent each component of $\boldsymbol{y}^t$ and the edges E^t represent conditional dependency relations between variables over time. More precisely, the edge $(i, j) \in E^t$ is present in the graph if the i and jth variables are conditionally dependent given all other variables. It follows in the Gaussian case that conditional independence $y_i^t \perp y_j^t \mid y_{-\{i,j\}}^t$ is satisfied if and only if the corresponding entries of the precision matrix are zero, that is, $\Theta_{ij}^t = \Theta_{ji}^t = 0$ (Lauritzen 1996). Therefore, since it encodes the conditional dependency graph, estimation of the precision matrix is of great interest.

Traditionally, estimation of GGM's is performed under the assumption of stationarity, that is, we have identically distributed draws from a Gaussian model. Letting $\boldsymbol{Y} = \left(\boldsymbol{y}^1, \ldots, \boldsymbol{y}^T\right)$ be a set of observations and assuming $\boldsymbol{y}^t \overset{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma})$, one can construct an estimator for $\boldsymbol{\Sigma}^{-1}$ by maximizing the log-likelihood: $\hat{\boldsymbol{\Theta}} := \arg\max_{\boldsymbol{X}}[\log(\det(\boldsymbol{X})) - \mathrm{tr}(\hat{\boldsymbol{S}}\boldsymbol{X})]$, where $\hat{\boldsymbol{S}} = \boldsymbol{Y}\boldsymbol{Y}^\top/2T$. In the case where the number of observations is greater than the number of variables ($T > P$), we can test for edge significance to find a GGM (Drton and Perlman 2004). However, in the nonidentical case, because only one datapoint may be observed at each node per time-step, the traditional empirical covariance estimator $\hat{\boldsymbol{S}}^t = \boldsymbol{y}^t (\boldsymbol{y}^t)^\top /2$ is rank deficient for $P > 1$. To this end, estimation of the precision matrix requires additional modeling assumptions. A strategy explored in several recent works (Ahmed and Xing 2009; Danaher, Wang, and Witten 2013; Gibberd and Nelson 2014; Monti et al. 2014) is to introduce priors in the form of regularized M-estimators, viz.,

$$\hat{\boldsymbol{\Theta}}^t := \underset{\boldsymbol{X}^t \in \{\boldsymbol{X}_+^t\}_{t=1}^T}{\arg\min} \left[\mathcal{L}\left(\left\{\boldsymbol{X}^t\right\}\right)\right], \tag{1}$$

with a convex cost function:

$$\mathcal{L}\left(\left\{\boldsymbol{X}^t\right\}\right) = \sum_{t=1}^{T} -L\left(\boldsymbol{X}^t, \boldsymbol{y}^t\right) + R_{\text{Shrink}}\left(\left\{\boldsymbol{X}^t\right\}\right) + R_{\text{Smooth}}\left(\left\{\boldsymbol{X}^t\right\}\right), \tag{2}$$

where $L(\boldsymbol{X}^t, \boldsymbol{y}^t)$ is proportional to the log-likelihood and follows from the normal distribution. The penalty terms $R_{\text{Shrink}}, R_{\text{Smooth}}$ correspond to prior shrinkage/smoothness assumptions. Typically, the smoothness term will be a function of the difference between estimates $\boldsymbol{X}^t - \boldsymbol{X}^{t-1}$, whereas the shrinkage term will act at specific time points, that is, directly on $\boldsymbol{X}^t$.

One popular approach that is relevant to GGM is to use these regularizers to place assumptions on the number of dependencies in a graph. For example, in the iid case, we need not consider conditional dependencies between all variables, but only a small subset of those which appear most dependent. This assumption of sparsity can be viewed as placing a prior on the parameters $\boldsymbol{\Theta}$ to induce zeros in the off-diagonal entries of the precision matrix. Akin to the Laplace prior associated with the least absolute shrinkage and selection operator (lasso) for linear regression (Tibshirani 1996), one can construct the graphical lasso estimator for the precision matrix as: $\hat{\Theta} := \arg\min_X[-L(\boldsymbol{X}, \hat{\boldsymbol{S}}) + \lambda\|\boldsymbol{X}\|_1]$, where $L(\boldsymbol{X}, \hat{\boldsymbol{S}}) = \log(\det(\boldsymbol{X})) - \mathrm{tr}(\hat{\boldsymbol{S}}\boldsymbol{X})$. This estimator, as examined by Banerjee, Ghaoui, and D'Aspremont (2007); Friedman, Hastie, and Tibshirani (2008), allows one to stabilize estimation of the precision matrix in the high-dimensional regime (where $P > T$) and estimate a sparse graph $\hat{G}$ (sparsity is controlled by λ). A full Bayesian treatment for the graphical lasso is given by Wang (2012) who investigate how representative the mode is of the full posterior.

Several approaches that incorporate dynamics in such graphical estimators have been suggested. Zhou et al. (2010) and Kolar and Xing (2011) used a local estimate of the covariance in the term $L(\boldsymbol{X}^t)$ by replacing $\hat{\boldsymbol{S}}$ in the graphical lasso with a time-sensitive weighted estimator

$$\hat{\boldsymbol{S}}^t = \sum_s w_s^t \boldsymbol{y}_s (\boldsymbol{y}_s)^\top / \sum_s w_s^t, \tag{3}$$

where $w_s^t = K(|s-t|/h)$ are weights derived from a symmetric nonnegative smoothing kernel function $K(\cdot)$ with width related to h. The resulting graphs $\hat{G}^t$ are now representative of some temporally localized data. By making some smoothness assumptions on the underlying covariance matrix such a kernel estimator can be shown to be risk consistent (Zhou et al. 2010). Kolar and Xing (2011) went further, and demonstrated that placing assumptions on the Fisher information matrix allows one to prove consistent estimation of graph structure in such dynamic GGM. In the next section, we discuss how one may adapt these assumptions to estimate piecewise constant GGM.

Group Fused Graphical Lasso (GFGL) We propose the group fused graphical lasso (GFGL) estimator for estimating piecewise constant GGM. The model assumes data are generated at time point $t = 1, \ldots, T$ according to $\mathbb{R}^P \ni \boldsymbol{y}^t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}^t)$, where the distribution is strictly stationary, that is, $\left\{\boldsymbol{\Sigma}^l = \boldsymbol{\Sigma}^m \mid \tau^k < l, m \le \tau^{k+1}\right\}$ between $k = 0, \ldots, K$ changepoints (note we set $\tau^0 = 0, \tau^{K+1} = T$). We propose to estimate the covariance and precision matrix at each time by minimizing a cost, as in Equation (1). The GFGL cost is given as

$$\begin{aligned}\mathcal{L}\left(\left\{\boldsymbol{X}^t\right\}\right) = &\underbrace{\sum_{t=1}^{T} \overbrace{\left(-\operatorname{logdet}\left(\boldsymbol{X}^t\right) + \operatorname{tr}\left(\hat{\boldsymbol{S}}^t \boldsymbol{X}^t\right)\right)}^{-L(\{X^t\},\{y^t\})}}_{\propto-\text{Likelihood}} \\ &+ \lambda_1 \underbrace{\overbrace{\sum_{t=1}^{T} \left\|\boldsymbol{X}_{-ii}^t\right\|_1}^{R_{\text{Shrink}}}}_{\ell_1\text{shrinkage}} + \lambda_2 \underbrace{\overbrace{\sum_{t=2}^{T} \left\|\boldsymbol{X}_{-ii}^t - \boldsymbol{X}_{-ii}^{t-1}\right\|_F}^{R_{\text{Smooth}}}}_{\text{group } \ell_{2,1} \text{ smoothing}},\end{aligned} \tag{4}$$

where $\left\|X_{-ii}^t\right\|_1 = \sum_{i \neq j} \left|X_{i,j}^t\right|$ is the matrix ℓ_1 norm with the diagonal entries removed.

In the remainder of this article, we describe how one can efficiently solve the GFGL problem and demonstrate two key properties, namely,

1. Estimated precision matrices encode a sparse dependency structure whereby many of the off axis entries are exactly zero, that is, $\hat{\Theta}_{i,j}^t = 0$.

2. Precision matrices maintain a piecewise constant structure where changepoints tend to be grouped across the precision matrix, such that for many edges indexed by (i, j) and (l, m) the estimated changepoints for the two edges are the same, viz., $\hat{\mathcal{T}}_{i,j} = \hat{\mathcal{T}}_{l,m}$, where $\hat{\mathcal{T}}_{i,j} = \left\{\hat{\tau}^1_{ij}, \ldots, \hat{\tau}^{\hat{K}_{ij}}_{ij}\right\}$ represents the set of $\hat{K}_{ij}$ estimated changepoints τ^k_{ij} on the i, jth edge.

[...]"

2.2.2 TVGL — LTGL

Hallac et al. (2017): "Many important problems can be modeled as a system of interconnected entities, where each entity is recording time-dependent observations or measurements. In order to spot trends, detect anomalies, and interpret the temporal dynamics of such data, it is essential to understand the relationships between the different entities and how these relationships evolve over time. In this paper, we introduce the time-varying graphical lasso (TVGL), a method of inferring time-varying networks from raw time series data. We cast the problem in terms of estimating a sparse time-varying inverse covariance matrix, which reveals a dynamic network of interdependencies between the entities. [...]

Consider a sequence of multivariate observations in $\boldsymbol{R}^p$ sampled from a distribution $x \sim N(0, \Sigma(t))$. Observations come in at times $0 \leq t_1 \leq \cdots \leq t_T$, where at each time t_i, there are $n_i \geq 1$ different observation vectors over the readings of all the nodes. (For now we assume that the readings are synchronous, where $t_i - t_{i-1}$ is a constant value for all i, but in Section 3.1 we extend this approach to asynchronous observations.) With these samples, we aim to estimate the underlying covariance matrix $\Sigma(t)$, which can change over time. That is, given a changing underlying distribution, we attempt to estimate it based on a sequence of observations. Here, we formulate a convex optimization problem to infer a sequence of sparse inverse covariance matrices, $\Theta_i = \Sigma(t_i)^{-1}$, one for each t_i. These estimates are based on local empirical observation vector(s), as well as coupling constraints with neighboring timestamps' covariance estimates. Recall that a sparse inverse covariance allows us to encode conditional independence between different variables [16].

GGM We first consider static inference, which is equivalent to the graphical lasso problem [2,7,35], and then build on it to extend to dynamic networks. In the static case, $\Sigma(t)$ is constant for all t. Given a series of multivariate readings, this can be written as

$$\text{minimize} \ - l(\Theta) + \lambda \|\Theta\|_{\text{od},1}, \tag{1}$$

where $\|\Theta\|_{\text{od},1}$ is a seminorm of Θ, the off-diagonal ℓ_1-norm, $\sum_{i \neq j} \left|\Theta_{ij}\right|$. This lasso penalty enforces element-wise sparsity in our solution for Θ, regulated by the trade-off parameter $\lambda \geq 0$. In Problem (1), $l(\Theta)$ is the log likelihood of Θ (up to a constant and scale) [29,35],

$$l(\Theta) = n(\log\det\Theta - \operatorname{Tr}(S\Theta)),$$

where Θ must be symmetric positive-definite $\left(\mathrm{S}_{++}^{p}\right)$, S is the empirical covariance $\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T$, n is the number of observations, and x_i are the different samples. When S is invertible, $l(\Theta)$ encourages Θ to be close to S^{-1}.

TVGL In order to infer a time-varying sequence of networks, we extend the above approach and allow $\Sigma(t)$ to vary over time. To do so we set up a sequence of graphical lasso problems, each coupled together in a chain to penalize deviations in the estimations, which we call the time-varying graphical lasso (TVGL). We solve for $\Theta = (\Theta_1, \ldots, \Theta_T)$, our estimated inverse covariance matrices at times $t_1, \ldots, t_T$,

$$\underset{\Theta \in \mathrm{S}_{++}^{p}}{\text{minimize}} \quad \sum_{i=1}^{T} -l_i(\Theta_i) + \lambda \|\Theta_i\|_{\text{od},1} + \beta \sum_{i=2}^{T} \psi(\Theta_i - \Theta_{i-1}). \tag{2}$$

Here, $l_i(\Theta_i) = n_i(\log\det\Theta_i - \mathrm{Tr}(S_i\Theta_i))$, $\beta \geq 0$, and $\psi(\Theta_i - \Theta_{i-1})$ is a convex penalty function, minimized at $\psi(0)$, which encourages similarity between Θ_{t-1} and Θ_t. We examine in Section 2.1 how different ψ 's enforce different behaviors. Note that Problem (2) can be viewed as an optimization problem defined on a chain graph, where each element solves for a different Θ_i, as shown in Figure 2.

The log-likelihood l_i depends on S_i, the empirical covariance at time t_i. In high-dimensional settings, where the number of dimensions p is larger than the number of observations n_i, S_i will be rank deficient and therefore non-invertible. However, our method is well-suited to overcome this problem of an insufficient number of observations. By enforcing structural similarity, each Θ_i borrows strength from the fact that neighboring network estimates should be similar, or even identical, across time. In the extreme case, we are able to estimate a network at a time where there is only one observation. The empirical covariance, $x_i x_i^T$, is rank 1 , but between the sparsity and the temporal consistency penalties, our approach can still infer an accurate estimate of the network at that snapshot in time.

Regularization Parameters λ and β. Problem (2) has two parameters, λ and β, which define important values on two trade-off curves. λ determines the sparsity level of the network: small values will better match the empirical data, but will lead to very dense networks, which are less interpretable and often overfit. β determines how strongly correlated neighboring covariance estimations should be. A small β will lead to Θ's which fluctuate from estimate-to-estimate, whereas large β's lead to smoother estimates over time. As $\beta \to \infty$, the temporal deviation penalty gets so large that Problem (2) turns into the original graphical lasso, since a constant Θ will be the solution across the entire time series.

Different types of penalty functions ψ allow us to enforce different behaviors in the evolution of the network structure. In particular, if we have an expectation about how the underlying network may change over time, we are able to encode it into ψ. Here, we define several common temporal patterns and their associated penalty functions. In Sections 6 and 7, we use these to analyze the time-varying dynamics of multiple real and synthetic datasets. For these penalties, note that ψ can sometimes be split into a sum of column-norms. As such, we refer to the j-th column of a matrix X as $[X]_j$.

- **A few edges changing at a time** - Setting $\psi(X) = \sum_{i,j} \left|X_{i,j}\right|$, an element-wise

ℓ_1 penalty, encourages neighboring graphs to be identical [6,34]. When the ij-th edge of the network "breaks", or is different at two neighboring times, this penalty still encourages the rest of the graph to remain exactly the same. As a result, this penalty is best used in cases where we expect only a handful of edges—at most—to change at a time.

- **Global restructuring** - Setting $\psi(X) = \sum_j \left\|[X]_j\right\|_2$, a group lasso ℓ_2 penalty, causes the entire graph to restructure at a select few timestamps, while at all other times, the graph remains piecewise constant [6,10]. This is useful for event detection and time-series segmentation, as it finds exact times where there is a regime-change, or shift, in the underlying covariance matrix.
- **Smoothly varying over time** - Setting $\psi(X) = \sum_{i,j} X_{i,j}^2$, a Laplacian penalty, causes smooth transitions of the graphical model from timestamp to timestamp. Adjacent graphs will often differ by small amounts, but severe deviations are largely penalized so they rarely occur [30]. This penalty is best used when we want to come up with a smoothly varying estimate over time.
- **Block-wise restructuring** - Setting $\psi(X)$ to an ℓ_∞ penalty, $\psi(X) = \sum_j \left(\max_i \left|X_{i,j}\right|\right)$, implies that, when an element in the inverse covariance matrix changes by ϵ at a given time, other elements are free to change by up to that same amount with no additional penalty (except for the original $\ell_{\text{od},1}$ sparsity condition). This is best used when a cluster of nodes suddenly changes its internal edge structure, the rest of the network does not change.
- **Perturbed node** - Setting ψ to the row-column overlap penalty [18], defined as $\psi(X) = \min_{V:V+V^T=X} \sum_j \left\|[V]_j\right\|_2$, yields an interesting behavior. When node i has a reweighting of one edge at time t, this says that the node can rewire all its edges at that same time with a minimal penalty. However the rest of the graph is strongly encouraged to remain the exact same. It is best used in situations where we are looking for single nodes re-locating themselves to a new set of neighbors within the network.

Here, we develop three extensions of the basic time-varying graphical lasso. First, we update our approach to allow for asynchronous observations of the data. Then, we derive a method of inferring an estimate at any time, even if there is no temporally-local data. Finally, we extend the algorithm so it can be deployed in an application with streaming data and real-time update constraints. [...]"

(Tomasi et al., 2018): "[...] A model for the inference of Θ including the sparse prior is the graphical lasso [15]:

$$\underset{\Theta}{\text{minimize}} - \ell(S, \Theta) + \alpha\|\Theta\|_{od,1}, \tag{1}$$

where ℓ is the Gaussian log-likelihood (up to a constant and scaling factor) defined as $\ell(S, \Theta) = \log\det(\Theta) - \operatorname{tr}(S\Theta)$ for Θ positive definite and $S = \frac{1}{n}X^\top X$ is the empirical covariance matrix. $\|\cdot\|_{od,1}$ is the off-diagonal ℓ_1-norm, which promotes sparsity in the precision matrix (excluding the diagonal).

Latent Variables Often, real-world observations do not conform exactly to a sparse GGM. This is due to global hidden factors that influence the system, which introduce spurious dependencies between observed variables [10, 11]. For this reason, GGMs can be extended by introducing latent variables able to represent factors which are not observed in the data. These latent variables are not principal components, since they do not provide a low-rank approximation of the graphical model. On the contrary, such factors are added to the model in order to condition the statistics of the observed variables. In particular, we consider both latent and observed variables to have a common domain [11].

Let latent variables be indexed by a set H, and observed variables by a set O. The precision matrix Θ of the joint distribution of both latent and observed variables may be partitioned into four blocks:

$$\Theta = \Sigma^{-1} = \left[\begin{array}{c|c} \Theta_H & \Theta_{HO} \\ \hline \Theta_{OH} & \Theta_O \end{array}\right].$$

Such blocks represent the conditional dependencies among latent variables (Θ_H), observed variables (Θ_O), between latent and observed (Θ_{HO}), and viceversa (Θ_{OH}). The marginal precision matrix Σ_O^{-1} of the observed variables is given by the Schur complement w.r.t. the block Θ_H [8]:

$$\hat{\Theta}_O = \Sigma_O^{-1} = \Theta_O - \Theta_{OH}\Theta_H^{-1}\Theta_{HO}. \tag{2}$$

Θ_O specifies the precision matrix of the conditional statistics of the observed variables given the latent variables, while $\Theta_{OH}\Theta_H^{-1}\Theta_{HO}$ is a summary of the effect of marginalisation over the latent variables. Such matrix has a small rank if the number of latent variables is small compared to the observed variables. Note that the rank is an indicator of the number of latent variables H [9].

The effect of the marginalisation is scattered over many observed variables, in such a way not to confound it with the true underlying conditional sparse structure of Θ_O [8]. Typically $\hat{\Theta}_O$ is not sparse due to the low-rank term, while, with the addition of the latent factors contribution, we can recover the true sparse GGM. For this reason, the graphical lasso in Equation (1) has been extended with the latent variable graphical lasso that includes the inference of a low-rank term, using the form (2), as follows [8]:

$$\tilde{\Theta}, \tilde{L} = \underset{\substack{(\Theta, L) \\ L \succcurlyeq 0}}{\operatorname{argmin}} - \ell(S, \Theta - L) + \alpha\|\Theta\|_{od,1} + \tau\|L\|_* \tag{3}$$

Here $\tilde{\Theta}$ provides an estimate of Θ_O (precision matrix of the observed variables) while $\tilde{L}$ provides an estimate of $\Theta_{OH}\Theta_H^{-1}\Theta_{HO}$ (marginalisation over the latent variables). Note that S is the empirical covariance matrix computed only on observed variables, since no information on the latent ones is available.

TVGL Time-varying network inference. Problems (1) and (3) aim at recovering the structure of the system at fixed time (static network inference). However, complex systems have temporal dynamics that regulate their overall functioning [1, 15]. Hence, the modelling of such complex systems requires a dynamical network inference, where the states of the network are co-dependent. This naturally leads to the idea of temporal consistency, which assumes similarities between consecutive states of the network. In fact, we can assume that, for sufficiently close time points, a system shows negligible differences. During the inference of a dynamical network, temporal consistency may translate into the imposition of similarities among temporally close networks [16]. In particular, graphical lasso with temporal consistency results in time-varying graphical lasso [17]:

$$\underset{(\Theta_1,\dots,\Theta_T)}{\text{minimize}} \sum_{i=1}^{T} \left[-\ell\left(S_i, \Theta_i\right) + \alpha \left\|\Theta_i\right\|_{od,1} \right] + \beta \sum_{i=1}^{T-1} \Psi\left(\Theta_{i+1} - \Theta_i\right) \tag{4}$$

where the inference of a network at a single time point i is guided by the states at adjacent time points. The network is encoded in a sequence of precision matrices $(\Theta_1, \dots, \Theta_T)$ which model the system at each time point $i = 1, \dots, T$. The type of similarity imposed to consecutive time points and its strength are specified by the penalty function Ψ and the parameter β, respectively. Options when choosing Ψ include [...]

LTGL In this work we propose a new statistical model for the inference of networks that change consistently in time under the influence of latent factors. We call such model latent variable time-varying graphical lasso (LTGL). LTGL infers the dynamical network of complex systems by decomposing the problem into two parts—to what has been done in [8] for static network inference. We consider two components of the dynamical network: a true underlying structure on the observed variables and the contribution of latent factors. This allows to factor out the contribution of hidden variables, favouring a reliable modelling of the dynamical system. The novelty of our method is the simultaneous inference of a dynamical network with latent factors that exploits the imposition of behavioural consistency on both observed variables interactions and latent influence through the use of penalisation terms. This allows for an easier interpretation of the evolution of the dynamical system while, at the same time, improving its graphical modelling. The two separate (while closely related) components at each time point are obtained by integrating the network inference with the information coming from temporally different states of the network. Formally, let $X_i \in \mathbb{R}^{n_i \times d}$, for $i = 1, \dots, T$, be a set of observations measured at T different time points composed by n_i samples of d observed variables. (Note that, for each time point i, samples are assumed to be drawn from the probability distribution on the observed variables conditioned on the latent ones.) Let $S_i = \frac{1}{n_i} X_i^\top X_i$ be the empirical covariance matrix at time i. The goal is to retrieve a set of sparse matrices $\boldsymbol{\Theta} = (\Theta_1, \dots, \Theta_T)$ and a set of low-rank matrices $\boldsymbol{L} = (L_1, \dots, L_T)$ such that, at each time point i, Θ_i encodes the conditional independences between the observed variables, while L_i provides the summary of marginalisation over latent variables on the observed ones.

Consider Equation (3) at a specific time i. Here we want to impose continuity between the structure and the hidden variables contribution in time, therefore we enforce the difference between consecutive graphs to abide certain constraints by adding two penalisation terms. Our LTGL model takes the following form:

$$\underset{\substack{(\Theta,L)\\ L_i\succcurlyeq 0}}{\text{minimize}} \sum_{i=1}^{T}\left[-\ell\left(S_i,\Theta_i-L_i\right)+\alpha\left\|\Theta_i\right\|_{od,1}+\tau\left\|L_i\right\|_*\right] \\ +\beta\sum_{i=1}^{T-1}\Psi\left(\Theta_{i+1}-\Theta_i\right)+\eta\sum_{i=1}^{T-1}\Phi\left(L_{i+1}-L_i\right) \tag{5}$$

where Ψ and Φ are both penalty functions that force the structure of the network to change over time according to a certain behaviour, by acting on Θ and L, respectively. Temporal consistency of both the structure of the network and latent factors contribution is guaranteed by the use of such penalty functions, which benefits the network inference in particular in presence of few available observations of the system. Possible choices for Ψ and Φ are listed in Section 2. Their choice is arbitrary and it is based on the prior knowledge on the respective components evolution in the system. Also, note that Ψ and Φ are independent, which allows LTGL to model a wide range of dynamical behaviours of complex systems."

(Tomasi et al., 2021): "In this paper we address the problem of possible non-Markovian temporal relationships by exploiting similarity kernels to infer a dynamical network. We identified two distinct cases that need to be addressed separately:

1. when the similarity pattern is known a priori (e.g., smooth, periodic) we use such knowledge by incorporating a suitable kernel in the model;
2. when the similarity is unknown or cannot easily be modelled by kernel functions we resort to the simultaneous inference of the kernel and the network structure.

To solve the first scenario we propose an extension of existing time-varying network inference methods to incorporate similarity kernels among time points. We present such extension on two state-of-the-art methods that consider totally observed and partially observed (latent) multivariate time-series data [7], [9]. Following the original naming we refer to our models as time-varying graphical lasso with temporal kernel (TGL$_\kappa$) and latent variable time-varying graphical lasso with temporal kernel (LTGL$_\kappa$), respectively. Leveraging on these methods, we address the second scenario by integrating TGL$_\kappa$ and LTGL$_\kappa$ with a network clustering approach. We simultaneously identify the network structure and the similarity patterns occurring at different time points. We call such methods time-varying graphical lasso with pattern inference (TGL$_p$) and latent variable time-varying graphical lasso with pattern inference (LTGL$_p$).

[...] Prior knowledge on the behaviour of the network may drive the analysis towards a reliable inference of the underlying system. Here we focus on behaviour of the network that can be modelled through stationary kernels, meaning that the kernel function that generates them only depends on the distance between two points and not on their identity. Such kernels, as previously introduced, are coupled with a distance function Ψ that encodes the system behaviour over time.

Given a multivariate time series, our goal is to infer a set of precision matrices Θ that encodes the dynamical graph. The task can be tackled as T jointly graphical lasso problems [20], [21] in which we maximise the Gaussian log-likelihood on each precision matrix Θ_t starting from the related empirical covariance matrix $S_t = \frac{1}{n_t}\sum_{i=1}^{n_t} X_t^\top X_t$. In addition, we impose a penalty $P_{\Psi,\kappa}(\Theta)$, defined as

$$\begin{aligned} P_{\Psi,\kappa}(\mathbf{\Theta}) &= \sum_{t'>t}^{T} \kappa^{\Psi}(t,t')\,\Psi(\Theta_{t'} - \Theta_t) \\ &= \sum_{m=1}^{T-1}\sum_{t=1}^{T-m} \kappa^{\Psi}(m,t)\Psi(\Theta_{m+t} - \Theta_t)\,. \end{aligned} \tag{1}$$

The penalty encloses the consistency type Ψ and its strength pattern κ—both assumed to be known a priori. Such penalty may, in principle, be applied to any method that considers the joint estimate of multiple graphical models. In this paper we instantiate two state-of-the-art methods to the use of such penalty.

TGLκ Temporal graphical lasso with stationary kernel: First, we extend the time-varying graphical lasso (TGL) [7] with TGL$_\kappa$, which takes the form

$$\underset{\Theta_t \in \mathcal{S}^d_{++}}{\text{minimize}} \sum_{t=1}^{T}\left[-n_t \ell(S_t, \Theta_t) + \alpha \|\Theta_t\|_{\text{od},1}\right] + P_{\Psi,\kappa}(\mathbf{\Theta}), \tag{2}$$

where $\ell(S_t,\Theta_t) = \log\det(\Theta_t) - \operatorname{tr}(S_t\Theta_t)$ is the Gaussian log-likelihood (up to a constant and scaling factor), $\mathcal{S}^d_{++}$ is the cone of positive definite matrices, and $\|\cdot\|_{od,1}$ is the off-diagonal ℓ_1-norm, which promotes sparsity in the precision matrix (excluding the diagonal).

LTGLκ Temporal latent graphical lasso with stationary kernel: Secondly, we extend latent variable time-varying graphical lasso (LTGL) [9] in order to be able to include latent variables in the model. The functional of LTGL$_\kappa$ is

$$\begin{aligned} \underset{\Theta_t \in \mathcal{S}^d_{++}, L_t \in \mathcal{S}^d_{+}}{\text{minimize}} \sum_{t=1}^{T}\left[-n_t \ell(S_t, \Theta_t - L_t) + \alpha \|\Theta_t\|_{\text{od},1} + \tau \|L_t\|_*\right] \\ + P_{\Psi,\kappa}(\mathbf{\Theta}) + P_{\Phi,\kappa}(\boldsymbol{L}), \end{aligned} \tag{3}$$

where $\|\cdot\|_*$ is the trace norm, which promotes sparsity in the eigenvalues of the latent contribution, enforcing the matrix to be low-rank (hence enforcing hidden factors to be much less than observed ones) [19]. Note that, following the original form [9], we require two distinct penalty functions (based on Ψ and Φ) that act on precision matrices and latent contribution, respectively. Indeed, the use of separate penalty functions allow to independently model the behaviour of latent factors with respect to observed variables. [...]

(L)TGL$_p$ Prior information required for the direct use of TGL$_\kappa$ and LTGL$_\kappa$ can be a limitation in real-world cases in which the underlying process that generated the data may not be available. Here we relax the dependency on a particular kernel κ by allowing for an automatic discovery of the similarity between time points, i.e., automatic kernel inference. Indeed, we formulate our second contribution as a two-step procedure in which first we learn the similarities between the networks, and then we use them to infer the structure in multiple iterative steps directly exploiting the methods described in the previous section.

The related methods, TGL$_P$ and LTGL$_p$, are deeply based on a clustering procedure that identifies the similarity patterns of precision matrices in time. The resulting clusters provide information on how the time points are related to each other at possibly distant times. Therefore, if two time points belong to the same cluster we will impose a stronger penalty Ψ on them, in order to make their structure to be more similar (but not necessarily identical). To exploit the temporal structure of the data, we also pair the learning of cluster-based kernels with a smooth RBF kernel. This imposition guides consecutive precision matrices to belong to the same cluster, which reduces noise and penalises the jump across different clusters. [...]"

2.2.3 TAGM — HMM

Tozzo et al. (2021): "Time Adaptive Gaussian Model (TAGM), that combines Hidden Markov Models (HMMs) and Gaussian Graphical Models (GGMs). [...] TAGM is based on the assumption that the system under analysis is non-stationary, i.e., the underlying distribution of variables at each time point may change in time. One of the possible methods to analyse sequential non-stationary data is a Hidden Markov Model (HMM) [2]. HMMs assume that the series of observations is generated by a certain number of (hidden) internal states connected through a Markov chain of latent variables. If we consider Figure 2, the latent variables are modelled as $z_1, \ldots, z_6$. Such variables may assume possible $K = 2$ different states, each of these gets associated to a possibly different distribution. The family of such distributions depends on the type of data in analysis, in this paper, for simplicity, we consider continuous data and we assume underlying multivariate Gaussian distributions. Nonetheless, adopting the same idea for other distribution assumptions is straightforward from the model definition. Differently from standard approaches, instead of directly estimating the empirical means and covariances of such distributions from observations, we associate a graphical model where the lack of an edge explicitly encodes the conditional independencies between a pair of variables. We exploit Gaussian Graphical Models (GGMs) [29] where the precision matrices (Θ_1 and Θ_2) are the inverse of the covariance matrices and they can be interpreted as the adjacency matrices of a graph. This switch of perspective still allows us to estimate a multivariate Gaussian distribution, while allowing for directly imposing sparsity on the adjacency matrices [18]. Sparsity is fundamental as: it provides a higher stability to noise in presence of fewer samples; it grants higher interpretability as it allows for identifying the most relevant dependencies (i.e. the edges) while removing spurious correlations that would be captured by the empirical covariance matrix; and, it allows us to extend this method to perform prediction as well as understand causality patterns (more details about this will be provided in Section 3 and 4).

Consider N sequential (temporal) observations, $\boldsymbol{x}_n \in \mathbb{R}^D$ for $n = 1, \dots, N$ on D variables. We assume that such observations follow a non-stationary process, and thus are generated by possibly more than one underlying distribution. The number of such distributions could be ideally infinite [4], but, for the sake of simplicity, we here assume them to be fixed to K. In order to map each observation $\boldsymbol{x}_n$ to one distribution, HMM pairs them to a hidden (latent) variable $\mathrm{z}_n \in \{0, 1\}^K$, that has only one non-zero component at position k if the n-th observation is associated to the k-th distribution. In the rest of the paper, we will use the notation $z_{n,k}$ to indicate the k-th positional value of the vector z_n.

The sequence of latent variables follows a Markov chain process, meaning that $z_{n+1} \perp\!\!\!\perp z_{n-1} \mid z_n$. As a consequence, if we condition on the latent variables, the observations $\boldsymbol{x}_n$ and $\boldsymbol{x}_{n+1}$ become independent thus allowing to freely use them to infer the corresponding underlying distributions (see Figure 2).

The joint distribution on observed $\boldsymbol{X} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \dots, \boldsymbol{x}_N)$ and latent variables $\boldsymbol{Z} = (z_1, z_2, \dots, z_N)$ is given by

$$p(\mathrm{X}, \mathrm{Z} \mid \pi, A, \phi) = p(\mathrm{z}_1 \mid \pi) \left[\prod_{n=2}^{N} p(\mathrm{z}_n \mid \mathrm{z}_{n-1}, A) \right] \prod_{n=1}^{N} p(\mathrm{x}_n \mid \mathrm{z}_n, \phi), \tag{1}$$

where [2]:

- the probability $p(z_1 \mid \pi) = \prod_{k=1}^{K} \pi_k^{z_{1,k}}$, with $\sum_k \pi_k = 1$, is the initial latent node z_1 probability, which differs from the other states as there is no parent node;
- the probability $p(z_n \mid z_{n-1,A}) = \prod_{k=1}^{K} \prod_{j=1}^{K} A_{j,k}^{z_{n-1,j} z_{n,k}}$ is the transition probability of moving from one state to the other. Here, $A \in [0, 1]^{K \times K}$ is the transition matrix that we assume to be constant in time and it is defined as

  $$A_{j,k} = p\left(z_{n,k} = 1 \mid z_{n-1,j} = 1\right)$$

  with $0 \leq A_{j,k} \leq 1$ and $\sum_k A_{j,k} = 1$.
- the probabilities $p(\boldsymbol{x}_n \mid z_n, \phi) = \prod_{k=1}^{K} p(\boldsymbol{x}_n \mid \phi_k)^{z_{n,k}}$, are the emission probabilities where $\phi = \{\phi_1, \dots, \phi_K\}$ is a set of K different parameters governing the distributions, one for each of the possible K states.

Combining HHMs with GGMs is achieved by setting the emission probabilities to

$$p(\boldsymbol{x}_n \mid z_n, \phi) = \prod_{k=1}^{K} \mathcal{N}\left(\boldsymbol{x}_n \mid \mu_k, \Theta_k^{-1}\right)^{z_{n,k}}$$

in such a way to have an explicit correspondence between the distribution of each state and a graph, modelled through the precision matrix Θ_k. Moreover, we impose sparsity by coupling the emission probabilities with a Laplacian prior on the precision matrices.

The posterior of the final model is defined as

$$p(\pi, A, \mu, \Theta \mid \boldsymbol{X}, \boldsymbol{Z}) \propto p(z_1 \mid \pi) \left[\prod_{n=2}^{N} p(z_n \mid z_{n-1}, A) \right]$$
$$\prod_{n=1}^{N} \prod_{k=1}^{K} \mathcal{N}\left(\boldsymbol{x}_n \mid \mu_k, \Theta_k^{-1}\right)^{z_{n,k}} e^{-\frac{\lambda}{2}\|\Theta_k\|_{1,od}}. \tag{2}$$

Optimization. The optimization of parameters $\theta = \{\pi, A, \mu, \boldsymbol{\Theta}\}$ in functional (2) is performed through a Maximum A Posteriori approach. In particular, we employ the Baum's version of the expectation maximization (EM) algorithm [3]. In short, it consists in an alternating minimization procedure, in which the parameters π, A, μ are updated as in a standard HMM, while the inference of the $\boldsymbol{\Theta}$ reduces to solving a Graphical Lasso [18]

$$\Theta_k = \underset{\Theta > 0}{\operatorname{argmin}} \operatorname{tr}(\Theta \, \mathrm{S_k}) - \log\det(\Theta) + \lambda \|\Theta\|_{1,\,\mathrm{od}} \tag{3}$$

where S_k is the empirical covariance matrix of the observations that are associated to the k-th hidden state, and if we denote such observations as X_k, it is defined as

$$\mathrm{S}_k = \frac{1}{\sum_{n=1}^{N} z_{nk}} \sum_{n=1}^{N} z_{nk} \, (\boldsymbol{x}_n - \hat{\mu}_k) \, (\boldsymbol{x}_n - \hat{\mu}_k)^{\mathrm{T}}, \quad \hat{\mu}_k = \frac{1}{\sum_{n=1}^{N} z_{nk}} \sum_{n=1}^{N} z_{nk} \boldsymbol{x}_n.$$

The objective value $\operatorname{tr}(\Theta S_k) - \log\det(\Theta)$ is the negative log likelihood of the multivariate normal distribution and $\|\cdot\|_{1,\,\mathrm{od}}$ is the off-diagonal ℓ_1-norm that imposes sparsity on the precision matrix Θ without considering the diagonal elements. More details on the optimization of this model can be found in [11].

Higher order and online extensions. TAGM could benefit from two extensions to better handle real-world scenarios. The first extension is an online learning variation, Incremental TAGM or IncTAGM, that could be useful in presence of high-frequency data. Indeed, in realworld contexts where new observations arrive at a high rate (e.g. every second) one may want to be able to fine tune the model online in order to consider such observations instantaneously instead of re-fitting on the complete time series. We derive such extension following the idea in [10]. In practice IncTAGM starts as a standard TAGM on the available data, as a new time point becomes available the set of parameters is updated accordingly and the new point gets assigned to a state. Such update can be achieved by a variation in the E step of the EM algorithm, more details in [11].

The second extension is a higher-order Markov chain version, Memory TAGM or MemTAGM. It is based on the idea that latent states could have higher order dependencies, which are not captured if we consider a Markov chain of order one. In short, the main idea is to allow the emission probability of $\boldsymbol{x}_n$ to depend not only on z_n but also from the previous $m \in \mathbb{Z}^+$sequence of states $p(\boldsymbol{x}_n \mid \{\boldsymbol{x}_\ell\}_{\ell<n}, \{z_\ell\}_{\ell \le n}) = p\left(\boldsymbol{x}_n \mid \{z_\ell\}_{\ell=n-(m-1)}^{n}\right)$. Each observation is conditionally independent of the previous ones and of the state sequence history, given the current and the preceding $m-1$ states. For the derivation of the related optimization algorithm we followed [20]. The main drawback is a much higher computational time as the transition matrix and the corresponding initial state dimensions increase, more details in [11]."

3 Ising Model (IM) / Item Response Theory (IRT)

- Extended-Rasch Model (E-RM)
- Marginal-Rasch Model (M-RM)
- Multidimensional two-parameter Logistic Model (MD-2PL)

Ising Model — IRT: Epskamp, Maris, et al. (2018), Kruis and Maris (2016), Marsman et al. (2018), and Marsman et al. (2015)

R-Package: R-Package 'irt', 'IsingFit', 'IsingSampler'

3.1 IM — Curie-Weiss

Marsman et al. (2018): "[...] the study of phase transitions with the Ising model proved to be very difficult and enticed the study of phase transitions in even simpler models. Most notable is the Curie-Weiss model (Kac, 1968; Kochmański et al., 2013), also known as the fully connected Ising model:

$$p(\boldsymbol{x}) = \frac{\exp\left\{\sum_{i=1}^{n}\mu_i x_i + \sum_{i=1}^{n}\sum_{j=1}^{n}\sigma x_i x_j\right\}}{\sum_{\boldsymbol{x}}\exp\left\{\sum_{i=1}^{n}\mu_i x_i + \sum_{i=1}^{n}\sum_{j=1}^{n}\sigma x_i x_j\right\}} = \frac{\exp\left\{\sum_{i=1}^{n}\mu_i x_i + \sigma x_+^2\right\}}{\sum_{\boldsymbol{x}}\exp\left\{\sum_{i=1}^{n}\mu_i x_i + \sigma x_+^2\right\}}, \tag{2}$$

in which $x_+ = \sum_{i=1}^{n} x_i$ refers to the sum of the node states, that is, it is the analog of a total test score in psychometrics. Whereas the Ising model allows the interaction strength σ_{ij} to vary between node-pairs, most notably that $\sigma_{ij} = 0$ whenever node i and j are no direct neighbors [...], the Curie-Weiss model assumes that there is a constant interaction $\sigma > 0$ between each pair of nodes [...]"

3.1.1 Curie-Weiss — E-RM

Marsman et al. (2018): "[...] it is interesting to observe that the Curie-Weiss model is closely related to the extended-Rasch model (E-RM; Cressie & Holland, 1983; Tjur, 1982), a particular item response model originating from educational and psychological measurement (Anderson, Li, & Vermunt, 2007; Hessen, 2011; Maris, Bechger, & San Martin, 2015). In the E-RM, the distribution of n binary variables $\boldsymbol{x}$—typically the scores of items on a test—is given by

$$p(\boldsymbol{x}) = \frac{\prod_{i=1}^{n}\beta_i^{x_i}\lambda_{x_+}}{\sum_{\boldsymbol{x}}\prod_{i=1}^{n}\beta_i^{x_i}\lambda_{x_+}}, \tag{3}$$

where the β_i relate to the difficulties of items on the test and the λ_{x_+} to the probability of scoring $2x_+ - n$ out of n" items correctly. It can be shown that the E-RM consists of two parts (e.g., Maris et al., 2015): a marginal probability $p\,(x_+ \mid \boldsymbol{\beta}, \boldsymbol{\lambda})$ characterizing the distribution of total scores x_+ (e.g., scores achieved by pupils on an educational test, or number of symptoms of subjects in a clinical population), and a conditional probability $p(\boldsymbol{x} \mid x_+, \boldsymbol{\beta})$ characterizing the distribution of the item scores/symptomatic states given that the total score was x_+. Comparing the expression for the Curie-Weiss model in Equation (2) and that for the E-RM in Equation (3), we see that they are equivalent subject to the constraints:

$$\log \beta_i = \mu_i \qquad \log \lambda_{x_+} = \sigma x_+^2$$

That is, whenever a quadratic relation holds between the total scores x_+ and the log of the λ_{x_+} parameters in the E-RM, the E-RM is equivalent to the Curie-Weiss model."

3.1.2 E-RM — M-RM — Curie-Weiss

Marsman et al. (2018): "It seems that we have made little progress, as the expressions for both the Curie-Weiss model in Equation (2) and the E-RM in Equation (3) do not involve latent variables. This is not the case, however, as the E-RM has been studied in the psychometric literature mostly due to its connection to a latent variable model known as the marginal-Rasch model (M-RM; Bock & Aitken, 1981; Glas, 1989). Consider again Equation (2), and suppose that we obtain a latent variable that explains all connections in the Curie-Weiss model, such that the observations are independent given the latent variable:

$$p(\boldsymbol{x} \mid \theta) = \prod_{i=1}^{n} p(x_i \mid \theta).$$

A particular example of such a conditional distribution is an IRT model known as the Rasch model (Rasch, 1960):

$$p(\boldsymbol{x} \mid \theta) = \prod_{i=1}^{n} \frac{\exp\{x_i(\theta - \delta_i)\}}{\exp\{\delta_i - \theta\} + \exp\{\theta - \delta_i\}},$$

where the δ_j are commonly referred to as item difficulties. In the M-RM, the distribution of the n binary variables $\boldsymbol{x}$ is then expressed as

$$\begin{aligned} p(\boldsymbol{x}) &= \int_{-\infty}^{\infty} p(\boldsymbol{x} \mid \theta) f(\theta) d\theta \\ &= \int_{-\infty}^{\infty} \prod_{i=1}^{n} \frac{\exp\{x_i(\theta - \delta_i)\}}{\exp\{\delta_i - \theta\} + \exp\{\theta - \delta_i\}} f(\theta) d\theta \end{aligned} \tag{4}$$

where $f(\theta)$ is a population or structural model for the latent variable θ (Adams et al., 1997). This structural model is typically assumed to be a normal distribution. It can be shown that the marginal probability $p(\boldsymbol{x})$ given by the M-RM consists of the same two parts as the E-RM (e.g., de Leeuw & Verhelst, 1986), except that the marginal probability $p(x_+ \mid \boldsymbol{\beta}, \boldsymbol{\lambda})$ is modeled with a latent variable in the M-RM. Importantly, the E-RM simplifies to an MRM if and only if the λ_s constitute a sequence of moments (Cressie & Holland, 1983; see Theorem 3 in Hessen, 2011, for the moment sequence of a normal structural model). This suggests that there exists an explicit relation between a one factor model (the unidimensional Rasch model) and the fully connected network in Figure 5.

Whereas the connection between the E-RM and the M-RM has been known for many decades, the connection between the Curie-Weiss model and the M-RM was only recently observed (Epskamp et al., in press; Marsman et al., 2015). The relation stems from an application of the Gaussian integral:

$$\exp\left\{a^2\right\} = \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} \exp\left\{2a\theta - \theta^2\right\} d\theta$$

Kac (1968) realized that the exponential of a square can be replaced by the integral on the right-hand side, and applied this representation to $\exp\left(\sigma x_+^2\right)$ in the expression for the Curie-Weiss model. We then only need a bit of algebra to obtain the M-RM in Equation (4), with $\delta_i = -\mu_i$. This is an important result as it implies the statistical equivalence of the network approach in Figure 5 and the latent variable approach in Figure 6.

The structural model that results from the derivation is not the typically used normal distribution, but the slightly peculiar:

$$g(\theta) = \sum_{\boldsymbol{x}} g(\boldsymbol{x}, \theta) = \sum_{\boldsymbol{x}} p(\boldsymbol{x}) g(\theta \mid \boldsymbol{x}),$$

where the posterior distribution $g(\theta \mid \boldsymbol{x})$ is a normal distribution with mean $2\sigma x_+$and variance 2σ. That is, $g(\theta)$ reduces to a mixture of $n + 1$ different normal (posterior) distributions; one for every possible test score. In the graphical modeling literature, the joint distribution $g(\boldsymbol{x}, \theta)$ is known as a homogenous conditional Gaussian distribution (HCG; Lauritzen & Wermuth, 1989), a distribution that was first studied by Olkin and Tate (1961, see also; Tate, 1954, 1966). Some instances of the structural model are shown in Figure 7, which reveals a close resemblance to a mixture of two normal distributions with equal variances and their respective means placed symmetrically about zero. For an interaction effect σ that is sufficiently small, Figure 7 reveals that $g(\theta)$ is close to the typically used normal model.

The relation between the Curie-Weiss model and the M-RM with structural model $g(\theta)$ has been established before in the psychometric literature on log-multiplicative association models (LMA; Anderson & Vermunt, 2000; Anderson & Yu, 2007; Anderson et al., 2007; Holland, 1990). However, it was not realized that the marginal distribution $p(\boldsymbol{x})$ (i.e., the LMA model) was a Curie-Weiss model. In return, the Gaussian integral representation that was introduced by Kac (1968) has been used in physics to study the Curie-Weiss model (e.g., Emch & Knops, 1970), and has been independently (re-)discovered throughout the statistical sciences (Lauritzen & Wermuth, 1989; McCullagh, 1994; Olkin & Tate, 1961). Here, however, it was not realized that the conditional distribution $p(\boldsymbol{x} \mid \theta)$ is a Rasch model."

3.2 IM — MD-2PLM

Marsman et al. (2018): "The theory that we have developed for the Curie-Weiss model carries over seamlessly to the more general Ising model. Let us first reiterate the Ising model:

$$p(\boldsymbol{x}) = \frac{\exp\left\{\sum_{i=1}^{n} \mu_i x_i + \sum_{i=1}^{n}\sum_{j=1}^{n} \sigma_{ij} x_i x_j\right\}}{\sum_{\boldsymbol{x}} \exp\left\{\sum_{i=1}^{n} \mu_i x_i + \sum_{i=1}^{n}\sum_{j=1}^{n} \sigma_{ij} x_i x_j\right\}},$$

where we have used the interaction parameters σ_{ij} to encode the network structure. For instance, for the lattice in Figure 2 we have that σ_{ij} is equal to some constant σ if nodes i and j are direct neighbors on the lattice and that σ_{ij} is equal to zero when nodes i and j are no direct neighbors. Since we use the ± 1 notation for the spin random variables x_i, we observe that the terms $\sigma_{ii} x_i^2 = \sigma_{ii}$ cancel in the expression for the Ising model, as these terms are found in both the numerator and the denominator (something similar occurs in the "usual" $\{0, 1\}$ notation).

We will use the eigenvalue decomposition of the so called connectivity matrix $\boldsymbol{\Sigma} = \left[\sigma_{ij}\right]$ to relate the Ising model to a multidimensional IRT model. However, since the elements σ_{ii} cancel in the expression of the Ising model, the diagonal elements of the connectivity matrix are undetermined. This indeterminacy implies that we have some degrees of freedom regarding the eigenvalue decomposition of the connectivity matrix. For now we will use:

$$\boldsymbol{\Sigma}^* = \boldsymbol{\Sigma} + c\boldsymbol{I} = \boldsymbol{Q}(\boldsymbol{\Lambda} + c\boldsymbol{I})\boldsymbol{Q}^\top = \boldsymbol{A}\boldsymbol{A}^\top$$

where $\boldsymbol{Q}$ is the matrix of eigenvectors, $\boldsymbol{\Lambda}$ a diagonal matrix of eigenvalues and we have used $\boldsymbol{A} = \boldsymbol{Q}(\boldsymbol{\Lambda} + c\boldsymbol{I})^{\frac{1}{2}}$ to simplify the notation. The constant c serves to ensure that $\boldsymbol{\Sigma}^*$ is a positive semi-definite matrix by translating the eigenvalues to be positive. Observe that this translation does not affect the eigenvectors but it does imply that only the relative eigenvalues—the *eigen spectrum*—are determined.

With the eigenvalue decomposition, we can write the Ising model in the convenient form:

$$p(\boldsymbol{x}) = \frac{\exp\left\{\sum_{i=1}^{n} \mu_i x_i + \sum_{r=1}^{n}\left(\sum_{i=1}^{n} a_{ir} x_i\right)^2\right\}}{\sum_{\boldsymbol{x}} \exp\left\{\sum_{i=1}^{n} \mu_i x_i + \sum_{r=1}^{n}\left(\sum_{i=1}^{n} a_{ir} x_i\right)^2\right\}},$$

where we have used r to index the columns of $\boldsymbol{A} = [a_{ir}]$. Applying Kac's integral representation to each of the factors $\exp\left(\sum_i a_{ir} x_i\right)^2$ reveals a multivariate latent variable expression for the Ising model, for which the latent variable model $p(\boldsymbol{x} \mid \boldsymbol{\theta})$ is known as the multidimensional two-parameter logistic model (MD-2PL; Reckase, 2009). The MD-2PL is closely related to the factor analytic model for discretized variables (Takane & de Leeuw, 1987), which is why we will refer to $\boldsymbol{A}$ as a matrix of factorloadings. This formal connection between Ising network models and multidimensional IRT models proves the assertion of Molenaar (2003), who was the first to note this correspondence, and shows that to each Ising model we have a statistically equivalent IRT model."

3.3 IM — Blume-Capel

van der Maas et al. (2026): "The minimal extension of the Ising model is to allow for nodes with a ternary structure (e.g., three-state system: $-1/0/1$). A practical reason to consider ternary spin models is that psychological item scales often have an option that is scored as zero. There are many instantiations of these zero options, such as 'don't know', 'not applicable', absence of symptomatology, skipping, 'do not want to reveal', and so forth"

⇒ **Blume-Capel (BC)** model (Blume, 1966; Capel, 1966)

van der Maas et al. (2026): "Ternary spin models, with three possible states $-1/0/1$, add extra terms to the Hamiltonian to deal with the zero state of nodes. The simple extension is the BC model:

$$H(\boldsymbol{x}) = -\sum_{\langle i,j\rangle} x_i x_j - \sum_{i}^{n} \tau x_i + \sum_{i}^{n} \alpha^2 x_i^2. \tag{5}$$

Any nonzero node increases the energy with the factor α^2. The last term is a punishment for not being in the zero state. Because α is squared, this term can only be positive. This seemingly simple extension introduces additional complexity in the mean field behavior (Kaufman & Kanner, 1990). If $\alpha = 0$, the BC's dynamics reduces to that of the basic Ising model. Given that the network is balanced and sufficiently dense, the dynamics are cusp-like. At high temperatures, nodes randomly flip between three states instead of two in the Ising model, but we still have divergence along the temperature axis and hysteresis along the external field axis (as in Figure 1). But if α is large (and the temperature is not too high), nodes tend to be 0. Along this α axis, as shown in Figure 2, we observe a new phenomenon: the butterfly pitchfork bifurcation [...]"

3.4 IM — Potts

Finnemann et al. (2026): "The fourth base model is the Potts model, which characterizes networks comprising elements with *categorical states*, such as ethnicity or nominal answer options in questionnaires. It uses variables x that assume positive integer category values c, where $c \in \{1, 2, 3, ..., h\}$. The discrete nature of states is modeled using the indicator function $I(x_i = x_j)$, which yields 1 if x_i and x_j are identical, otherwise 0:

$$H(\boldsymbol{x}) = -\sum_{\langle i,j\rangle} \omega_{ij} I\left(x_i = x_j\right) - \sum_i \sum_c \alpha_c I\left(x_i = c\right). \tag{11}$$

This energy function counts how many elements are in the same state and aligned with their category-specific external field, α_c. [...] when $h = 2$, the model acts like an Ising model where we see a pitchfork bifurcation [...]"

4 Random Graphs (Erdős–Rényi)

- Stochastic Block Model (SBM; Holland et al., 1983)
- Stochastic Block Ising Model (SBIM)
- Generalized Random Graph (GRG)

- Exponential Random Graph Model (ERGM)
- Temporal ERGM (TERGM)
- Kinetic IM (KIM)

Drobyshevskiy and Turdakov (2019): "What is called a random graph? In almost all cases it is meant by default the Erdős–Rényi (ER) model which refers to one of two very similar classic models: $G(n, m)$ [52] introduced by Erdős and Rényi, and $G(n, p)$ suggested by Edgar Gilbert [61]. $G(n, m)$ gives equal probabilities to all graphs with n nodes and m edges, while in $G(n, p)$ each possible edge on n nodes appears independently with constant probability p. These two models are mostly used in applications and are extensively developed.

[...] one can encounter diverse notions behind the term random graph. [...] In general, 'random graph' can refer to any model, wherein is specified a probability distribution over a set of graphs."

Frieze and Karoński (2015): "The study of random graphs in their own right began in earnest with the seminal paper of Erdős–Rényi [276]. This paper was the first to exhibit the threshold phenomena that characterize the subject.

Let $\mathcal{G}_{n,m}$ be the family of all labeled graphs with vertex set $V = [n] = \{1, 2, \ldots, n\}$ and exactly m edges, $0 \le m \le \binom{n}{2}$. To every graph $G \in \mathcal{G}_{n,m}$, we assign a probability

$$\mathbb{P}(G) = \binom{\binom{n}{2}}{m}^{-1}.$$

Equivalently, we start with an empty graph on the set $[n]$, and insert m edges in such a way that all possible $\binom{\binom{n}{2}}{m}$ choices are equally likely. We denote such a random graph by $\mathbb{G}_{n,m} = ([n], E_{n,m})$ and call it a uniform random graph.

We now describe a similar model. Fix $0 \le p \le 1$. Then for $0 \le m \le \binom{n}{2}$, assign to each graph G with vertex set $[n]$ and m edges a probability

$$\mathbb{P}(G) = p^m(1-p)^{\binom{n}{2}-m},$$

where $0 \le p \le 1$. Equivalently, we start with an empty graph with vertex set $[n]$ and perform $\binom{n}{2}$ Bernoulli experiments inserting edges independently with probability p. We call such a random graph, a binomial random graph and denote it by $\mathbb{G}_{n,p} = \big([n], E_{n,p}\big)$. This was introduced by Gilbert [367]"

4.1 Encompassing Network Model — Idiographic IM

Marsman et al. (2023)

Marsman and Huth (2023): "It was almost fifty years ago that Fortuin and Kasteleyn observed several unexplained relations between the physical phenomena studied in two disparate areas of research. On the one hand, percolation research studies how particles trickle through a porous object, such as how water drips through coffee grounds. On the other hand, in research into magnetism, one studies the way that elementary particles such as electrons may interact to produce a magnet. Fortuin and Kasteleyn set out to discover if the unexplained relations observed between these two research fields were more than just coincidence (Grimmet, 2006) and worked out how the theories of magnetism and percolation relate (Fortuin, 1972a, 1972b; Fortuin & Kasteleyn, 1972). Fortuin and Kasteleyn's approach to consolidate the two scientific theories will be the basis of our idiographic network characterizations. [...]

In a series of papers, Fortuin and Kasteleyn worked out how the theories of percolation and magnetism relate (Fortuin, 1972a, 1972b; Fortuin & Kasteleyn, 1972). They linked the two theories by formulating an encompassing network model that simultaneously describes the states of the variables in the network and the relations between them:

$$\begin{aligned} p(\boldsymbol{X} = \boldsymbol{x}, \boldsymbol{W} = \boldsymbol{w}) &= p(\boldsymbol{X} = \boldsymbol{x} \mid \boldsymbol{W} = \boldsymbol{w}) p(\boldsymbol{W} = \boldsymbol{w}) \\ &= p(\boldsymbol{W} = \boldsymbol{w} \mid \boldsymbol{X} = \boldsymbol{x}) p(\boldsymbol{X} = \boldsymbol{x}). \end{aligned}$$

We refer to this model as the encompassing network because it encompasses both types of network models: A random graph model is used to model the network's topology and a graphical model is used to model the network's node states. The random graph model is the edge-marginal of the encompassing network model,

$$p(\boldsymbol{W} = \boldsymbol{w}) = \sum_{\boldsymbol{x}} p(\boldsymbol{X} = \boldsymbol{x}, \boldsymbol{W} = \boldsymbol{w}),$$

and the graphical model is its node-marginal,

$$p(\boldsymbol{X} = \boldsymbol{x}) = \sum_{\boldsymbol{w}} p(\boldsymbol{X} = \boldsymbol{x}, \quad \boldsymbol{W} = \boldsymbol{w}),$$

where the sums are with respect to all possible node states and edge states, respectively. The idea of this encompassing network model is thus that both nodes and edges are random variables. [...]

Fortuin and Kasteleyn used an encompassing network approach to show that the probability of observing a pattern of node states $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$ in the Ising model can be characterized as follows (see Appendix A for a proof),

$$\overbrace{p(\boldsymbol{X} = \boldsymbol{x})}^{\text{Ising model}} = \sum_{\boldsymbol{w}} \overbrace{p(\boldsymbol{X} = \boldsymbol{x} \mid \boldsymbol{W} = \boldsymbol{w})}^{\text{graph coloring}} \; \overbrace{p(\boldsymbol{W} = \boldsymbol{w})}^{\text{random-cluster model}} \tag{3}$$

where a random graph model $p(\boldsymbol{w})$ known as the random-cluster model is used to express the probability of observing a topological structure $\boldsymbol{w} = (w_{12}, \ldots, w_{n(n-1)})^\top$, and the conditional distribution $p(\boldsymbol{x} \mid \boldsymbol{w})$ expresses the probability of observing the pattern of variable states $\boldsymbol{x}$ given the topology $\boldsymbol{w}$. Since the node states are arbitrarily labeled, the labels are sometimes called colors, and the process of assigning values to the nodes is then called coloring. Thus, $p(\boldsymbol{x} \mid \boldsymbol{w})$ describes the process of coloring the nodes of the graph that is induced by $\boldsymbol{w}$.

[...] Two aspects make Fortuin and Kasteleyn's characterization of the Ising model valuable for modeling psychological phenomena. First, when we assume that the network relations can differ between persons, the persons are characterized by idiographic topologies. Modeling a population of topologies leads us to consider random graph models. Second, we will discuss how the graph coloring model in Equation (3) stipulates a direct connection between the links in the idiographic networks and their correlations in the cross-sectional model. We believe that this connection is essential because it shows us how established psychometric phenomena can emerge from models of the individual."

4.1.1 Graph Coloring

Marsman and Huth (2023): "The following deterministic rule underlies the encompassing network approach: If two variables are connected, then they must be in the same state:

$$\left(W_{ij} = 1\right) \Rightarrow \left(X_i = X_j\right).$$

This rule states that an edge or link between variables causes their values to align, and offers a simplified contact process—a model for a disease spreading about the graph's nodes (Grimmett, 2018)—in which two nodes infect each other, and align their values, when they are in direct contact or connected. Van Borkulo et al. (2017) used similar ideas to model the symptom dynamics underlying depression. While the rule does not imply that two aligned variables are connected, we can use it to learn about the graph's topology based on node alignment (i.e., how the disease has spread). Since, by consequence of the above rule, if two variables are in different states, then they must be disconnected:

$$\left(X_i \neq X_j\right) \Rightarrow \left(W_{ij} = 0\right).$$

We can thus establish that certain direct links are impossible by tracing which nodes are affected and which nodes are not.

In the contact process, the graph's topology is an essential aspect of a disease's dynamics. In Fortuin and Kasteleyn's encompassing network, this is no different. The deterministic rule underlying the encompassing network identify open paths or clusters as a focal point in modeling the latent topologies. Clusters are groups of directly and indirectly related nodes that, by definition, align their values. This could be a group of co-existing symptoms or attitudes or a set of co-related item responses. The graph coloring model assumes that the network's clusters are independently colored, and is equal to

$$\begin{aligned} p(\boldsymbol{X} = \boldsymbol{x} \mid \boldsymbol{w}) &= \prod_{c=1}^{\kappa(\boldsymbol{w})} P\left(\boldsymbol{X}_c = \mathbf{1}\right)^{\frac{x_c+1}{2}} P\left(\boldsymbol{X}_c = -\mathbf{1}\right)^{1-\frac{x_c+1}{2}} \\ &= \frac{1}{2^{\kappa(\boldsymbol{w})}} \end{aligned} \tag{4}$$

where $\boldsymbol{X}_c$ denotes the observables in a cluster c and x_c denotes their collective state. The sampling mechanism inspired by this graph coloring model is relatively simple: If the nodes are in the same cluster, then each node in that cluster is set equal to +1 with probability 1/2 or set equal to -1 with probability 1/2. Fortuin and Kasteleyn showed that this graph coloring model corresponds to an Ising model without main effects.

Cioletti and Vila (2016) have recently shown how to extend Fortuin and Kasteleyn's formulation to Ising models that include main effects. Their generalization shows that the Ising model's main effects impact the weight assigned to the cluster's values. The coloring probabilities in Cioletti and Vila's generalization become,

$$P(\boldsymbol{X}_c = \mathbf{1}) = \frac{\exp\left(\sum_{i\in \mathrm{V}_c} \mu_i\right)}{\exp\left(\sum_{i\in \mathrm{V}_c} \mu_i\right) + \exp\left(-\sum_{i\in \mathrm{V}_c} \mu_i\right)},$$

where V_c denotes the set of indices of vertices in cluster c. In this model, there is a greater probability of generating a positive than generating a negative value for the cluster's nodes if their main effects are positive. In contrast, there is a smaller probability of generating a positive than generating a negative value for the cluster's nodes if their main effects are negative. In psychopathology, for example, the main effects are typically negative to cover the overall absence of symptoms in the population. In education, on the other hand, the main effects are typically positive to cover the overall ability to solve test items.

The graph coloring model introduced by Fortuin and Kasteleyn and generalized by Cioletti and Vila opens up a direct connection between the idiographic topologies and cross-sectional correlations."

4.1.2 Random Cluster Model — ER — Divide and Color

Divide and Color Model (Häggström, 2001)
Marsman and Huth (2023): "In their seminal work, Fortuin and Kasteleyn (1972) showed that the edge-marginal in Equation (3) is an Ising model if the topologies are samples from a ran-dom-cluster model. The random-cluster model is defined by the following probability distribution over topological structures $\boldsymbol{w}$:

$$p(\boldsymbol{W} = \boldsymbol{w}) = \frac{1}{\mathrm{Z}_R} \overbrace{\prod_{i=1}^{n-1} \prod_{j=i+1}^{n} \theta_{ij}^{w_{ij}} \left(1 - \theta_{ij}\right)^{1-w_{ij}}}^{\text{Erdős–Rényi model}} \lambda^{\kappa(\boldsymbol{w})},$$

where $\kappa(\boldsymbol{w})$ denotes the number of connected components or clusters in the topological structure that is implied by $\boldsymbol{w}$, λ is a positive clustering weight, and Z_R is the model's normalizing constant. Observe that the random-cluster model has the same kernel as the Erdős–Rényi model model in Equation (1), but differs from the Erdős-Rényi model in terms of the weight it assigns to the network's clusters. The random-cluster model tends to favor networks with fewer clusters when the weight is smaller than one (i.e., $\lambda < 1$) and tends to favor networks with more clusters when the weight is larger than one (i.e., $\lambda > 1$). The random-cluster model with a unit clustering weight coincides with the Erdös-Rényi model in Equation (1). Fortuin and Kasteleyn showed that the characterization in Equation (3) coincides with an Ising model without main effects if the topologies are samples from a random-cluster model with a clustering weight of 2.

[...] The generalization of Cioletti and Vila (2016) shows that the main effects impact the clustering weights of the random-cluster model, which evolve to

$$\begin{aligned}\lambda_c &= 2\cosh\left(\sum_{i\in V_c}\mu_i\right)\\ &= \exp\left(\sum_{i\in V_c}\mu_i\right)+\exp\left(-\sum_{i\in V_c}\mu_i\right)\geq 2.\end{aligned} \tag{6}$$

Thus, by including main effects, the random-cluster model assigns different weights to the different clusters,

$$p(\boldsymbol{W}=\boldsymbol{w}) = \frac{1}{Z_R}\prod_{i=1}^{n-1}\prod_{j=i+1}^{n}\theta_{ij}^{w_{ij}}\left(1-\theta_{ij}\right)^{1-w_{ij}}\prod_{c=1}^{\kappa(\boldsymbol{w})}\lambda_c.$$

Since the main effects in empirical applications typically do not sum to zero, their inclusion thus leads to even more fragmentation in the idiographic networks. In psychopathology, for example, the main effects are typically negative to cover the overall absence of symptoms in the population. In education, on the other hand, the main effects are typically positive to cover the overall ability to solve test items. In both applications, the clustering weights will then be strictly larger than two.

[...] there might be reasons that persuade us to consider other population models for the latent topologies. One reason might be the random-cluster model's analytic intractability. The normalizing constant

$$\mathrm{Z}_R = \sum_{w}\prod_{i=1}^{n-1}\prod_{j=i+1}^{n}\theta_{ij}^{w_{ij}}\left(1-\theta_{ij}\right)^{1-w_j}\prod_{c=1}^{\kappa(w)}\lambda_c,$$

involves a sum over all $2^{\frac{1}{2}n(n-1)}$ possible topological states $\boldsymbol{w}$, and may be difficult to evaluate in practice. Another reason might be the empirical fit of the random-cluster model—and thus also the Ising model. One solution is to use the Erdős–Rényi model rather than the full random-cluster model for the distribution of the latent topologies. The Erdős–Rényi model is analytically tractable and offers great modeling flexibility. The application of an Erdős–Rényi model to describe the distribution of random graphs in Equation (3) is known as the divide and color model (Häggström, 2001)."

$$\overbrace{p(\boldsymbol{X}=\boldsymbol{x})}^{\text{Divide and Color model}} = \sum_{\boldsymbol{w}}\overbrace{p(\boldsymbol{X}=\boldsymbol{x}\mid\boldsymbol{W}=\boldsymbol{w})}^{\text{graph coloring}}\ \overbrace{p(\boldsymbol{W}=\boldsymbol{w})}^{\text{Erdős–Rényi model}} \tag{7}$$

—The relation between idiographic topologies and cross-sectional phenomena—

The divide and color model does not coincide with an Ising model. However, Häggström (2001) showed that the model does share several features with the particular Ising model in Fortuin and Kasteleyn's formulation. In particular, that the population correlation between any two observables X_i and X_j in the network is non-negative (Fortuin et al., 1971). This observation reveals a significant restriction on both the divide and color model and the particular Ising model that emerges from Fortuin and Kasteleyn's theory. However, the finding that observables non-negatively correlate in the population is consistent with other latent variable approaches (Holland & Rosenbaum, 1986), and with the robust finding of a positive manifold in psychometric research (e.g., Caspi et al., 2014; van der Maas et al., 2006).

The result that observables non-negatively correlate in the divide and color model in Equation (7) and the Fortuin-Kasteleyn model in Equation (3) reveals a deep connection between observables and underlying topologies. This relationship between two observables in the network and their connection in the latent topologies can be made precise (c.f. Grimmet, 2006, Theorem 1.16; Steif & Tykesson, 2017, Eq. 1.1)

$$P\left(X_i = X_j\right) = \frac{1}{2} + \frac{1}{2} \quad P\left(X_i \leftrightarrow X_j\right) \geq \frac{1}{2}, \tag{8}$$

where $P\left(X_i \leftrightarrow X_j\right)$ denotes the probability that the two variables end up in the same cluster. The two-point correlation $\tau\left(X_i, X_j\right) = P\left(X_i = X_j\right) - \frac{1}{2}$ then equals $\frac{1}{2} P\left(X_i \leftrightarrow X_j\right)$, which confirms the non-negative population correlations. Since the states of observables align when they are in the same cluster, their cross-sectional correlation will be high if they end up in the same cluster for many individuals. Thus, the way that the latent topologies cluster has a tremendous impact on observed correlations. The relation in Equation (8) holds for every encompassing network model that uses Fortuin and Kasteleyn's graph coloring model, i.e., for every random graph model $p(\boldsymbol{w})$ stipulated on the latent topologies. However, Equation (8) only pertains to scenarios that involve no main effects. Equation (1.1) in Steif and Tykesson (2017) and Theorem 2 in Cioletti and Vila (2016) offer the generalizations that are relevant for scenarios that involve main effects.

The topologies that are consistent with the Ising model are heavier fragmented than the topologies that underly the divide and color model. This difference in fragmentation is the result of using a smaller clustering weight in the divide and color model, c.f. Figure 4. Since the edge inclusion probabilities are higher when there is less fragmentation, the two-point population correlations will be higher for the divide and color model than for the Fortuin and Kasteleyn model. In Figure 5 we show that, for models with the same edge inclusion probabilities, the correlations are higher for the divide-and-color than for Fortuin and Kasteleyn's idiographic Ising model. The graph generating mechanisms of the random-cluster model and the Erdös-Rényi model are thus fundamentally different and may lead to divergent predictions and interventions on underlying topologies.

That population correlations in Fortuin and Kasteleyn's theory are a function of the edge inclusion probabilities of the variable pairs hints at a relation between associations in the Ising model and edge probabilities in the random-cluster model in Equation (3). This relationship indeed exists for the Fortuin-Kasteleyn model,

$$\sigma_{ij} = -\frac{1}{2} \log\left(1 - \theta_{ij}\right). \tag{9}$$

We illustrate the relation in Figure 6. Observe that the relation is only valid for non-negative associations and is consistent with the restriction to non-negative population correlations. The generalization of the relation to negative associations has been the topic of several investigations that we will not pursue further here (e.g., C. Newman, 1991; Swendsen & Wang, 1987).

The relationship in Equation (9) yields the bridge between group-level associations and links in idiographic networks. On the one hand, it allows us to interpret group-level associations in terms of the proportion of observed edges between variables in the idiographic networks. One substantial result, for example, is that absent associations at the group-level imply the absence of a link between two variables in all idiographic networks. Thus, the idiographic networks mirror the conditional independence property of the Ising model. On the other hand, the relation in Equation (9) also underscores the difference between group-level associations and links at the individual level. At the individual level, two variables will be entirely dependent if they are in the same cluster and independent otherwise. At the group-level, however, this can still generate a wide variety of associations. Moreover, the fact that patterns in observed correlations are reflections of patterns in the underlying edge inclusion probabilities provides us with an alternative view on psychometric models. Savi et al. (2019), for example, characterizes the bi-factor structure, which is ubiquitous in psychometrics, in terms of an underlying community or stochastic block structure. This connection between psychometric models and models from network science offers another exciting avenue for future research (Marsman et al., 2022)."

4.2 ER — SBM

Marsman and Huth (2023): "In general, Erdős–Rényi model can be used to model the distribution of edges between pairs of nodes in a network. With W_{ij} denoting a binary random variable that, if it is equal to one, indicates the presence of a link between nodes i and j, and if it is equal to zero indicates the absence of that link, the Erdős–Rényi model for an n-variable network is characterized by the following probability distribution

$$
\begin{aligned}
p\Big(\boldsymbol{W} = \begin{pmatrix} w_{12}, & \dots, & w_{(n-1)n} \end{pmatrix}^\top\Big) \\
= \prod_{i=1}^{n-1} \prod_{j=i+1}^{n} \theta_{ij}^{w_{ij}} \left(1 - \theta_{ij}\right)^{1-w_{ij}} \qquad (1)
\end{aligned}
$$

where θ_{ij} denotes the probability that the random variable W_{ij} is equal to one. Thus, the Erdős–Rényi model comprises a set of independent Bernoulli variables, one for every pair of nodes in the network. Where the original Erdős–Rényi model has a constant edge inclusion probability θ, our formulation allows for distinct edge inclusion probabilities θ_{ij} for the separate edges w_{ij}.

[...] The Erdős–Rényi model is thus a model for the topology of the network, as it describes a probability distribution over different realizations of the network's structure. [...]

Our formulation of the Erdős–Rényi model model—a product of independent Bernoulli distributions—covers the stochastic block model as a special case (Holland et al., 1983). The stochastic block model is a popular variant of the Erdős–Rényi model that groups nodes into blocks or communities (i.e., it differentiates the inclusion probabilities of links between nodes that are part of the same community from the inclusion probabilities of links between nodes from different communities)."

S. Wang (2021): "In this section, I summarize the posteriori blockmodeling of networks introduced by Snijders and Nowicki (1997) and Nowicki and Snijders (2001). In the stochastic blockmodels, the probability of a connection between two nodes depends on their block (group) memberships. Suppose there is a network written as an adjacency matrix:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,N} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \dots & x_{N,N} \end{bmatrix}, \tag{1}$$

where $x_{a,b}$ represents the presence of an edge (binary network) or the degree of association (weighted network) between nodes a and $b, a, b = 1, \dots, N$ and $b \neq a$, and N is the total number of nodes. Let us define the discrete latent variable u_a as the latent block membership for node $a, u_a \in 1, 2, \dots, K$, where K is the total number of blocks or groups. Conditional on the latent variable, the probability of two nodes forming an edge is independent of all the other edges in the network:

$$P(\boldsymbol{X} \mid \boldsymbol{U}) = \prod_{a=1}^{N} \prod_{b=1, b\neq a}^{N} p\left(x_{a,b} \mid u_a, u_b\right), \quad p\left(x_{a,b} \mid u_a, u_b\right) = m_{u_a,u_b}, \tag{2}$$

where m_{u_a,u_b} is the probability of a connection between the corresponding blocks."

Liu (2017): "Stochastic block model (SBM), also known as planted partition model, is one of the most commonly used generative network model. In this model, every node $i \in V = \{1, 2, \cdots, n\}$ is assigned a latent type (community label) σ_i with probability π_{σ_i}. Conditioned on node types, the connection between nodes is independent of each other. For every two nodes i and j, the conditional connection probability is $q_{\sigma_i \sigma_j}$, which depends on the types of the two nodes. Denote the SBM defined by q, π as $\mathrm{SBM}(q, \pi)$. When the connection probability is a constant, the model becomes the Erdős–Rényi model $\mathcal{G}(n, q)$. The clustering (community detection) problem is to infer the latent types from the network structure. This is an important problem in many areas [...]"

4.2.1 SBM — SBIM — IM

Sekulovski et al. (2025): "Graphical models facilitate the representation of psychological variables as complex systems of interacting variables structured as a network. However, their existing statistical analyses overlook the assumption of clustering—the grouping of subsets of variables that are more densely connected within the network—despite its central role in many psychological theories. We address this gap by proposing the use of the Stochastic Block Model (SBM) as a prior distribution on the network structure of a graphical model for binary and ordinal data. The SBM assumes that variables belong to latent clusters, where the probability of an edge depends on the cluster membership of the nodes. Embedding this prior in a Bayesian graphical modeling framework allows researchers to formally incorporate theoretical expectations about clustering, test hypotheses about the number of clusters, and estimate cluster assignments from cross-sectional data."

Zhang et al. (2023): "In this paper, we propose a novel stochastic block Ising model (SBIM) to incorporate the inter-layer dependence for community detection in multilayer networks. It models the homogeneous community structure with the SBM model, and employs the popular Ising model (Cheng et al., 2014; Ravikumar et al., 2010) to characterize the inter-layer dependence."

M. Ye (2021): "In this paper, we propose a natural composition of SBM and the Ising model, which we call the Stochastic Ising Block Model (SIBM). First we use SBM to generate a graph $G = ([n], E(G))$ with vertex set $[n]$ and edge set $E(G)$ based on an (unknown) partition of the vertex set $[n]$. Next we use G as the underlying dependency graph of the Ising model and draw m i.i.d. samples from it. The objective is to exactly recover the partition of the vertex set in SBM from the samples generated by the Ising model, without observing the graph G."

Zhao et al. (2021): "In this paper, we study the phase transition property of an Ising model defined on a special random graph—the stochastic block model (SBM)"

Berthet et al. (2019): "In this work, we introduce the Ising blockmodel in order to combine the notions of the stochastic blockmodel and that of graphical model by assuming that we observe independent copies of a vector of spins $\sigma = \left(\sigma_1, \ldots, \sigma_p\right) \in \{-1, 1\}^p$ distributed according to an Ising model with a block structure analogous to the one arising in the stochastic blockmodel.

[2.1. Definition.] Let $p \geq 2$ be an even integer and let $S \subset [p] := \{1, \ldots, p\}$ be a subset of size $|S| = m = p/2$. For any partition $(S, \bar{S})$, where $\bar{S} = [p]\backslash S$ denotes the complement of S, write $i \sim j$ if $(i, j) \in S^2 \cup \bar{S}^2$ and $i \nsim j$ if $(i, j) \in [p]^2 \backslash \left(S^2 \cup \bar{S}^2\right)$. Fix $\beta, \alpha \in \mathbb{R}$ and let $\sigma \in \{-1, 1\}^p$ have density $f_{S,\alpha,\beta}$ with respect to the counting measure on $\{-1, 1\}^p$ given by

$$f_{S,\alpha,\beta}(\sigma) = \frac{1}{Z_{\alpha,\beta}} \exp\left[\frac{\beta}{2p} \sum_{i \sim j} \sigma_i \sigma_j + \frac{\alpha}{2p} \sum_{i \nsim j} \sigma_i \sigma_j\right], \tag{2.1}$$

where

$$Z_{S,\alpha,\beta} := \sum_{\sigma \in \{-1,1\}^p} \exp\left[\frac{\beta}{2p} \sum_{i \sim j} \sigma_i \sigma_j + \frac{\alpha}{2p} \sum_{i \nsim j} \sigma_i \sigma_j\right] \tag{2.2}$$

is a normalizing constant traditionally called partition function. Let $\mathbb{P}_{S,\alpha,\beta}$ denote the probability distribution over $\{-1, 1\}^p$ that has density $f_{S,\alpha,\beta}$ with respect to the counting measure on $\{-1, 1\}^p$. We call this model the Ising Blockmodel (IBM). In the sequel, we write simply $f_{\alpha,\beta}$ and $\mathbb{P}_{\alpha,\beta}$ to emphasize the dependency on α, β and simply $\mathbb{P}_S$ to emphasize the dependency on S. We study in this work the statistical problem of identifying the partition ($S, \bar{S}$) based on n independent draws from $\mathbb{P}_{\alpha,\beta}$, when $\alpha < \beta$.

[2.2. Link with the Curie-Weiss model.] When $\alpha = \beta > 0$, the model (2.1) is the mean field approximation of the (ferromagnetic) Ising model and is called the Curie-Weiss model (without external field)."

4.3 ER — GRG

Shi and Qian (2009): "A generalization of the classical Erdős–Rényi (ER) random graph is introduced and investigated. A generalized random graph (GRG) admits different values of probabilities for its edges rather than a single probability uniformly for all edges as in the ER model. In probabilistic terms, the vertices of a GRG are no longer statistically identical in general, giving rise to the possibility of complex network topology. Depending on their surrounding edge probabilities, vertices of a GRG can be either 'homogeneous' or 'heterogeneous'. [...]

A standard graph with N vertices can be completely specified by the adjacency matrix E : a symmetric $N \times N$ matrix consisting of 0's and 1's with all diagonal entries being 0. A matrix entry e_{ij} with value 1 represents an edge between vertex pair (i, j). A generalized random graph is defined as follows: a Bernoulli random variable $\boldsymbol{e}_{ij}\left(p_{ij}\right)$ is assigned to the vertex pair (i, j). A Bernoulli random variable $\boldsymbol{e}(p)$ has $\Pr\{\boldsymbol{e} = 1\} = p$ and $\Pr\{\boldsymbol{e} = 0\} = 1 - p$. The matrix $\boldsymbol{M} = \left[\boldsymbol{e}_{ij}\left(p_{ij}\right)\right]_{N \times N}$, therefore, is a random matrix, which corresponds to a random graph ensemble. Note that each realization of $\boldsymbol{M}$ should be symmetric with all the diagonal entries being 0. Theoretically, this is the universal representation for all random graphs; we shall denote it by $\mathcal{G}(N, \boldsymbol{M})$. In the present paper, we assume that the entries $\boldsymbol{e}_{ij}$ in matrix $\boldsymbol{M}$ are mutually independent unless otherwise noted. Random graphs with dependent $\boldsymbol{e}_{ij}$'s exhibit greater topological complexity which will be the subject of forthcoming publications. From $\mathcal{G}(N, \boldsymbol{M})$, different graph realizations could be obtained by randomly sampling the Bernoulli random variable $\boldsymbol{e}_{ij}$ on each vertex pair (i, j). All possible graph realizations can be characterized by their adjacency matrices $E_1, E_2, \ldots, E_m$, where $m = 2^{N(N-1)/2}$. We use $G(N, E)$ to denote the graph realization G with adjacency matrix $E = \left[e_{ij}\right]_{N\times N}$. The following two facts hold:

FACT 2.1.

$$\Pr\{G(N, E) \mid \boldsymbol{M}\} = \prod_{i,j} p_{ij}^{e_{ij}} \left(1 - p_{ij}\right)^{1-e_{ij}}$$

. [...]"

→ the same as in (Marsman & Huth, 2023)?

4.4 ER — ERGM — GLM

R-Package: 'ergm', 'tergm', 'bigergm'

Kolaczyk and Csárdi (2020): "Exponential random graph models (ERGMs) are designed in direct analogy to the classical generalized linear models (GLMs). They are formulated in a manner that is intended to facilitate the adaptation and extension of well-established statistical principles and methods for the construction, fitting, and comparison of models. Nevertheless, the appropriate specification and fitting of ERGMs can be decidedly more subtle than with standard GLMs. Moreover, much of the standard inferential infrastructure available for GLMs, resting on asymptotic approximations to appropriate chi-square distributions, has yet be formally justified in the case of ERGMs. As a result, while this class of models arguably has substantial potential, in practice it must be used with some care.

Consider $G = (V, E)$ as a random graph. Let $Y_{ij} = Y_{ji}$ be a binary random variable indicating the presence or absence of an edge $e \in E$ between the two vertices i and j in V. The matrix $\boldsymbol{Y} = \left[Y_{ij}\right]$ is thus the (random) adjacency matrix for G. Denote by $\boldsymbol{y} = \left[y_{ij}\right]$ a particular realization of $\boldsymbol{Y}$. An exponential random graph model is a model specified in exponential family form for the joint distribution of the elements in $\boldsymbol{Y}$. The basic specification for an ERGM is a model of the form

$$\mathbb{P}_\theta(\boldsymbol{Y} = \boldsymbol{y}) = \left(\frac{1}{\kappa}\right) \exp\left\{\sum_H \theta_H g_H(\boldsymbol{y})\right\}, \tag{6.2}$$

where

1. each H is a *configuration*, which is defined to be a set of possible edges among a subset of the vertices in G;

2. $g_H(\boldsymbol{y}) = \prod_{y_{ij}\in H} y_{ij}$, and is therefore either one if the configuration H occurs in $\boldsymbol{y}$, or zero, otherwise;

3. a non-zero value for θ_H means that the Y_{ij} are dependent for all pairs of vertices $\{i, j\}$ in H, conditional upon the rest of the graph; and

4. $\kappa = \kappa(\theta)$ is a normalization constant,

$$\kappa(\theta) = \sum_{\boldsymbol{y}} \exp\left\{\sum_{H} \theta_H g_H(\boldsymbol{y})\right\}. \tag{6.3}$$

Note that the summation in (6.2) is over all possible configurations H. Importantly, given a choice of functions g_H and their coefficients θ_H, this implies a certain (in)dependency structure among the elements in $\boldsymbol{Y}$, which is, of course, appealing, given the inherently relational nature of a network. Generally speaking, such structure typically can be described as specifying that the random variables $\left\{Y_{ij}\right\}_{(i,j)\in\mathscr{A}}$ are independent of $\left\{Y_{i'j'}\right\}_{(i',j')\in\mathscr{B}}$, conditional on the values of $\left\{Y_{i'',j''}\right\}_{(i'',j'')\in\mathscr{C}}$, for some given index sets $\mathscr{A}$, $\mathscr{B}$, and $\mathscr{C}$. Conversely, we can begin with a collection of (in)dependence relations among subsets of elements in $\boldsymbol{Y}$ and try to derive the induced form of the (g_H, θ_H) pairs.

The ERGM framework allows for a number of variations and extensions. For example, directed versions of ERGMs are also available. In addition, in defining ERGMs for either undirected or directed graphs, it is straightforward to include, if desired, information on vertices beyond their connectivity, such as actor attributes in a social network or known functionalities of proteins in a network of protein interactions. Given a realization $\boldsymbol{x}$ of a random vector $\boldsymbol{X}$ on the vertices in G, we simply specify an exponential form for the conditional distribution $\mathbb{P}_\theta(\boldsymbol{Y} = \boldsymbol{y} \mid \boldsymbol{X} = \boldsymbol{x})$ that involves additional statistics $g(\cdot)$ that are functions of both $\boldsymbol{y}$ and $\boldsymbol{x}$.

[...] We have already seen an example of what is arguably the simplest ERGM, in the form of the Bernoulli random graph model of Sect. 5.2. To see this, suppose we specify that, for a given pair of vertices, the presence or absence of an edge between that pair is independent of the status of possible edges between any other pairs of vertices. Then $\theta_H = 0$ for all configurations H involving three or more vertices. As a result, the ERGM in (6.2) reduces to

$$\mathbb{P}_\theta(\boldsymbol{Y} = \boldsymbol{y}) = \left(\frac{1}{\kappa}\right) \exp\left\{\sum_{i,j} \theta_{ij} y_{ij}\right\}. \tag{6.4}$$

Furthermore, if we assume that the coefficients θ_{ij} are equal to some common value θ (typically referred to as an assumption of homogeneity across the network), then (6.4) further simplifies to

$$\mathbb{P}_\theta(\boldsymbol{Y} = \boldsymbol{y}) = \left(\frac{1}{\kappa}\right) \exp\{\theta L(\boldsymbol{y})\}, \tag{6.5}$$

where $L(\boldsymbol{y}) = \sum_{i,j} y_{ij} = N_e$ is the number of edges in the graph. The result is equivalent to a Bernoulli random graph model, with $p = \exp(\theta)/[1 + \exp(\theta)]$."

Schweinberger et al. (2020): "Some notable special cases of ERGMs include:

- "network" GLMs (McCullagh and Nelder, 1983), including network logistic regression and other forms of network regression (Krackhardt, 1988);
- Bernoulli random graphs (Gilbert, 1959; Erd˝os and R´enyi, 1959, 1960);
- categorical data models (e.g., U|MAN models, Wasserman and Faust, 1994);
- β-models (Chatterjee, Diaconis and Sly, 2011);
- p_1 models (Holland and Leinhardt, 1981);
- canonical exponential-family models with Markov dependence (Frank and Strauss, 1986), which are related to Ising models in physics (Ising, 1925), Markov logic networks in artificial intelligence (Richardson and Domingos, 2006), Markov random fields in spatial statistics (Besag, 1974; Cressie, 1993; Stein, 1999), and undirected graphical models (e.g., Lauritzen, 1996; Ravikumar, Wainwright and Lafferty, 2010);
- and curved exponential-family models (Snijders et al., 2006; Hunter and Handcock, 2006; Hunter, Goodreau and Handcock, 2008).

" The advantages of exponential-family representations of random graph models are many. Some of the more important ones are:

1. *Language.* Exponential families provide a convenient language for formulating ideas about network data and dependencies therein. In so doing, the language of exponential families facilitates the construction of models for complex and dependent network data.
2. *Unifying statistical framework.* [...]
3. *Computational advantages.* [...]
4. *Theoretical advantages.* [...]
5. *Practical advantages.* Among the practical advantages is the fact that ERGMs include GLMs for independent network data—e.g., logistic regression models—as special cases and may be viewed as GLMs for independent as well as dependent network data. Many network scientists are familiar with logistic regression models, which facilitates the interpretation of results, in that estimates of parameters may be interpreted in terms of conditional log odds and log odds ratios (e.g., Hunter, 2007; Lusher, Koskinen and Robins, 2013; Stewart et al., 2019).

In addition, exponential-family models can be used as building blocks to construct more complex models, e.g., by using them as building blocks in stochastic block models to capture transitivity and other complex dependencies within blocks (Schweinberger and Handcock, 2015)."

4.5 ERGM — TERGM — KIM

Krivitsky and Handcock (2014): "Temporal ERGMs (TERGMs) are a natural elaboration of the traditional ERGM framework. They are essentially stepwise ERGMs in time."

Campajola et al. (2022): "A network, defined by a set of M nodes and a set of links between pairs of nodes, can be described by an $M \times M$ binary adjacency matrix $G \in \{0, 1\}^{M\times M}$, where $G_{ij} = 1$ if a link between nodes i and j is present and $G_{ij} = 0$ otherwise. When the relation described by the links is not directional, $G_{ij} = G_{ji}$ and the network is said to be undirected. We consider temporal networks where the number of nodes M is fixed across multiple time steps and indicate the adjacency matrix of the graph at time t by $G(t)$.

In order to use the KIM to model a temporal network, we map the elements of the adjacency matrix into spins, associating a present link to a spin $+1$ and an absent link to a spin -1. In this way we represent each adjacency matrix $G(t)$ as a vector $s(t) \in \{-1, 1\}^N$ where $N = M(M - 1)/2$, assuming the network to be undirected and without self loops. In light of this mapping, the matrix J now captures the tendency of links to influence each other at lag one—for example the diagonal terms can be interpreted as measuring link persistence, while the elements of h are associated with the idiosyncratic probability to observe a given link.

Interestingly, such a mapping highlights that the (standard) KIM can be seen as belonging to the Temporal Exponential Random Graph Model family, as we discuss in the SI. Moreover, it turns out that a large subset of possible specifications from this family can be mapped into a KIM. Hence, the score driven KIM that we propose here is an extension of the Temporal Exponential Random Graph Model allowing its parameters to evolve in time."

5 Time Series Analysis (Econometrics)

- AutoRegressive Moving-Average (ARMA) → Vector ARMA (VARMA)
- Vector Discrete AR (VDAR)
- AR Tensor (ART)
- Threshold AR (TAR) → hysteric TAR (HysTAR)
- Structural VAR (SVAR)
- unified SEM (uSEM)
- Group Iterative Multiple Model Estimation(GIMME)
- graphical VAR (gVAR or GVAR; Epskamp, Waldorp, et al., 2018)

- multilevel VAR (mlVAR)
- Dynamic Factor Model (DFM)
- Hidden Markov Model (HMM)
- Residual Structural Equation Model (RSEM)
- Dynamic Structural Equation Modeling (DSEM)
- Dynamic Latent Class Analysis (DLCA)
- Nonlinear Dynamic Latent Class - Structural Equation Modeling(NDLC-SEM)
- time-series latent variable gVAR (ts-lvgvar)

R-Package: 'arima2'

5.1 (AR —) VAR — VDAR

Holan et al. (2010): The multivariate generalization of an ARMA series is the vector autoregressive moving-average (VARMA or MARMA) series. A K-dimensional series $\{\boldsymbol{X}_t\}$ is said to be a VARMA series of autoregressive order p and moving-average order q if it is a solution to

$$\boldsymbol{X}_t - \boldsymbol{\Phi}_1 \boldsymbol{X}_{t-1} - \cdots - \boldsymbol{\Phi}_p \boldsymbol{X}_{t-p} = \boldsymbol{Z}_t + \Theta_1 \boldsymbol{Z}_{t-1} + \cdots + \Theta_q \boldsymbol{Z}_{t-q}, \tag{7.35}$$

with $\boldsymbol{\Theta}_p \neq \boldsymbol{0}$, $\boldsymbol{\Theta}_q \neq \boldsymbol{0}$, and $\{\boldsymbol{Z}_t\}$, defined for fixed t by $\boldsymbol{Z}_t = (Z_{1,t}, \ldots, Z_{K,t})'$, is zero mean multivariate white noise with covariance matrix $E\left[\boldsymbol{Z}_t \boldsymbol{Z}_t'\right] = \boldsymbol{\Sigma}$. The coefficient matrices $\boldsymbol{\Phi}_i$, $i = 1, \ldots, p$, and $\boldsymbol{\Theta}_j$, $j = 1, \ldots, q$, are $K \times K$ matrices.

The autoregressive and moving-average polynomials $\boldsymbol{\Phi}(\cdot)$ and $\boldsymbol{\Theta}(\cdot)$ are

$$\boldsymbol{\Phi}(z) = \boldsymbol{I} - \boldsymbol{\Phi}_1 z - \cdots - \boldsymbol{\Phi}_p z^p \quad \text{and} \quad \boldsymbol{\Theta}(z) = \boldsymbol{I} + \boldsymbol{\Theta}_1 z + \cdots + \boldsymbol{\Theta}_q z^q$$

for a complex-valued z; here, $\boldsymbol{I}$ is the $K \times K$ identity matrix. Using the backshift operator, (7.35) becomes

$$\boldsymbol{\Phi}(B)\boldsymbol{X}_t = \boldsymbol{\Theta}(B)\boldsymbol{Z}_t.$$

Causality and invertibility of the VARMA model imposes that solutions to $\det(\boldsymbol{\Phi}(z)) = 0$ and $\det(\boldsymbol{\Theta}(z)) = 0$ lie outside the complex unit circle, respectively. A causal VARMA model has the linear representation

$$\boldsymbol{X}_t = \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \boldsymbol{Z}_{t-j}, \tag{7.36}$$

where the $\boldsymbol{\Psi}_j$ s are found by equating coefficients in the relationship

$$\boldsymbol{\Psi}(z) := \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j z^j = \boldsymbol{\Phi}(z)^{-1} \boldsymbol{\Theta}(z)$$

Causality implies that $\sum_{j=0}^{\infty} \left|\mathbf{\Psi}_j\right| < \infty$ in a component-by-component sense. Evaluating covariances via (7.36) provides

$$\mathbf{\Gamma}(h) = \sum_{j=0}^{\infty} \mathbf{\Psi}_{j+h} \mathbf{\Sigma} \mathbf{\Psi}_j'.$$

The spectral density matrix of the causal VARMA (p, q) process is

$$\boldsymbol{f}(\lambda) = \frac{1}{2\pi} \mathbf{\Psi}\left(e^{-i\lambda}\right) \mathbf{\Sigma} \mathbf{\Psi}\left(e^{-i\lambda}\right)'.$$

Mazzarisi et al. (2020): "a discrete autoregressive process of order p, namely DAR(p). The DAR(p) process, first introduced by Jacobs and Lewis (1978) , is the natural extension of the standard autoregressive process for binary time series. We consider a multivariate generalization of the DAR(p) process, namely VDAR(p), where the binary variable X describing the occurrence of an extreme event for the underlying time series x can be copied from its past values or from the past values of the extreme events of another time series y.

[...] In this section we propose the multivariate generalization of the DAR(p) process to describe jointly the dependence structure of N binary random variables which represent the extreme events of some underlying time series. The introduction of this modeling framework permits to test for the presence of non-zero terms of (lagged) interactions between the binary variables, which signal causal relations.

The discrete autoregressive model DAR(p) has been introduced for the first time by Jacobs and Lewis (1978) and recently applied to the modeling of temporal networks (Mazzarisi et al., 2020; Williams et al., 2019a; 2019b). In the most general setting, it describes the time evolution of a categorical variable which has p-th order Markov dependence and a multinomial marginal distribution. Here we are interested in the case of a binary random variable X, thus the marginal distribution is Bernoulli. If X_t is the realization of X at time t, we have

$$X_t = V_t X_{t-\tau_t} + (1 - V_t) Z_t \tag{2.1}$$

where $X_t \in \{0, 1\} \forall t, V_t \sim \mathcal{B}(\tilde{\nu})$ is a Bernoulli random variable with $\tilde{\nu} \in [0, 1], \tau_t \sim \mathcal{M}\left(\tilde{\gamma}_1, \ldots, \tilde{\gamma}_p\right)$ is a multinomial random variable with $\tilde{\gamma} \equiv \left\{\tilde{\gamma}_j\right\}_{j=1,\ldots,p}$ such that $\sum_{i=1}^{p} \tilde{\gamma}_i = 1$ and $Z_t \sim \mathcal{B}(\tilde{\chi})$ is a Bernoulli random variable with $\tilde{\chi} \in [0, 1]$. In other words, at each time, V_t determines whether X_t is copied from the past or sampled from the marginal. When X_t is copied from the past, the multinomial random variable τ_t selects the time lag and, accordingly, which past realization of X we copy.

The process defined by (2.1) has the property that the autocorrelation at any lag is larger than or equal to zero. This is by construction, since $\tilde{\nu}, \tilde{\gamma}_1, \ldots, \tilde{\gamma}_p$ are non-negative definite. Moreover, the Yule-Walker equations associated with (2.1) are formally equivalent to the ones of the standard AR(p) process, see Jacobs and Lewis (1978).

We consider the generalization of the process (2.1) to the case of a multivariate (binary) time series $\boldsymbol{X} \equiv \left\{X_t^i\right\}_{t=0,1,\ldots,T}^{i=1,\ldots,N}$. With respect to the univariate DAR(p), the difference is that when X_i^t is copied from the past it can be copied either from its own past values or from the past values of one of the other $N-1$ variables. Specifically, the binary random variable X_i^t is copied from the past with probability ν_i, and in this case it selects which variable j to copy according to the probabilities λ_{ij} and the time lag according to the multinomial τ_t^{ij}. Otherwise, it is sampled from its marginal Bernoulli distribution with parameter χ_i. We can write the evolution of X_i as

$$X_t^i = V_t^i X^{J_t^i}_{t-\tau_t^{ij_t^j}} + \left(1 - V_t^i\right) Z_t^i \tag{2.2}$$

where $X_t^i \in \{0,1\} \forall i,t$, $V_t^i \sim \mathcal{B}(\nu_i)$ with $\nu_i \in [0,1] \forall i$, $\tau_t^{ij} \sim \mathcal{M}\left(\gamma_{ij,1},\ldots,\gamma_{ij,p}\right) \forall i,j$ with $\boldsymbol{\gamma}_{ij} \equiv \left\{\gamma_{ij,s}\right\}_{s=1,\ldots,p}$ such that $\sum_{s=1}^{p} \gamma_{ij,s} = 1$, $J_t^i \sim \mathcal{M}(\lambda_{i1},\ldots,\lambda_{iN})$ is a multinomial random variable with $\lambda_i \equiv \left\{\lambda_{ij}\right\}_{j=1,\ldots,N}$ such that $\sum_{j=1}^{N} \lambda_{ij} = 1 \forall i$, and $Z_t^i \sim \mathcal{B}(\chi_i)$ with $\chi_i \in [0,1], \forall i$. For notational simplicity, let us define $\boldsymbol{\gamma} \equiv \left\{\boldsymbol{\gamma}_{ij}\right\}_{i,j=1,\ldots,N}$ and $\boldsymbol{\lambda} \equiv \{\boldsymbol{\lambda_i}\}_{i=1,\ldots,N}$. We refer to (2.2) as the Vector Discrete AutoRegressive VDAR(p) process."

5.2 VAR — ART

R-Package: 'tensorTS'

Billio et al. (2023): "Many modern datasets in applied science have a complex and multidimensional structure which is naturally represented by multidimensional arrays, or tensors [...]

We exploit the contracted product, an operator that generalizes the Cayley matrix multiplication to tensors (Ji and Wei 2018; Behera, Nandi, and Sahoo 2020;Wang,Du, and Ma 2020), to introduce a new autoregressive tensor model (ART) which generalizes the existing tensor regression frameworks along two lines. First, the ART model introduces dynamics in linear tensor regression and provides the tools for analyzing shock propagation in multidimensional dynamical systems. Second, we allow for both tensor-valued outcomes and covariates, a more general framework encompassing existing tensor as well as multivariate linear models (e.g., vector autoregressions, or VARs). [...]

Let $\mathcal{Y}_t$ be a $(I_1 \times \cdots \times I_N)$-dimensional tensor of endogenous variables, $\mathcal{X}_t$ a $(J_1 \times \cdots \times J_M)$-dimensional tensor of covariates, and $S_y = \times_{j=1}^{N} \left\{1,\ldots,I_j\right\} \subset \mathbb{N}^N$ and $S_x = \times_{j=1}^{M} \left\{1,\ldots,J_j\right\} \subset \mathbb{N}^M$ sets of n-tuples of integers. We define the autoregressive tensor model of order p, with exogenous variables, ART(p), as the system of equations

$$\mathcal{Y}_{\mathbf{i,t}} = \mathcal{A}_{\mathbf{i,0}} + \sum_{j=1}^{p} \sum_{\mathbf{k} \in \mathbf{S_y}} \mathcal{A}_{\mathbf{i,k,j}} \mathcal{Y}_{\mathbf{k,t-j}} + \sum_{\mathbf{m} \in \mathbf{S_x}} \mathcal{B}_{\mathbf{i,m}} \mathcal{X}_{\mathbf{m,t}} + \mathcal{E}_{\mathbf{i,t}},$$
$$\mathcal{E}_{\mathbf{i,t}} \sim^{iid} \mathcal{N}\left(0, \sigma_{\mathbf{i}}^2\right), \quad (2)$$

$t \in \mathbb{Z}$, with given initial conditions $\mathcal{Y}_{-p+1}, \ldots, \mathcal{Y}_0 \in \mathbb{R}^{I_1 \times \cdots \times I_N}$, where $\mathbf{i} = (\mathbf{i_1}, \ldots, \mathbf{i_N}) \in \mathbf{S_y}$ and $\mathcal{Y}_{\mathbf{i,t}}$ is the $\mathbf{i}$th entry of $\mathcal{Y}_t$. The general model in Equation (2) allows for measuring the effect of all the cells of the exogenous variable $\mathcal{X}_t$ and of the lagged values of $\mathcal{Y}_t$ on each endogenous variable. When exogenous variables are not included, the model is denoted by ART(p)."

5.3 VAR — (Hys)TAR

R-Package: `'hystar'`, 'tsDyn'

de Jong et al. (2023): "conventional statistical models like the autoregressive model of order 1 (AR(1), Box et al., 2015; Chatfield & Xing, 2019). These models are often too restrictive to capture the inherently complex dynamics underpinning psychological phenomena, since they imply a symmetric distribution with a constant mean, variance and temporal associations between measurements (Haslbeck et al., 2022; Bringmann et al., 2017). However, it is evident that many psychological time series data exhibit nonstandard features, such as skewness (Haslbeck et al., 2023; Peeters et al., 2006; Von Klipstein et al., 2022), time-varying temporal associations (Koval & Kuppens, 2012; Bringmann et al., 2017) and bimodality (Haslbeck et al., 2023; Van der Maas & Molenaar, 1992). [...]

One class of models that was developed to account for bimodality and large sudden changes is formed by threshold autoregressive (TAR) models (Tong & Lim, 1980). In these time series models, the variable of interest falls within a certain regime at each point in time. The value of a control variable determines which regime this is. When the control variable crosses a certain threshold, the system shifts to another regime. [...]

But TAR models fail to account for an interesting regime-switching phenomenon: hysteresis. Hysteresis is present when the switches between two regimes are associated with two different thresholds, one for each direction of the state transition (Gilmore, 1993; Strogatz, 2018). [...]

Fortunately, a straightforward adaptation to the TAR model that incorporates the hysteresis effect was recently developed in the field of econometrics (G. Li et al., 2015), resulting in the hysteretic TAR (HysTAR) model. In the current paper, we bring the HysTAR model to the psychological realm in an accessible manner, opening up the possibility to study how hysteretic processes unfold over time, using intensive longitudinal data.[...]

TAR The TAR model is a discrete time series model using piece-wise linearization to approximate non-linear features in the data, like bimodality and sudden jumps (Tong & Lim, 1980). This approach yields a very flexible, interpretable but still accurate description of the process at hand, which is one of the reasons why TAR models are so popular and widely applied in economics (Hansen, 2011). In psychology, the TAR model has been applied to capture sudden shifts in affect regulation within individuals (Hamaker et al., 2010; de Groot, 2022) and dyads (De HaanRietdijk et al., 2016; Hamaker et al., 2009). In the two-regime TAR model Tong & Lim (1980), an outcome variable y follows an autoregressive process according to one of the regimes R_t at each time $t = 0, 1, \ldots, T$:

$$y_t = \begin{cases} \phi_{00} + \phi_{01} y_{t-1} + \cdots + \phi_{0p_0} y_{t-p_0} + \sigma_0 \varepsilon_t & \text{if } R_t = 0 \\ \phi_{10} + \phi_{11} y_{t-1} + \cdots + \phi_{1p_1} y_{t-p_1} + \sigma_1 \varepsilon_t & \text{if } R_t = 1, \end{cases} \tag{1}$$

where the residuals ε_t have a mean of 0 and a variance of 1 , and are all uncorrelated with each other. Note that all AR parameters are allowed to differ between regimes: the intercept ϕ_{j0}, the autoregressive coefficients $\phi_{j1}, \ldots, \phi_{jp_j}$ and the residual standard deviations σ_j, where j denotes the regime and p_j denotes the order of the AR process, that is, the largest lag that is used in the prediction of y_t. In this way, regimes can differ in their mean, temporal dynamics, and residual variance. The long-run mean of y in regime j is given by $\phi_{j0} / \left(1 - \sum_{i=1}^{p_j} \phi_{ji}\right)$. The dynamics are described by the autoregressive coefficients, that indicate the extent to which current and past values of y are carried over to future observations. Additionally, regime differences in σ can indicate different associations between y and unmeasured variables, or different amounts of measurement error.

How the process switches between regimes is dictated by the following equation:

$$R_t = \begin{cases} 0 & \text{if } z_{t-d} \leq r \\ 1 & \text{if } z_{t-d} > r, \end{cases} \tag{2}$$

where z denotes the control variable, d is zero or a positive integer allowing for a delayed effect of z and r is the threshold parameter. Note that the delay parameter (also called threshold lag) is associated with the problem of identifying the control variable z, since the value of d changes the control variable. Also note that the control variable can be y itself, in which case the model is called a self-exciting TAR (SETAR) model. The left panel of Figure 1 illustrates how the regime is related to the control variable in the TAR model.

The switches between regime 0 and 1 have the same threshold, regardless whether the switch is from regime 0 to 1 or from 1 to 0. This may not always be realistic: Sometimes, a switch from 0 to 1 cannot be reversed directly, because the control variable has to move to a different threshold for a switch from 1 to 0. For example, a switch to a regime characterized by high levels of alcohol consumption (a relapse) may be triggered by increasing levels of negative affect. However, the restoration of negative affect will not necessarily bring about a switch back to a low drinking regime again, but may only be possible after decreasing negative affect to an even lower value (Hufford et al., 2003). Or as another example, consider the amplifying feedback loop between perceived threat and arousal producing panic attacks in individuals with panic disorder (D. Robinaugh et al., 2019). When perceived threat is elevated by some external event, an individual might experience a panic attack, and the feedback loops of the system will retain this state. To end the panic attack, perceived threat may have to decrease to a lower level than the level that triggered the attack in the first place. This phenomenon is called hysteresis. Unfortunately, the TAR model cannot capture this, because it has only one threshold parameter. To transfer the benefits of the TAR model into a model that can also account for hysteresis, G. Li et al. (2015) proposed the hysteretic TAR model.

HysTAR In the hysteretic TAR model (HysTAR, also: buffered AR or hysteretic AR model), the dynamics of the outcome variable y in each regime are modeled in the same way as in the TAR model (see Equation 1). The difference is in the regime switching equation, which can be expressed now as

$$R_t = \begin{cases} 0 & \text{if } z_{t-d} \leq r_0 \\ R_{t-1} & \text{if } r_0 < z_{t-d} \leq r_1 \\ 1 & \text{if } z_{t-d} > r_1. \end{cases} \tag{3}$$

In this equation, the regime R_t remains unchanged when z_{t-d} enters the hysteresis zone $((r_0, r_1]$, which leads to the hysteresis effect. In the Figure 1, it is illustrated how the HysTAR model differs from the TAR model. That is, a switch from regime 0 to regime 1 happens when z_{t-d} crosses r_1, but for a switch back from regime 1 to regime 0, z_{t-d} has to cross the lower threshold r_0. Note that the HysTAR model is equivalent to the TAR model if $r_0 = r_1$; hence, the TAR model is nested under the HysTAR model."

5.4 VAR — SVAR — uSEM — SEM

R-Package: 'svars'

A. Ye et al. (2021): "There recently has been growing interest in the study of psychological and neurological processes at an individual level. One goal in such endeavors is to construct person-specific dynamic assessments using time series techniques such as Vector Autoregressive (VAR) models. However, two problems exist with current VAR specifications: (1) VAR models are restricted in that contemporaneous relations are typically modeled either as undirected relations among residuals or directed relations among observed variables, but not both [...]

Granger Causality We begin with some definitions necessary for understanding VAR models. In some cases, a variable measured at a given time can explain variability in a future observation of itself; this relation is a lagged relation often referred to as an "autoregressive (AR) effect." In a multivariate case, a lagged effect from one variable to another at a later measurement is referred to as a "cross-lagged effect." They are the source for observed temporal dependency among repeated measures across time. When a cross-lagged effect is significant after taking into account the AR effect it is said that Granger causality (Granger, 1969) takes place. Granger causality is defined as follows: variable X is said to Granger cause variable Y if Y values are better predicted with values of X (either contemporaneously or lagged) than with past values of Y alone (Geweke, 1982; Granger, 1969; Lütkepohl, 2005). In either case, the direction of the effect (i.e., $X \xrightarrow{GC} Y$ or $Y \xrightarrow{GC} X$) discloses the information flow, with the possibility of both variables Granger causing each other. [...]

VAR The VAR model (Hamilton, 1994; Shumway & Stoffer, 2017) is a traditional time series technique for modeling person-specific processes using time series data. Let $Y_t = \left[y_{1t}, y_{2t}, \ldots, y_{pt}\right]'$ be a vector of a p-variate time series, $p > 1$, at a given time point t, with $t = 1, \ldots, T$. Suppose Y_t can be represented by a stationary linear time series (i.e., having a constant means and covariance function), defined by the following VAR model with lag order-m, VAR(m):

$$\boldsymbol{Y}_t = \boldsymbol{\Phi}_1 \boldsymbol{Y}_{t-1} + \boldsymbol{\Phi}_2 \boldsymbol{Y}_{t-2} + \cdots + \boldsymbol{\Phi}_m \boldsymbol{Y}_{t-m} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim N_p(\boldsymbol{0}, \boldsymbol{\Theta}) \tag{1}$$

where the sequence of (p, p) dimensioned Φ_k matrices are the lagged coefficient matrices at order $k, k = 1, \ldots, m$, with diagonal elements the AR(k) regression coefficients and the offdiagonal cross-lagged coefficients. For simplicity, the intercept is omitted here. The coefficients are assumed to be time invariant. Further, there is no auto or cross-lagged dependency in the error term ε_t, i.e., the temporal covariance at lag $u, \Theta_u = \operatorname{cov}\left[\varepsilon_t, \varepsilon_{t-u}'\right] = 0 \forall u \in 1 \ldots \infty$. This zero-mean p-variate ε_t may be correlated contemporaneously.

The simplest form is the first-order standard VAR model, or lag-1 VAR, or simply VAR(1). VAR(1) estimates temporal relations of only consecutive measurements:

$$\boldsymbol{Y}_t = \boldsymbol{\Phi}_1 \boldsymbol{Y}_{t-1} + \boldsymbol{\varepsilon}_t, \boldsymbol{\varepsilon}_t \sim N_p(\boldsymbol{0}, \boldsymbol{\Theta}). \tag{2}$$

VAR(1) model assumes a "memoryless" (i.e., Markov) process in which all previous information useful to predict the current values of Y_t is contained in Y_{t-1}. [...]

SVAR — uSEM As pointed out by Econometricians (e.g., Shapiro & Watson, 1988; Sims, 1981), VAR is a "reduced form" of a more general class of models referred to as the Structural VAR (SVAR; Chen et al., 2011). Instead of only identifying AR and cross-lagged coefficients, SVAR incorporates directed contemporaneous associations. A fully saturated SVAR model includes free parameters for all possible direct effects among observed variables and among residuals. However, the fully saturated model is not identifiable. In practice, restriction is imposed to reduce it to a testable model, typically with only one form of contemporaneous relations allowed and the lower diagonal of the corresponding matrix estimated (Lütkepohl, 2005). The uSEM approach represents a reformulation of a type of SVAR model where the directed contemporaneous relations occur among the observed variables. The uSEM is written as (Gates et al., 2010):

$$\boldsymbol{Y}_t = \boldsymbol{A} \boldsymbol{Y}_t + \boldsymbol{\Phi}_1^* \boldsymbol{Y}_{t-1} + \boldsymbol{\zeta}_t, \quad \zeta_t \sim N(\boldsymbol{0}, \boldsymbol{\Psi}). \tag{3}$$

The $p \times p$ $\boldsymbol{A}$ matrix contains the contemporaneous estimates for relations among the component series of Y_t, with independent white noise ζ_t (or "innovations") having a diagonal covariance matrix $\boldsymbol{\Psi}$. Two coefficients, a_{ij} and a_{ji}, capture the presence and directionality of the relation between two given variables at time y_{ti} and y_{tj}, with a_{ij} representing the opposite directionality of a_{ji}. Note that estimates in the lagged relation matrix now differ: $\Phi \neq \Phi^*$."

5.4.1 uSEM — SEM

Kim et al. (2007): "[...] we analyze the fMRI multivariate time series data for each subject individually via a "unified" SEM approach, which includes longitudinal as well as contemporaneous relations. Specifically, longitudinal temporal relations are defined as relationships between brain regions involving different time points, and are represented in the form of a multivariate autoregressive model (MAR). Conversely, contemporaneous relations reflect relationships between brain regions at the same time point, and involve conventional SEMs."

5.4.2 uSEM — GIMME

R-Package: 'gimme', 'GIMMEgVAR'

Gates and Molenaar (2012): "Group Iterative Multiple Model Estimation (GIMME) estimates both the unified SEM (uSEM; Kim et al., 2007) and extended unified SEM (euSEM; Gates et al., 2011). The uSEM assuages problems encountered when solely lagged or contemporaneous relations are modeled with fMRI data (Gates et al., 2010) and is useful for block designs (when the researcher would like separate maps for each block) and resting-state analysis. The euSEM builds from uSEM by allowing for direct and modulating (i.e., bilinear) experimental manipulation effects (much like the DCM) for block and event-related designs. The logic behind GIMME follows directly from well-established principles. GIMME first identifies a group model (i.e. a connectivity map common to most individuals in the sample) by selecting paths which will improve the majority of individuals' maps in an iterative forward selection procedure. Next, individual-level paths that will optimally improve that model are opened (Gates et al., 2010)."

Beltz et al. (2016): "The nomothetic approach (i.e., the study of interindividual variation) dominates analyses of clinical data, even though its assumption of homogeneity across people and time is often violated. The idiographic approach (i.e., the study of intraindividual variation) is best suited for analyses of heterogeneous clinical data, but its person-specific methods and results have been criticized as unwieldy. Group iterative multiple model estimation (GIMME) combines the assets of the nomothetic and idiographic approaches by creating person-specific maps that contain a grouplevel structure. The maps show how intensively measured variables predict and are predicted by each other at different time scales. [...]

GIMME generates a graph for each participant that can be conceptualized as a personspecific network or connectivity map. The graphs, networks, or maps can be behavioral (e.g., explaining associations among self-reported personality facets), biological (e.g., explaining links between cortisol and substance use), or neural (e.g., explaining connections among brain regions). They take advantage of the temporal ordering of measurements to show how one variable linearly affects or is affected by another.

GIMME was developed to map functional neuroimaging data, explaining how activity (usually the blood oxygen level–dependent signal collected via functional magnetic resonance imaging, or fMRI) in one neural region of interest was explained by activity in other region of interests (Gates & Molenaar, 2012). [...] it maps both lagged and contemporaneous relations among variables. Variables are predicted by other variables measured at the same time point and by variables measured at previous time points. GIMME accomplishes this by implementing a uSEM (Wright, Beltz, et al., 2015) at both the group- and individual-levels (Gates et al., 2010; Kim et al., 2007). A uSEM is a structural vector autoregressive model (Lütkepohl, 2005) that is estimated by simultaneously fitting both lagged and contemporaneous weighted relations to the data. [...]

In matrix notation, GIMME with constant mean (fixed at zero) is specified as

$$\boldsymbol{\eta}_i(t) = (\mathbf{A}_i + \mathbf{A}^g)\,\boldsymbol{\eta}_i(t) + \left(\boldsymbol{\Phi}_{1,i} + \boldsymbol{\Phi}_1^g\right)\boldsymbol{\eta}_i(t-1) + \boldsymbol{\zeta}_i(t),$$

where $\eta_i(t)$ is the p-variate series to be explained at time points $t = 1, 2, \ldots T$, with T the total number of time points; A is the (p, p)-dimensional matrix of contemporaneous relations among variables; $\boldsymbol{\Phi}_1$ is the (p, p)-dimensional matrix of lagged relations among variables; $\boldsymbol{\zeta}_i$ is the p-variate error matrix, lacking sequential dependences and having zero mean and a diagonal covariance matrix; superscript g indicates group-level relations; subscript i indicates individual-level relations. GIMME is fit using a special sequential procedure that is described in the Supplemental Materials (see also Gates & Molenaar, 2012). This procedure involves Lagrange multiplier testing combined with Wald trimming and provides results consistent with those obtained through likelihood ratio testing (Chou & Bentler, 1990).

There are two assumptions of GIMME that must be satisfied for each data set. One is that the solution produced for each participant (that contains group- and individual-level relations) is unique; there are no other networks that similarly fit the data. This may not always be the case because the A matrix of contemporaneous relations is essentially a crosssectional path model (e.g., SEM) embedded within GIMME, and multiple solutions are a characteristic of cross-sectional path models (MacCallum, Wegener, Uchino, & Fabrigar, 1993). To solve this potential problem, GIMME for multiple solutions (GIMME-MS; Beltz & Molenaar, in press) can be used to generate all possible solutions so that an optimal one can be selected.

The other assumption is that there are no temporal dependencies in the error term, that is, that the residuals are white noise. GIMME is currently implemented with a lag of 1, meaning that each variable has the opportunity to be explained by the other variables (and itself) at the prior measurement (and concurrently), but each variable may also be related to other variables (and itself) from two measurements ago, three measurements ago, and so on. If this is the case, then a lag of 1 will be insufficient for the data (and residuals will not be white noise), and a lag of 2, 3, and so on must be included in the model (in $\boldsymbol{\Phi}_2$, $\boldsymbol{\Phi}_3$, and so on matrices). A posteriori model validation can be used to test the white noise assumption and to subsequently modify individual-level maps when lags greater than 1 are necessary (Beltz & Molenaar, 2015)."

5.5 VAR — ALS VAR

Bulteel et al. (2016): "In this paper, we propose a clusterwise extension of the VAR(1) model. Starting from the key idea behind clusterwise linear regression (Späth, 1979, 1982; DeSarbo et al., 1989; Brusco et al., 2008), we will cluster persons according to their VAR(1) regression weights [i.e., each person $i(i = 1, \ldots, I)$ is assigned to one particular cluster $k(k = 1, \ldots, K)$], and simultaneously, fit a shared VAR(1) model to all persons within a cluster. In terms of the taxonomy of Liao (2005), who distinguishes clustering time series based on (a) the raw data, (b) features extracted from the data, and (c) model parameters, our clustering approach is based on model parameters to ensure that we identify groups of persons with similar time dynamics.

The model formula is the following:

$$\boldsymbol{y}_{it} = \sum_{k=1}^{K} \mathrm{p}_{ik} \left(\boldsymbol{c}_k + \boldsymbol{\Phi}_k \boldsymbol{y}_{i,t-1} + \boldsymbol{u}_{kt} \right). \tag{7}$$

The $M \times 1$ vectors $\boldsymbol{y}_{it}$ and $\boldsymbol{y}_{i,t-1}$ now contain the scores of person i on the M variables at time points t and $t-1$ $(t = 1, \ldots, T_i)$. Importantly, the number of time points does not have to be equal across persons, as is shown by the subscripted i in T_i. p_{ik} denotes an element of the $I \times K$ partition matrix. When p_{ik} equals 1 , person i belongs to cluster k; p_{ik} equals 0 if this is not the case. The VAR(1) regression coefficients $\boldsymbol{c}_k$ and $\boldsymbol{\Phi}_k$ differ across the clusters of persons as indicated by the subscript k. The same assumptions as described above need to be met for the clusterwise extension, but now also across persons. More specifically, this means that intervals between the measurements should be equal for all persons. In addition, the mean of each variable should be constant across persons. Person-mean centering can be applied if this assumption does not hold. It is also assumed that the innovation covariation matrix is equal for all participants within a cluster. Note that a similar assumption is made in case of a multilevel extension of the VAR model (Bringmann et al., 2013), which focuses on quantitative rather than qualitative differences in model parameters.

Regarding the interpretation of the clusterwise VAR(1) model, the above mentioned approaches—forecasting and computing R^2—can be used to compare the clusters. It is especially interesting to see how the predictions given a specific state (i.e., a particular set of variable scores) differ across the clusters. Inspecting possible differences in R^2 values may also be useful, as they indicate whether the variable scores are more predictable in one cluster than in another, and for which variables this holds."

5.6 VAR — multi-VAR

R-Package: `'multivar'`

Fisher et al. (2022): " We focus our attention on the multivariate time series, $\{\boldsymbol{X}_t\}_{t\in\mathbb{Z}} = \left\{ \left(X_{j,t} \right)_{j=1,\ldots,d} \right\}_{t\in\mathbb{Z}}$. $\boldsymbol{X}_t$ is considered to follow a vector autoregressive model of order p, VAR(p), if

$$\boldsymbol{X}_t = \boldsymbol{\Phi}_1 \boldsymbol{X}_{t-1} + \cdots + \boldsymbol{\Phi}_p \boldsymbol{X}_{t-p} + \boldsymbol{\varepsilon}_t, \quad t \in \mathbb{Z}, \tag{1}$$

for some $d \times d$ matrices $\boldsymbol{\Phi}_1, \ldots, \boldsymbol{\Phi}_p$ and a white noise series $\{\boldsymbol{\varepsilon}_t\}_{t\in\mathbb{Z}} \sim \mathrm{WN}(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon})$ characterized by $\mathbb{E}(\boldsymbol{\varepsilon}_t) = 0$ and $\mathbb{E}(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_s') = 0$ for $s \neq t$. For simplicity, we assume $\boldsymbol{X}_t$ is of zero mean. Generally, a unique causal stationary solution to (1) can be ensured by satisfying the stability condition given by $\det(\boldsymbol{\Phi}(z)) \neq 0$, for $|z| \leq 1$, $z \in \mathbb{C}$, where $\boldsymbol{\Phi}(z) = \boldsymbol{I}_d - \boldsymbol{\Phi}_1 z - \cdots - \boldsymbol{\Phi}_p z^p$."

It is common to estimate (1) using ordinary least squares (OLS) regression,

$$\left(\widehat{\boldsymbol{\Phi}}_1, \ldots, \widehat{\boldsymbol{\Phi}}_p\right) = \underset{\boldsymbol{\Phi}_1, \ldots, \boldsymbol{\Phi}_p}{\operatorname{argmin}} \sum_{t=p+1}^{T} \left\| \boldsymbol{X}_t - \boldsymbol{\Phi}_1 \boldsymbol{X}_{t-1} - \cdots - \boldsymbol{\Phi}_p \boldsymbol{X}_{t-p} \right\|_2^2, \tag{2}$$

A drawback of the unrestricted VAR model is the large number of parameters that must be estimated. [...] when $(T-p)d < pd^2 + d(d-1)/2$, estimation via OLS is not possible.

As a consequence of the dimensionality issues surrounding unrestricted VAR estimation, much attention has been paid to methods for reducing the VAR parameter space. [...] Several data-driven approaches have been presented in the literature to overcome this issue of high-dimensionality (Basu & Michailidis, 2015a; Han & Liu, 2013). In our current approach, we assume sparsity of $\boldsymbol{\Phi}$ and use penalized estimation to recover the model parameters.

VAR LASSO Estimation To set up the least absolute shrinkage and selection operator (LASSO; Tibshirani, 1996), the VAR model and associated data are expressed in a regression form

$$\underbrace{\begin{bmatrix} \boldsymbol{X}'_{p+1} \\ \boldsymbol{X}'_{p+2} \\ \vdots \\ \boldsymbol{X}'_T \end{bmatrix}}_{\mathcal{Y}} = \underbrace{\begin{bmatrix} \boldsymbol{X}'_p & \cdots & \boldsymbol{X}'_1 \\ \boldsymbol{X}'_{p+1} & \cdots & \boldsymbol{X}'_2 \\ \vdots & \ddots & \vdots \\ \boldsymbol{X}'_{T-1} & \cdots & \boldsymbol{X}'_{T-p} \end{bmatrix}}_{\mathcal{X}} \underbrace{\begin{bmatrix} \boldsymbol{\Phi}'_1 \\ \boldsymbol{\Phi}'_2 \\ \vdots \\ \boldsymbol{\Phi}'_p \end{bmatrix}}_{\mathcal{B}^*} + \underbrace{\begin{bmatrix} \boldsymbol{\varepsilon}'_{p+1} \\ \boldsymbol{\varepsilon}'_{p+2} \\ \vdots \\ \boldsymbol{\varepsilon}'_T \end{bmatrix}}_{\mathcal{E}} \tag{3}$$

or, equivalently,

$$\operatorname{vec}(\mathcal{Y}) = (\mathrm{I}_d \otimes \mathcal{X}) \operatorname{vec}(\mathcal{B}^*) + \operatorname{vec}(\mathcal{E}) \tag{4}$$

$$\underbrace{\boldsymbol{Y}}_{Nd\times 1} = \underbrace{\boldsymbol{Z}}_{Nd\times q} \underbrace{\boldsymbol{B}^*}_{q\times 1} + \underbrace{\boldsymbol{E}}_{Nd\times 1} \tag{5}$$

where the star $*$ indicates the true parameter, $N = T - p$ and $q = pd^2$. Here, we assume that $\boldsymbol{B}^*$ is s-sparse (i.e., $\Sigma_{i=1}^{p} \|\operatorname{vec}(\boldsymbol{\Phi}_i)\|_0 = \|\boldsymbol{B}\|_0 = \sum_{i=1}^{q} 1_{\{B_i \neq 0\}} = s$ where $\|\cdot\|_0$ is the ℓ_0-norm). With this structure in place, we can write the LASSO estimator as

$$\hat{\boldsymbol{B}} = \underset{\boldsymbol{B} \in \mathbb{R}^q}{\operatorname{argmin}} \frac{1}{N} \|\boldsymbol{Y} - \boldsymbol{Z}\boldsymbol{B}\|_2^2 + \lambda \|\boldsymbol{B}\|_1, \tag{6}$$

where $\|\boldsymbol{B}\|_1 = \Sigma_{i=1}^{q} |B_i|$ for $\boldsymbol{B} = \left(B_1, \ldots, B_q\right)'$ and $\lambda > 0$ is the regularization penalty parameter. In (6), the scaling constant $\frac{1}{N}$ (corresponding to λ) sometimes takes the values $\frac{1}{2N}$, 1 and 2 depending on the convention. Here, $N = T - p$ refers to the time series length of a given individual in the sample. Changing the scaling context corresponds to a reparameterization of λ and does not impact the estimation of (6). Large values of λ typically correspond to sparser solutions.

multi-VAR Up to this point, we have presented the VAR model and optimization problem in terms of a single multivariate time series. This was useful for describing the estimators; however, the majority of ILD and many psychophysiological applications involve observing the same variables across multiple subjects. With multivariate repeated measurements collected from multiple subjects, we are now interested in estimating the sparse parameter vectors $\hat{\boldsymbol{B}}_1, \ldots, \hat{\boldsymbol{B}}_K$ for K individuals. Rarely, if ever are the relationships among items strictly the same across any two individuals in the sample. However, it is certainly possible and maybe even expected that certain qualitative aspects of a dynamic process are similar across individuals. For this reason, strategies involving the estimation of K separate LASSO problems are generally suboptimal. To overcome this limitation, we propose the multi-VAR modeling framework for multivariate time series data collected from multiple subjects.

The approach described herein relies on the following decomposition of $\boldsymbol{B}_k^*$:

$$\boldsymbol{B}_k^* = \boldsymbol{\mu}^* + \boldsymbol{\Delta}_k^*, \quad k = 1, \ldots, K, \tag{7}$$

where $\boldsymbol{\mu}^* \in \mathbb{R}^q$ corresponds to the common effects across K individuals and $\boldsymbol{\Delta}_k^* \in \mathbb{R}^q$ corresponds to the effects unique to individual k. Now, considering the regularization parameters λ_1 and $\lambda_{2,k}$, $k = 1, \ldots, K$, which govern the cross-sectional heterogeneity in our solution, we can write the revised optimization problem as

$$\begin{aligned}\left(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Delta}}_1, \ldots, \hat{\boldsymbol{\Delta}}_K\right) = \underset{\boldsymbol{\mu}, \boldsymbol{\Delta}_1, \ldots, \boldsymbol{\Delta}_K}{\operatorname{argmin}} \frac{1}{N} \sum_{k=1}^{K} \left\| \boldsymbol{Y}^{(k)} - \boldsymbol{Z}^{(k)} \left(\boldsymbol{\mu} + \boldsymbol{\Delta}_k\right) \right\|_2^2 \\ + \lambda_1 \|\boldsymbol{\mu}\|_1 + \sum_{k=1}^{K} \lambda_{2,k} \|\boldsymbol{\Delta}_{\boldsymbol{k}}\|_1 .\end{aligned} \tag{8}$$

As mentioned previously, we prefer a sensible approach to handling multivariate time series data arising from multiple individuals, specifically in the case where the cross-sectional heterogeneity of the individual processes is unknown. If the individuals share very little in common, in terms of their time series, this approach should return essentially independent solutions. That is the results should be similar to what would be obtained from estimating K separate VAR models. In (6), larger values of the penalty parameter λ will increasingly drive the corresponding coefficient matrix $\boldsymbol{B}$ to zero. Similarly, in (8), large enough values of λ_1 will shrink the common effect matrix, $\hat{\boldsymbol{\mu}}$ toward zero, and we will be left with the individual-specific effects $\hat{\boldsymbol{B}}_k = \hat{\boldsymbol{\Delta}}_k$. This would essentially produce results similar to those obtained from estimating K independent VAR models.

Likewise, if the individual-level processes are essentially homogenous, a sensible approach would return results similar to estimating a single pooled model for all individuals in the sample. In this case, we would expect that large enough values of $\lambda_{2,k}$ would drive the individual-specific transition matrices, $\hat{\boldsymbol{\Delta}}_k$, toward zero, leaving only the common effect transition matrix to explain the individual-level results, $\hat{\boldsymbol{B}}_k = \hat{\boldsymbol{\mu}}$. Finally, if an individual's process has both common and unique components, a sensible approach would attempt to balance these contributions. In this case, the penalty parameters, λ_1 and $\lambda_{2,k}$, are selected to optimally govern the contribution of both the common ($\hat{\boldsymbol{\mu}}$) and unique $\left(\hat{\boldsymbol{\Delta}}_k\right)$ effects to each individual's dynamics $\left(\hat{\boldsymbol{B}}_k\right)$.

5.7 VAR — gVAR — GGM

R-Package: 'graphicalVAR'

Epskamp, Waldorp, et al. (2018): "The simplest way to deal with temporal ordering of cases is to incorporate the effect between consecutive measurements (Chatfield, 2016; Hamilton, 1994; Shumway & Stoffer, 2010). This is called a Lag-1 model because it includes both measurements at the current time point t as well as measurements from the previous time point $t-1$. We will focus our discussion on Lag-1 models, noting that everything below also generalizes to more complicated models (e.g., Lag-2 models). In intraindividual analysis, VAR (Brandt & Williams, 2007; Rosmalen, Wenting, Roest, de Jonge, & Bos, 2012) has gained substantive footing in visualizing temporal information through networks. The lag-1 VAR model can be denoted as a regression model on the previous measurement occasion:

$$\begin{aligned} \boldsymbol{y}_t &= \boldsymbol{B}\boldsymbol{y}_{t-1} + \boldsymbol{\varepsilon}_t \\ \boldsymbol{\varepsilon}_T &\sim N(\mathbf{0}, \boldsymbol{\Theta}). \end{aligned} \tag{7}$$

The model matrix $\boldsymbol{B}$ encodes temporal predictive effects from variables on variables in the next measurement occasion, and can be used to obtain a directed network, which we term the *temporal network*. The variance-covariance matrix $\boldsymbol{\Theta}$ can be inverted $\left(\boldsymbol{K}^{(\boldsymbol{\Theta})} = \boldsymbol{\Theta}^{-1}\right)$ to obtain a GGM modeling effect within the same measurement occasion, after controlling for temporal effects. These can be displayed again as a network, which we term the *contemporaneous network*.

Temporal networks, encoded by $\boldsymbol{B}$, have grown popular in recent psychological literature (e.g., Bringmann et al., 2013, 2015; Bos et al., 2017; Klippel et al., 2017; Snippe et al., 2017; Wigman et al., 2015). A temporal network is formed by combining a lagged variable y_{t-1} and current variable y_t into a single node, connected with directed edges which are weighted according to the regression parameters contained in $\boldsymbol{B}$. Thus, an edge in the temporal network indicates that a node predicts another node (or itself in the common case of self-loops) at the next measurement occasion, after controlling for all other variables at the previous measurement occasion. Temporal prediction is central to the concept of *Granger causality* in the economic literature (Eichler, 2007; Granger, 1969), and it satisfies at least the temporal requirement for causation (i.e., the cause must precede the effect). Temporal networks may thus highlight potential causal pathways. While temporal networks are typically cyclic, they can also be interpreted as summarizing a DAG unfolding over time.

gVAR (N=1) In addition to temporal effects, VAR analyses also include contemporaneous effects, which can be modeled as a GGM. We will term this modeling framework (a VAR model with contemporaneous effects explicitly modeled and portrayed as a GGM) graphical VAR (GVAR; Wild et al., 2010). A useful equivalent way to denote a GVAR model is by using a conditional Gaussian distribution:

$$\boldsymbol{y}_T \mid \boldsymbol{y}_{T-1} = \boldsymbol{y}_{t-1} \sim N\left(\boldsymbol{B}\boldsymbol{y}_{t-1}, \boldsymbol{\Theta}\right).$$

Which is equivalent to Equation (7). Now, it becomes evident that if consecutive cases can be assumed to be independent, and thus $\boldsymbol{B} = \mathbf{0}$, the GVAR model is exactly the same as the GGM model described above for independent cases. Thus, the GVAR model can be seen as a generalization of the GGM model to temporally ordered data. GVAR only differs from regular VAR in that the contemporaneous structure is modeled and represented as a GGM, instead of being saturated. This leads to a strikingly different interpretation of the VAR model; the VAR model can be seen as an inclusion of temporal effects on a GGM. [...]

gVAR (N=Large) A type of data that is increasingly common due to the emergence of ESM studies is time series of multiple subjects (e.g., Bringmann et al., 2013, 2015; Mõttus et al., 2017; Schmiedek, Lövdén, & Lindenberger, 2010; Wigman et al., 2015). Such data sets pose a promising gateway to study both intraindividual dynamics and between-subjects overlap as well as their differences. Here, we assume that the number of time points might differ per person and that measurement occasions are nested in people. We can model the temporal data of every person with an individual GVAR model:

$$\begin{aligned}
\boldsymbol{y}_{[t,p]} &= \boldsymbol{\mu}_p + \boldsymbol{B}_p\left(\boldsymbol{y}_{[t-1,p]} - \boldsymbol{\mu}_p\right) + \boldsymbol{\varepsilon}_{[t,p]} \\
\boldsymbol{\varepsilon}_{[T,p]} &\sim N\left(\boldsymbol{0}, \boldsymbol{\Theta}_p\right) \\
\boldsymbol{\Theta}_p^{-1} &= \boldsymbol{K}_p^{(\boldsymbol{\Theta})}
\end{aligned}$$

in which $\boldsymbol{\mu}_p$ indicates the stationary mean vector of subject p (which enters the model because we can no longer assume within-subject means are zero without loss of generality), $\boldsymbol{B}_p$ encodes the person-specific temporal network, and $\boldsymbol{K}_p^{(\boldsymbol{\Theta})}$ encodes the person-specific contemporaneous GGM.

To gain insight in the general network structure over subjects, we can investigate the individual networks at a second level. Doing so is termed multilevel modeling, explained in more detail in Section 2.1 of the supplementary materials. Let $\boldsymbol{B}_*$ and $\boldsymbol{K}_*^{(\boldsymbol{\Theta})}$ encode the expected temporal and contemporaneous network when selecting a person at random. Furthermore, we can assume without loss of generality that data are grand-mean centered. We then obtain:

$$\begin{aligned}
\mathcal{E}\left(\boldsymbol{\mu}_p\right) &= \boldsymbol{0} \\
\mathcal{E}\left(\boldsymbol{B}_p\right) &= \boldsymbol{B}_* \\
\mathcal{E}\left(\boldsymbol{K}_p^{(0}\right) &= \boldsymbol{K}_*^{(\boldsymbol{\Theta})}
\end{aligned}$$

Here, $\boldsymbol{B}_*$ and $\boldsymbol{K}_*^{(\boldsymbol{\Theta})}$ now encode the average parameters in the population: the *fixed effects*. Deviations from these fixed effects, such as $\boldsymbol{B}_p - \boldsymbol{B}_*$, are often called *random effects*. Besides the individual network structures, researchers often aim to estimate the structure and parameters of these fixed effects because these tell us something about the average intraindividual effect. Researchers also aim to estimate the variance-covariance structure of the random effects because it tells us something about individual differences (Bringmann et al., 2013).

The random effects can be modeled by assuming a second level normal distribution on all the parameters. This can be complicated, however, especially when modeling partial correlation coefficients in such a way (e.g., any hierarchical model for $\boldsymbol{K}^{(\boldsymbol{\Theta})}$ needs to take into account that this matrix must remain positive definite). The interpretation of, for example, correlations between different temporal or contemporaneous edges is also difficult. Therefore, we only focus here on a subset of the parameters where we can easily interpret the second-level model: the mean structure. As a result, if a multivariate normal is assumed for all parameters, then it is also assumed for the marginal distribution of the means—regardless of other parameters:

$$\boldsymbol{\mu}_P \sim N(\mathbf{0}, \boldsymbol{\Omega}).$$

Again, we can invert the variance-covariance matrix to obtain a GGM,

$$K^{(\Omega)} = \Omega^{-1},$$

which we will term the *between-subjects network*, a network between stationary means of different subjects. As such, estimating the GVAR model on $n > 1$ time-series analysis allows for the separation of variance into three distinct network structures: temporal networks, contemporaneous networks, and the between-subjects network."

5.8 VAR — mlVAR – DSEM

R-Package: `'mlVAR'`

Y. Li et al. (2022): "The multilevel vector autoregressive (mlVAR) model, also regarded as a special case of the dynamic structural equation model (DSEM) (see, e.g., Asparouhov et al., 2018; Chow et al., 2010b; Hamaker et al., 2018; Schuurman et al., 2016), is one of such methods known for its merits to simultaneously model reciprocal linkages between dynamic processes and individual differences. Its simpler variant, the vector autoregressive (VAR) model, is a key constituent of dynamic network models that have seen increasing uptake in areas such as clinical psychopathology (Bringmann et al., 2013), neuroimaging (Gates & Molenaar, 2012), emotion regulation (Bringmann et al., 2018; Chow et al., 2010a), and personality dynamics (Beck & Jackson, 2020; Wright et al., 2019). A quick trend analysis of the combined keywords, "autoregression" and "network" using the Web of Science portal revealed a 10-fold increase in the annual number of relevant publications between 2010 and 2020, in close parallel to the growing prevalence of ILD in recent years.

[...] However, fitting mlVAR in a single step is a computationally challenging problem (Epskamp et al., 2018) [...]

Before introducing the mlVAR model formally, we begin first by introducing one of its simplest and most commonly known univariate special case, the autoregressive (AR) model a model describing how a variable in a system predicts itself at the next time point. The AR(p) model is expressed as:

$$y_{i,t} = \phi_0 + \sum_{l=1}^{p} \phi_l \left(y_{i,t-l} - \phi_0\right) + u_{i,t}, \quad u_{i,t} \sim N\left(0, \sigma^2\right), \tag{1}$$

where i represents the i th person $(i = 1, 2, \ldots, N), t$ represents the t th time point $(t = 1, 2, \ldots, T_i)$, and p is the order of the AR process. Conceptually, an AR(p) model can be viewed as a linear regression model in which the explanatory variables consist of $y_{i,t-l}(l = 1, \ldots, p)$, namely, the lags of the dependent variable, $y_{i,t}$. The intercept and regression coefficients are denoted by ϕ_0 and $\phi_l(l = 1, \ldots, p)$ respectively. The innovation term (also called process noise), $u_{i,t}$, reflects unmeasured sources that affect the dynamics of $y_{i,t}$, following a normal distribution with zero mean and variance σ^2. An AR(p) process is stationary (Lütkepohl, 2005) when the modulus of all the roots of $1 - \sum_{l=1}^{p} \phi_l B^l = 0$ are greater than 1 , where B is the lag operator such that $By_t = y_{t-1}$. Model parameters from any AR-type models can be properly estimated if and only if the process is stationary.

In many applications of the AR model, data have been demeaned and often detrended prior to model fitting, thus the intercept term ϕ_0 can be omitted from the above model. In the present study, we choose to include ϕ_0 because its value reflects an individual's baseline around which the process of interest fluctuates and thus can be of explicit interest. When $p = 1$, the AR(1) model has only one regression coefficient ϕ_1, which has been referred to as the "inertia" of a dynamic process given that a high positive value of ϕ_1 reflects a construct's resistance to change (Brose et al., 2015; Koval et al., 2012; Kuppens et al., 2010). For instance, when modeling affect dynamics using the AR(1) model, a high positive ϕ_1 indicates that the level of individual's affect is less likely to change or it may take long to recover after a change caused by external events, thus indicating weak affect regulation.

If the relationship between dynamics of multiple outcome variables are of interest, then the VAR model of lag order p (VAR(p) model) can be used to account for the reciprocal linkages among dynamic processes as:

$$\begin{aligned} \boldsymbol{y}_{i,t} = \boldsymbol{\phi}_0 + \boldsymbol{\phi}_1 (\boldsymbol{y}_{i,t-1} - \boldsymbol{\phi}_0) + \ldots + \boldsymbol{\phi}_p \left(\boldsymbol{y}_{i,t-p} - \boldsymbol{\phi}_0\right) + \boldsymbol{u}_{i,t}, \\ \boldsymbol{u}_{i,t} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}), \end{aligned} \tag{2}$$

where $\boldsymbol{y}_{i,t}$ is a $D \times 1$ (D is the number of outcome variables) matrix of outcome variables; $\boldsymbol{\phi}_0$ is a $D \times 1$ matrix of intercepts; $\boldsymbol{\phi}_l(l = 1, \ldots, p)$ is a $D \times D$ coefficients matrix; $\boldsymbol{u}_{i,t}$ is a $D \times 1$ matrix of innovation terms following a multivariate normal distribution with a fixed covariance matrix $\boldsymbol{\Sigma}$. A VAR(p) process is stationary when all the roots of the determinant of matrix $\boldsymbol{I} - \boldsymbol{\phi}_1 B - \ldots - \boldsymbol{\phi}_p B^p$ have moduli greater than 1 , where B has been defined above.

The specifications above assume that model parameters are the same for each individual, which may not be the case in reality - individuals may have different baseline equilibria as well as patterns of dynamics. One possible way of representing such individual differences is to extend the VAR(p) model in Equation (2) into an mlVAR(p) model in which all level-1 or *within-level* parameters are person-specific, labeled with a subscript i, and come from joint level-2 or *between-level* distributions. The within-level model is specified as:

$$\begin{aligned} &\boldsymbol{y}_{i,t} = \boldsymbol{\phi}_{0,i} + \boldsymbol{\phi}_{1,i} (\boldsymbol{y}_{i,t-1} - \boldsymbol{\phi}_{0,i}) + \ldots + \boldsymbol{\phi}_{p,i} \left(\boldsymbol{y}_{i,t-p} - \boldsymbol{\phi}_{0,i}\right) + \boldsymbol{u}_{i,t}, \\ &\boldsymbol{u}_{i,t} \sim N(\boldsymbol{0}, \boldsymbol{\Sigma}_i) \end{aligned} \tag{3}$$

where $\boldsymbol{\phi}_{0,i}$ is a $D \times 1$ matrix of *person-specific* intercepts; $\boldsymbol{\phi}_{l,i}(l = 1, \ldots, p)$ is a $D \times D$ *person-specific* coefficients matrix; and $\boldsymbol{u}_{i,t}$ is a $D \times 1$ matrix of random innovations following a multivariate normal distribution with a *person-specific* covariance matrix $\boldsymbol{\Sigma}_i$. The between-level model is expressed as:

$$g(\boldsymbol{\theta}_i) = \boldsymbol{B}\boldsymbol{x}_i + \boldsymbol{\zeta}_i, \quad \boldsymbol{\zeta}_i \sim N(\mathbf{0}, \boldsymbol{\Psi}), \tag{4}$$

where $\boldsymbol{\theta}_i$ is an $n_p \times 1$ matrix that consists of all person-specific parameters that appear in $\phi_{0,i} - \phi_{p,i}$ and Σ_i in Equation (3); $g(\cdot)$ is a function used for parameter transformation (see, e.g., the section below for parameter transformations adopted in the present article). Variable $\boldsymbol{x}_i$ is an $n_x \times 1$ matrix of exogenous predictors hypothesized to explain some of the inter-individual differences in $\boldsymbol{\theta}_i$, with the first entry being unity to define an intercept term, and $\boldsymbol{B}$ is the corresponding $n_p \times n_x$ matrix of "between-level" regression coefficients (or fixed effects). Finally, $\boldsymbol{\zeta}$ i is an $n_p \times 1$ matrix of random effects which represent person i's deviations in the values of $\boldsymbol{\theta}_i$ not accounted for by the exogenous variables; these are distributed with zero means and the random effect covariance matrix Ψ."

5.9 VAR — fitlandr

R-Package: 'fitlandr', 'simlandr'

Cui et al. (2023): "The availability of smart devices has made it possible to collect intensive longitudinal data (ILD) from individuals, providing a unique opportunity to study the complex dynamics of psychological systems. Existing time-series methods often have limitations, such as assuming linear interactions or having restricted forms, leading to difficulties in capturing the complex nature of these systems. To address this issue, we introduce fitlandr, a method with implementation as an R package that integrates nonparametric estimation of the driftdiffusion function and stability landscape. The drift-diffusion function is estimated using the multivariate kernel estimator (MVKE; Bandi & Moloche, 2018), and the stability landscape is estimated through Monte-Carlo estimation of the steady-state distribution (Cui et al., 2021; Cui, Lichtwarck-Aschoff, et al., 2023). Using a simulated emotional system, we demonstrate that fitlandr can effectively recover bistable dynamics from data, even in the presence of moderate noise, and that it primarily relies on dynamic information from the system instead of distributional information. [...]

The vector autoregression (VAR) model is the most widely used approach for multivariate time series analysis and serves as a foundation for longitudinal network analysis (Bringmann et al., 2013). In a VAR model, the state of the system from the previous time point, represented as a vector, is used to predict the state of the system at the next time point through a regression model. Despite its simplicity, VAR provides a powerful way to gain insights from ILD. Yet, VAR models assume constant linear interactions of the variables discretely, thus fail to capture complex features such as nonstationarity and multistability (Haslbeck et al., 2022; Haslbeck & Ryan, 2022; Olthof et al., 2020).

To overcome this issue, many variants of the original VAR model have been developed, seeking to capture more complex dynamics in the system. For example, the time-varying VAR (TV-VAR) model allows for smooth change in the parameters of the VAR model, accounting for nonstationary behavior in the data (Bringmann et al.,2017; Haslbeck et al., 2021). Threshold-VAR (De Haan-Rietdijk et al., 2016; Tong

& Lim, 1980) and Markov-Switching-VAR (Hamilton, 1989; Haslbeck & Ryan, 2022) are models that apply different regression equations based on a threshold variable or a latent variable that indicates which phase the system is in. Although these variants improve upon the VAR model, they do not explicitly model nonlinear dynamics. More concretely, even though these models allow the linear interaction coefficient to change nonlinearly over time, they still do not permit the function form to be nonlinear (i.e., the influence of any variable on itself or other variables is always linear). As a result, the system at each time point is a linear system, which means that at any given moment, when a higher x corresponds to a higher y, it follows that an even greater x will consistently result in a proportionally higher y, without any exceptions. This limits the ability to capture nonlinear interactions among variables. Moreover, they represent nonstationary and multistable behaviors with one or more variables (such as the time-varying coefficients in the TV-VAR model, one or more assessed variables in the Threshold-VAR model, and latent states in the Markov-Switching-VAR model). These variables can undergo discrete or smooth changes, making the model undergo transformations, but they are not explanans per se. As a result, the model does not inherently account for the reasons behind their changes. Last, if the transition between different phases of the system is the primary focus rather than the specific dynamics, a hidden Markov model (HMM) can also be used, which assumes that the data come from multiple probability distributions (Haslbeck & Ryan, 2022; Visser, 2011). However, in HMM, only the specific transition probabilities between hidden states are defined, whereas within each hidden state, the time points are considered independent from each other, lacking any temporal dependencies. Consequently, HMM primarily focuses on capturing the transition possibilities between hidden states rather than providing a detailed description of the specific dynamics.

Another branch of methods for analyzing ILD is based on the drift-diffusion process, in which the moving trajectory of a system is governed by two types of forces: the drift part, which represents the deterministic forces, and the diffusion part, which represents the stochastic forces. The drift-diffusion process is modeled using stochastic differential equations instead of linear regressions, thus the models based on the drift-diffusion process are inherently continuous. [...]

Step 1: Estimating System Dynamics The first step of the method is to estimate the dynamics of the system from raw data. In fitlandr, we employ the general drift-diffusion form of stochastic differential equations to describe the dynamics of the system, written as follows:

$$\mathrm{d}\boldsymbol{X} = \boldsymbol{\mu}(\boldsymbol{X})\mathrm{d}t + \boldsymbol{\sigma}(\boldsymbol{X})\mathrm{d}\boldsymbol{W}, \tag{1}$$

in which the first term in the right-hand side, $\boldsymbol{\mu}(\boldsymbol{X})$, represent the deterministic force, and the second term represents the noise of the system (the diffusion matrix, $\boldsymbol{a} = \boldsymbol{\sigma}\boldsymbol{\sigma}^T$, is more commonly used). In fitlandr, we use a flexible kernel algorithm, namely the MVKE (Bandi & Moloche, 2018) to estimate the drift-diffusion equation. Assuming $\boldsymbol{x}_i (i = 1, 2, 3, \ldots, n)$ are n observations of $\boldsymbol{X}_t$ in $t = 1, 2, 3, \ldots, n$, the kernel estimators for $\boldsymbol{\mu}(\boldsymbol{x})$ and $\boldsymbol{a}(\boldsymbol{x})$ are given by (Bandi & Moloche, 2018):

$$\hat{\boldsymbol{\mu}}(\boldsymbol{x}) = \frac{\sum_{i=1}^{n-1} K(\boldsymbol{x}_i - \boldsymbol{x})(\boldsymbol{x}_{i+1} - \boldsymbol{x}_i)}{\sum_{i=1}^{n} K(\boldsymbol{x}_i - \boldsymbol{x})}, \tag{2}$$

$$\hat{\boldsymbol{a}}(x) = \frac{\sum_{i=1}^{n-1} K(\boldsymbol{x}_i - \boldsymbol{x})(\boldsymbol{x}_{i+1} - \boldsymbol{x}_i)(\boldsymbol{x}_{i+1} - \boldsymbol{x}_i)^T}{\sum_{i=1}^{n} K(\boldsymbol{x}_i - \boldsymbol{x})}, \tag{3}$$

in which $K(\boldsymbol{x})$ is a product kernel function, as follows in our implementation:

$$K(\boldsymbol{x}) = \frac{1}{h^D} \prod_{j=1}^{D} e^{\frac{-x_j}{h}}, \tag{4}$$

where x_j is the jth element of $\boldsymbol{x}$, and h is a user-specified parameter controlling the width of the kernel estimator. Intuitively, MVKE is a weighted average of the dynamics from the data. The weight of a data point depends on its distance with the evaluation point, $K(\boldsymbol{x}_i - \boldsymbol{x})$. The closer the evaluation point is to a data point, the more similar the dynamics is.

Due to the use of a kernel algorithm, the output cannot be represented as parameter estimates. Instead, the force of the system in different directions is shown through a vector field. A vector field plot displays multiple arrows in a grid, representing the direction and tendency of how the system evolves from a given point. However, it is not as straightforward to see how variables influence one another. In a VAR model, a positive coefficient of variable X_1 on variable X_2, $c_{1,2}$ indicates that a higher value of X_1 at time t predicts a higher value of X_2 at time $t+1$. In a vector field, however, the influence of X_1 to X_2 may not be monotonous. A higher value of X_1 may predict a higher value of X_2 at the next time point in a certain range, but the relationship may be reversed in another range, and the relationship may also depend on the value of X_2. It is exactly this flexibility that we see as the key advantage of the current method. Whereas previous time-series methods such as VAR and OUM can also be depicted with a vector field, they limit the shape of the vector field (see Appendix A). In the proposed method, we do not impose a specific form on the drift part. Instead, we use kernel methods to estimate it in a flexible manner."

5.10 VAR — latent VARMA — DFM — P-Technique

Valsiner et al. (2009): "The vector ARMA or VARMA model is a straightforward extension of the univariate ARMA model, which was discussed above. Let $\boldsymbol{y}_t$ be an M-variate observation at occasion t, which may be predicted from previous observations, and from unpredictable parts of previous observations. The VARMA (p, q) model is denoted as

$$\begin{aligned} \boldsymbol{y}_t &= \boldsymbol{\Phi}_0 + \boldsymbol{\Phi}_1 \boldsymbol{y}_{t-1} + \cdots + \boldsymbol{\Phi}_p \boldsymbol{y}_{t-p} + \boldsymbol{u}_t - \boldsymbol{\Theta}_1 \boldsymbol{u}_{t-1} - \cdots - \boldsymbol{\Theta}_q \boldsymbol{u}_{t-q} \\ &= \boldsymbol{\Phi}_0 + \sum_{j=1}^{p} \boldsymbol{\Phi}_j \boldsymbol{y}_{t-j} + \boldsymbol{u}_t - \sum_{j=1}^{q} \boldsymbol{\Theta}_j \boldsymbol{u}_{t-j} \end{aligned} \tag{18}$$

where $\mathbf{\Phi}_0$ is an M-variate vector with constants, and $\boldsymbol{u}_t$ is an M-variate vector with innovations. Although not strictly necessary, the elements of $\boldsymbol{u}_t$ are often assumed to be uncorrelated with each other.

The $M \times M$ matrices Φ contain the AR parameters on the main diagonal (i.e., the parameters that are used to regress the series upon itself at an earlier occasion), while the off-diagonal elements represent the cross-regression parameters. Thus, element $\phi_{ij,k}$ is used to regress the series i at occasion t on the series j at $t-k$. These matrices are not necessarily symmetric. For instance, series i may be regressed upon series j at previous occasions ($\phi_{ij,k} \neq 0$), whereas series j is not regressed on series $i\left(\phi_{ji,k} = 0\right)$. The $M \times M$ matrices $\mathbf{\Theta}$ contain the MA coefficients on the main diagonal. The off-diagonal elements are the parameters by which the unpredictable part of one series at a particular occasion may be predictive of the observation of another series at a later occasion. The model in Eq. (18) may be further simplified to a VAR model, in which case all Θ matrices are zero matrices (e.g., Hamilton, 1994, p. 291). Such models may be used to determine whether there are indications of causal relationships between two or more variables that were measured repeatedly. [...]

At first sight the VAR model may seem useful for modeling all kinds of data for which we assume one of the variables has a causal effect on the other (and possible vice versa). However, a VARMA model represents a stationary model and thus it requires the data to be stationary or to be rendered stationary by a suitable transformation. Suppose a researcher is interested in the effect of the empathy of a therapist on the depressive symptoms of a client. If the latter actually show a decline over time, the raw observations can not be modeled directly according to a VAR process. Rather, the researcher will have to make the series stationary, either by detrending the data or by differencing the data. Both approaches have disadvantages. [...]

The VARMA model can be extended to a model with multiple indicators measuring a reduced number of latent variables, which follow a VARMA process. Suppose we have a K-factor model with M observed variables. The factor loadings are restricted to be equal over time, such that the model can be represented as

$$\boldsymbol{y}_t = \boldsymbol{\mu} + \mathbf{\Lambda}\boldsymbol{\eta}_t + \boldsymbol{e}_t \tag{20}$$

where $\boldsymbol{\mu}$ is an M-variate vector with means, $\mathbf{\Lambda}$ is an $M \times K$ matrix with factor loadings which do not depend on time, $\boldsymbol{\eta}_t$ is a K-variate vector with latent variables at occasion t, and $\boldsymbol{e}_t$ is an M-variate vector with measurement errors at occasion t. At the latent level, a VARMA model is specified, such that

$$\boldsymbol{\eta}_t = \mathbf{\Phi}_1\boldsymbol{\eta}_{t-1} + \cdots + \mathbf{\Phi}_p\boldsymbol{\eta}_{t-p} + \boldsymbol{u}_t - \mathbf{\Theta}_1\boldsymbol{u}_{t-1} - \cdots - \mathbf{\Theta}_q\boldsymbol{u}_{t-q} \tag{21}$$

where Φ and Θ are now $K \times K$ matrices, and $\boldsymbol{u}_t$ is a K-variate vector with innovations of this latent VARMA process. This model can be recognized as a special version of the more general dynamic factor model as discussed by Molenaar (1985). Moreover, when all the Φ and Θ matrices are zero matrices, this model becomes the P-technique model discussed by Cattell, Cattell, and Rhymer (1947). If $q = 0$, the model in Eqs. (20) and (21) reduces to a latent VAR(p) model, which is also known as the direct autoregressive factor score model (Nesselroade, McArdle, Aggen, & Meyers, 2002). This model has been compared to the white noise factor score model (Nesselroade et al., 2002), which is also a special version of the more general dynamic factor model discussed by Molenaar (1985). Although the white noise factor score model can not be conceived of as an extension of the ARMA model (because the lagged relationships are not modeled in an ARMA manner, but by use of lagged factor loadings instead), there are situation in which the the white noise factor score model can be rotated into a direct autoregressive factor score model (Molenaar & Nesselroade, 2001). Moreover the latent VMA(q) can be shown to be a special case of the white noise factor score model."

5.10.1 Dynamic Factor Model (DFM) — HMM

R-Package: 'dfms', 'sparseDFM'

Stock and Watson (2016): "The DFM is an example of the much larger class of state-space or hidden Markov models, in which observable variables are expressed in terms ofunobserved or latent variables, which in turn evolve according to some lagged dynamics with finite dependence (ie, the law ofmotion ofthe latent variables is Markov). What makes the DFM stand out for macroeconometric applications is that the complex comovements of a potentially large number ofobservable series are summarized by a small number ofcommon factors, which drive the common fluctuations of all the series. [...]

The DFM expresses a $N \times 1$ vector X_t of observed time series variables as depending on a reduced number q of unobserved or latent factors f_t and a mean-zero idiosyncratic component e_t, where both the latent factors and idiosyncratic terms are in general serially correlated. The DFM is,

$$X_t = \lambda(\mathrm{L}) f_t + e_t \tag{1}$$

$$f_t = \Psi(\mathrm{L}) f_{t-1} + \eta_t \tag{2}$$

where the lag polynomial matrices $\lambda(\mathrm{L})$ and $\Psi(\mathrm{L})$ are $N \times q$ and $q \times q$, respectively, and η_t is the $q \times 1$ vector of (serially uncorrelated) mean-zero innovations to the factors. The idiosyncratic disturbances are assumed to be uncorrelated with the factor innovations at all leads and lags, that is, $Ee_t\eta'_{t-k} = 0$ for all k. In general, e_t can be serially correlated. The i th row of $\lambda(\mathrm{L})$, the lag polynomial $\lambda_i(\mathrm{L})$, is called the dynamic factor loading for the i th series, X_{it}.

The term $\lambda_i(\mathrm{L}) f_t$ in (1) is the common component of the i th series. Throughout this chapter, we treat the lag polynomial $\lambda(\mathrm{L})$ as one sided. Thus the common component of each series is a distributed lag of current and past values of $f_t{}^c$

The idiosyncratic disturbance e_t in (1) can be serially correlated. If so, models (1) and (2) are incompletely specified. For some purposes, such as state-space estimation discussed later, it is desirable to specify a parametric model for the idiosyncratic dynamics. A simple and tractable model is to suppose that the i th idiosyncratic disturbance, e_{it}, follows the univariate autoregression,

$$e_{it} = \delta_i(\mathrm{L}) e_{it-1} + \nu_{it}, \tag{3}$$

where ν_{it} is serially uncorrelated."

5.11 VAR — CLPM — DSEM — DLCA — NDLC-SEM

5.11.1 VAR — CLPM — (R)DSEM

R-Package: 'crosslag'

Asparouhov and Muthén (2023): "The residual variables in a structural equation model (SEM) can be used to create a secondary structural model. This combination of the primary and the secondary structural models is what we call the Residual Structural Equation Model (RSEM). The RSEM model has been discussed previously in the context of the dynamic structural equation models (DSEM), see Asparouhov et al. (2018) and Asparouhov and Muthen (2020), and it is generally referred to as the residual dynamic structural equation model (RDSEM). The RSEM model has also been used in the context of longitudinal cross-lagged panel models and is the basis for the random intercept cross-lagged panel model (RI-CLPM), see Hamaker et al. (2015), and the latent curve model with structured residuals (LCM-SR), see Curran et al. (2014)."

5.11.2 Dynamic Structural Equation Modeling (DSEM)

R-Package: 'dsem'

Asparouhov et al. (2018): "The novel modeling framework that we present here is a direct extension of the cross-classified modeling framework for ILD, as described in Asparouhov and Muthén (2016): We simply add to that framework the ability to regress any variable, observed or latent, not only on other variables at that same time point, but also on itself and other variables at previous time points. This extended framework is referred to as dynamic structural equation modeling (DSEM), and it combines four different modeling techniques: multilevel modeling, time-series modeling, structural equation modeling (SEM), and time-varying effects modeling (TVEM). Each of these four techniques addresses different aspects of the data and is used to model different correlations that are found in such data. The multilevel modeling is based on correlations that are due to individual-specific effects. The time-series modeling is based on correlations due to proximity of observations. The SEM is based on correlations between different variables. The TVEM is based on correlations due to the same stage of evolution. The goal of the DSEM framework is to parse out and model these four types of correlations and thereby give us a fuller picture of the dynamics found in ILD.

The DSEM model described here can also be viewed as the multilevel extension of the dynamic factor models described in Molenaar (1985), Zhang and Nesselroade (2007), and Zhang, Hamaker, and Nesselroade (2008). [...]

The general DSEM model consists of three separate models. The most general model is the *cross-classified DSEM model*, which incorporates both individual and time-specific random effects. The second most general model is the *two-level DSEM model*, which incorporates individual-specific random effects only. This model could actually be the most popular and useful model, as it is easier to estimate, identify, and interpret. The third model is the *single-level DSEM model* for time-series data from a single individual (cf. Zhang & Nesselroade, 2007). In the latter model, there are no random effects. Here we describe the most general cross-classified DSEM model. The two-level and the single-level DSEM models are special cases of the cross-classified DSEM model.

Fundamental to the DSEM framework is the decomposition of the observed scores into three components. Let Y_{it} be a vector of measurements for individual i at time t, where the ith individual is observed at times $t = 1, 2, \ldots, T_i$. The cross-classified DSEM model begins with the following decomposition:

$$\boldsymbol{Y}_{it} = \boldsymbol{Y}_{1,it} + \boldsymbol{Y}_{2,i} + \boldsymbol{Y}_{3,t}, \tag{1}$$

where $Y_{2,i}$ and $Y_{3,t}$ are individual-specific and time-specific contributions and $Y_{1,it}$ is the deviation of individual i at time t. The two-level DSEM model only makes use of the first two components (i.e., $Y_{3,t}$ is omitted), and the single-level model is based on simply using $Y_{it} = Y_{1,it}$. All three components are latent normally distributed random vectors and are used to form three sets of structural equation models, one on each level. [...]

From the decomposition just presented, we obtain two between-level components; that is, the individual-specific component $Y_{2,i}$ and the time-specific component $Y_{3,t}$. Their structural equation models take the usual form of a *measurement equation* and a *structural equation*; that is,

$$\boldsymbol{Y}_{2,i} = \boldsymbol{\nu}_2 + \boldsymbol{\Lambda}_2\boldsymbol{\eta}_{2,i} + \boldsymbol{K}_2\boldsymbol{X}_{2,i} + \boldsymbol{\varepsilon}_{2,i} \tag{2}$$

$$\boldsymbol{\eta}_{2,i} = \boldsymbol{\alpha}_2 + \boldsymbol{B}_2\boldsymbol{\eta}_{2,i} + \boldsymbol{\Gamma}_2\boldsymbol{X}_{2,i} + \boldsymbol{\xi}_{2,i} \tag{3}$$

$$\boldsymbol{Y}_{3,t} = \boldsymbol{\nu}_3 + \boldsymbol{\Lambda}_3\boldsymbol{\eta}_{3,t} + \boldsymbol{K}_3\boldsymbol{X}_{3,t} + \boldsymbol{\varepsilon}_{3,t} \tag{4}$$

$$\boldsymbol{\eta}_{3,t} = \boldsymbol{\alpha}_3 + \boldsymbol{B}_3\boldsymbol{\eta}_{3,t} + \boldsymbol{\Gamma}_3\boldsymbol{X}_{3,t} + \boldsymbol{\xi}_{3,t} \tag{5}$$

The vector $\boldsymbol{X}_{2,i}$ is a vector of individual-specific, time-invariant covariates, and $\boldsymbol{X}_{3,t}$ is a vector of time-specific but individual-invariant covariates. Similarly, $\boldsymbol{\eta}_{2,i}$ is a vector of individual-specific, time-invariant latent variables, and $\boldsymbol{\eta}_{3,t}$ is a vector of time-specific, individual-invariant latent variables. The variables $\boldsymbol{\varepsilon}_{2,i}, \boldsymbol{\xi}_{2,i}, \boldsymbol{\varepsilon}_{3,t}, \boldsymbol{\xi}_{3,t}$ are zero mean residuals as usual and the remaining vectors and matrices in this equation are nonrandom model parameters.

Although the preceding equations do not include regressions among Y components, such regressions are typically achieved by creating a latent variable equal to the Y variable, that is, the Y variable would be a perfect error-free indicator for a latent variable. Once such latent variables are included in the model, the regression between the Y variables is specified as a regression between the corresponding latent variables using the structural equations. This is a simple way to reduce the number of matrices in these equations and is somewhat of a tradition in the SEM literature, but it has no implication for model specification or estimation.

In the preceding specification we did not include observed dependent variables, but such variables are easy to accommodate as well. That is, in addition to the latent decomposition parts of the variables $\boldsymbol{Y}_{it}$, the vectors $\boldsymbol{Y}_{2,i}$ and $\boldsymbol{Y}_{3,t}$ can also include observed variables that are subjectspecific or time-specific.

The within-level part of the DSEM model is described by the following equations, which now include time-series components:

$$
\begin{aligned}
\boldsymbol{Y}_{1,it} =& \boldsymbol{\nu}_1 + \sum_{l=0}^{L} \boldsymbol{\Lambda}_{1,l} \boldsymbol{\eta}_{1,l,t-l} + \sum_{l=0}^{L} \boldsymbol{R}_l \boldsymbol{Y}_{1,i,t-l} \\
& + \sum_{l=0}^{L} \boldsymbol{K}_{1,l} \boldsymbol{X}_{1,i,t-l} + \boldsymbol{\varepsilon}_{1,it} \qquad (6) \\
\boldsymbol{\eta}_{1,it} =& \boldsymbol{a}_1 + \sum_{l=0}^{L} \boldsymbol{B}_{1,l} \boldsymbol{\eta}_{1,j,t-l} + \sum_{l=0}^{L} \boldsymbol{Q}_l \boldsymbol{Y}_{1,i,t-l} \\
& + \sum_{l=0}^{L} \boldsymbol{\Gamma}_{1,l} \boldsymbol{X}_{1,i,t-l} + \boldsymbol{\xi}_{1,it}. \qquad (7)
\end{aligned}
$$

Here $\boldsymbol{X}_{1,jt}$ is a vector of observed covariates for individual i at time t and $\boldsymbol{\eta}_{1,ti}$ is a vector of latent variables for individual i at time t.

In the preceding equations the latent variables $\boldsymbol{\eta}$, the dependent variables $\boldsymbol{Y}$, and the covariates $\boldsymbol{X}$ at times $t, t-1, \ldots, t-L$ can be used to predict the latent variables $\boldsymbol{\eta}$ and the dependent variables $\boldsymbol{Y}$ at time t. Including the lagged predictors $\boldsymbol{X}$ in the preceding equations is somewhat inconsequential, and we do this mostly for completeness. The covariate $\boldsymbol{X}_{1,i,l-l}$ is not any different from the covariate $\boldsymbol{X}_{1,i,t}$ because the model does not include distributional assumptions about the covariates $\boldsymbol{X}$ and is essentially a model for the conditional distribution of $[\boldsymbol{Y} \mid \boldsymbol{X}]$. Including the lagged covariates $\boldsymbol{X}_{1,j,t-l}$ does not involve any special statistical consideration with one small exception of the initial unobserved values, which we address later.

[...] the ARMA models are a special case of the DSEM model. [...] For identification purposes, restrictions need to be imposed on the preceding general model. For example, mean structure parameters can exist only on one of the levels for most common situations; that is, $\boldsymbol{\nu}_j$ will be fixed to 0 on two out of the three levels. Other identifying restrictions need to be imposed along the lines of standard structural equation models. [...]

The RDSEM model can be viewed as a special case of the DSEM model where the residual variables $\hat{\boldsymbol{Y}}_{1,it}$ and $\hat{\boldsymbol{\eta}}_{1,it}$ are modeled as within-level latent variables."

5.11.3 Dynamic Latent Class Analysis (DLCA)

Asparouhov et al. (2017): "The goal of the novel modeling framework we describe here is to allow for the study of (a) a latent (or hidden) Markov model that accounts for the switching between different states (also referred to as latent classes or regimes), with (b) individual differences in transition probabilities modeled as random effects, and (c) the option of dynamic relationships within each state through time series analysis and multilevel extensions of this. [...] the combination of DSEM with the multilevel HMM, which implies that observed or latent continuous variables can be autocorrelated both through the latent class autocorrelation (i.e., the HMM), and directly (i.e., through the dependencies over time in the form of, for instance, autoregressive relationships using DSEM). [...]

Consider the DSEM model of lag L. Let $\boldsymbol{Y}_{it}$ be an observed vector of measurements for individual i at time t. We begin with the usual within-between decomposition

$$\boldsymbol{Y}_{it} = \boldsymbol{Y}_{1,it} + \boldsymbol{Y}_{2,i} \tag{1}$$

where $\boldsymbol{Y}_{2,i}$ is the individual-specific random effect and $\boldsymbol{Y}_{1,it}$ is the individual i deviation at time t. The two components are assumed to be normally distributed random vectors and are used to form two separate sets of structural equations-one on each level. The Level 2 structural equation model takes the usual form

$$\boldsymbol{Y}_{2,i} = \boldsymbol{\nu}_2 + \boldsymbol{\Lambda}_2 \boldsymbol{\eta}_{2,i} + \boldsymbol{\varepsilon}_{2,i} \tag{2}$$

$$\boldsymbol{\eta}_{2,i} = \boldsymbol{\alpha}_2 + \boldsymbol{B}_2 \boldsymbol{\eta}_{2,i} + \boldsymbol{\Gamma}_2 \boldsymbol{x}_{2,i} + \boldsymbol{\xi}_{2,i} \tag{3}$$

where $\boldsymbol{x}_{2,i}$ is a vector of individual-specific time-invariant covariates and $\boldsymbol{\eta}_{2,i}$ is a vector of individual-specific timeinvariant latent variables. The variables $\boldsymbol{\varepsilon}_{2,i}$ and $\boldsymbol{\xi}_{2,i}$ are zero mean residuals as usual and the remaining vectors and matrices in these equations are nonrandom model parameters.

The within-level structural equation model consists of two equations that are used to model the contemporaneous and lagged relationships; that is,

$$\boldsymbol{Y}_{1,it} = \sum_{l=0}^{L} \boldsymbol{\Lambda}_{1,i,l}\, \boldsymbol{\eta}_{l,i,t-l} + \boldsymbol{\varepsilon}_{1,it} \tag{4}$$

$$\boldsymbol{\eta}_{1,i,t} = \boldsymbol{\alpha}_{1,i} + \sum_{l=0}^{L} \boldsymbol{B}_{1,i,l}\, \boldsymbol{\eta}_{1,i,t-l} + \boldsymbol{\Gamma}_{1,i}\, \boldsymbol{x}_{1,it} + \boldsymbol{\xi}_{1,it}. \tag{5}$$

Here $\boldsymbol{x}_{1,it}$ is a vector of observed covariates for individual i at time t and $\boldsymbol{\eta}_{l,i,t}$ is a vector of latent variables for individual i at time t. The difference between the standard structural model and the model in Equations 4 and 5 is that all the elements in the observed and the latent vectors on the left side have the same time index, where as the latent variables on the right side are associated with both the same occasion, but also times $t-1, \ldots, t-L$, showing that these preceding latent variables can now be used as predictors for the observed and latent variables at time t. [...]

Let S_{it} be a categorical latent variable for individual i at time t; that is, S_{it} is a within-level latent class. In the time series literature such a latent class variable is more often referred to as a latent state variable; therefore, we use S as the variable name rather than the traditional C. Suppose that S_{it} takes values $1, 2, \ldots, K$ where K is the number of classes in (or states of) the model. The DSEM mixture model consists of Equations 1 through 5, however Equations 4 and 5 now depend on S_{it} as follows:

$$[\boldsymbol{Y}_{1,it} \mid S_{it} = s] = \boldsymbol{\nu}_{1,s} + \sum_{l=0}^{L} \boldsymbol{\Lambda}_{1,l,s} \boldsymbol{\eta}_{i,t-l} + \boldsymbol{\varepsilon}_{it} \tag{7}$$

$$[\boldsymbol{\eta}_{i,t} \mid S_{it} = s] = \boldsymbol{\alpha}_{l,s} + \sum_{l=0}^{L} \boldsymbol{B}_{1,l,s} \boldsymbol{\eta}_{i,t-l} + \boldsymbol{\Gamma}_{1,s} \boldsymbol{x}_{it} + \boldsymbol{\xi}_{it}. \tag{8}$$

The residual covariance matrices of ε_{it} and of ξ_{it} are also state specific, meaning they depend on s. [...]

The distribution of S_{it} is given by

$$P\left(S_{it} = s\right) = \frac{\text{Exp}\left(\alpha_{is}\right)}{\sum_{s=1}^{K} \text{Exp}\left(\alpha_{is}\right)} \tag{10}$$

where α_{is} are normally distributed random effects. For identification purposes $\alpha_{iK} = 0$. These random effects α_{is} are included as part of the between-level latent variable vector η_{2,i^*}. This implies also that individual-level predictors $x_{2,i}$ can be used to predict $P(S_{it} = s)$ through the structural Equation 3. [...]"

5.11.4 Nonlinear Dynamic Latent Class - Structural Equation Modeling (NDLC-SEM)

Kelava and Brandt (2019): "The proposed NDLC-SEM contains several submodels. In its comprehensive version, the NDLC-SEM incorporates both inter-individual and time-specific random effects. The individual specific random effects can be used to specify a DLCA (e.g., Asparouhov et al., 2017a), which models latent transitions processes using Markov chain models. A variety of models can be specified as special cases, such as two-level dynamic SEM, dynamic LCA models, single-level dynamic CFA and SEM, state-space models, and time-series models.

We start by decomposing the observed scores. Let Y_{it} be a $(J \times 1)$ dimensional vector of measurements for individual i at time t. Note that each individual i is observed at times $t = 1, 2, \ldots, T_i$ and that T_i might be individual specific (individuals might differ in the number of measurement occasions.). The decomposition in the general NDLCSEM is as follows:

$$Y_{it} = Y_{1it} + Y_{2i} + Y_{3t} \tag{1}$$

where Y_{2i} and Y_{3t} are $(J \times 1)$ dimensional individual-specific and time-specific parts, respectively. Y_{1it} is the deviation of an individual i at time t from Y_{2i} and Y_{3t}. Note that $Y_{1\text{it}}$ might depend on a latent state s of an individual i at time t, which leads to $[Y_{1it} \mid S_{it} = s]$ instead of Y_{1it}. For example, affective states could be measured repeatedly with different items. Y_{2i} then refers to the individual item score level across time, Y_{3t} refers to the average item score across subjects at each time point t, and Y_{1it} is the person- and time-specific deviation of a subject's score relative to the person's and time-related item levels.

In the next subsection, we describe both the individual-specific and time-specific (between-level) models. After this, we describe the within-level model for Y_{1it} and the Markov switching model for the latent states.

Individual-specific component The individual-specific component Y_{2i} is explained by a measurement equation and a structural equation ("between-level" in multilevel models). It models the interindividual differences in the developmental trajectories that are independent of time:

$$Y_{2i} = \nu_2 + \Lambda_2 \eta_{2i} + K_2 X_{2i} + \epsilon_{2i} \tag{2}$$

$$\eta_{2i} = \alpha_2 + B_2 \eta_{2i} + \Omega_2 h_2\left(\eta_{2i}\right) + \Gamma_2 X_{2i} + \zeta_{2i} \tag{3}$$

X_{2i} is a $(G_2 \times 1)$ dimensional vector of individual-specific, time-invariant (baseline) covariates (e.g., gender). η_{2i} is a $(M_2 \times 1)$ dimensional vector of individual-specific, time-invariant latent variables (e.g., representing latent constructs at baseline measure such as baseline depression).

$h_2 (\eta_{2i})$ is R_2 dimensional vector of functions of η_{2i}. For example, if interaction and quadratic effects are specified, $h(\eta_{2i}) = \text{vech}(\eta_{2i}\eta_{2i}{'})$ is a $(R_2 = M_2 (M_2 + 1) /2 \times 1)$ dimensional vector of products of latent variables. $\epsilon_{2i}(J \times 1)$ and $\zeta_{2i} (M_2 \times 1)$ are multivariate residuals with zero expectations. At this point, we do not assume a specific distribution of the residuals, which should rely on theoretical considerations. In the context of Bayesian estimation, even complicated multivariate distributions can be easily specified. In the frequentist context, practical consideration of available implementations might lead to the assumption of multivariate normal distributions. The same holds for the following equations with residuals. $\nu_2(J{\times}1), \quad \Lambda_2 (J \times M_2), \quad K_2 (J \times G_2), \quad \alpha_2 (M_2 \times 1), \quad B_2 (M_2 \times M_2), \Omega_2 M$ R_2, and $\Gamma_2 (M_2 \times G_2)$ are matrices of fixed effects (including intercepts and coefficients).

$h_2()$ can be used very flexibly such as for the specification of splines that approximate unknown relationships between the latent variables. Then, h_2 consists of unidimensional functions $h_{2m} (m = 1, \dots M_2)$ for each latent predictor η_{2mi} that are defined as (see, e.g., Guo, Zhu, Chow, & Ibrahim, 2012):

$$h_{2m} (\eta_{2mi}) = \sum_{v_m=1}^{N_m} h^*_{2mv_m} (\eta_{2mi}), \tag{4}$$

where $h^*_{2mv_m}$ are basis functions with dimension N_m, for example, cubic splines with N_m nodes (see details in Hastie, Tibshirani, & Friedman, 2009; Wood, 2017). Note that standard identification rules for splines apply (i.e., with regard to the intercept and linear effects in Equation (3) that need to be considered when including basis expansions, see, Guo et al., 2012).

Further, it is possible to add semiparametric interactions between two predictors using two-dimensional functions $h_{2mm'}, m' > m = 1 \dots M_2 - 1$ which are operationalized as tensor product bases of the form

$$h_{2mm'} (\eta_{2mi}, \eta_{2mi'}) = \sum_{v_m=1}^{N_m} \sum_{v_{m'}=1}^{N_{m'}} h^*_{2mv_m} (\eta_{2mi}) \, h^*_{2m'v_{m'}} (\eta_{2m'ik}). \tag{5}$$

Splines are a non-parametric regression technique that can be seen as extensions of linear regression models. They have the advantage that they can flexibly adapt to nonlinearities and interactions of variables, even if the functional form of the relationship between the variables is unknown. Compared with the strict parametric functional representation of nonlinear effects (i.e., $\Omega_2\eta_{2i}\eta_{2i}{'}$), splines come with less assumptions and can approximate even higher order nonlinear effects by piecewise polynomials. Recently, the extension of SEM to incorporate splines has received increasing attention (e.g., Feng, Wang, Wang, & Song, 2015; Feng, Wu, & Song, 2015, 2017b, Guo et al., 2012; Song, Li, Cai, & Ip, 2013) because they considerably extend the possibilities to model relationships between latent variables.

Note that the measurement model in Equation (2) is conceptualized as linear. By introducing additional restrictions in Equation (3), nonlinear measurement models can be obtained. Again, the same holds true for the following measurement models.

Time-specific component Similarly, the time-specific component Y_{3t} is explained by a measurement equation and a structural equation. It models the average time trend in the data which are independent of the individuals and do not contain information about inter-individual differences in the growth process:

$$Y_{3t} = v_3 + \Lambda_3 \eta_{3t} + K_3 X_{3t} + \epsilon_{3t} \tag{6}$$

$$\eta_{3t} = \alpha_3 + B_3 \eta_{3t} + \Omega_3 h_3 (\eta_{3t}) + \Gamma_3 X_{3t} + \zeta_{3t} \tag{7}$$

X_{3t} is a $(G_3 \times 1)$ dimensional vector of time-specific, individual-invariant/ independent covariates (e.g., situation effects such as hospital size where the treatment is conducted or season of the year when the study is conducted). η_{3t} is a $(M_3 \times 1)$ dimensional vector of time-specific, individual-invariant latent variables. $h_3 (\eta_{3t})$ is a $(R_3 \times 1)$ dimensional vector of (nonlinear) functions for the latent variables (i.e., interactions and quadratic effects). $\epsilon_{3t}(J \times 1)$ and $\zeta_{3t}\,(M_3 \times 1)$ are multivariate residuals with zero expectations. $v_3(J \times 1)$, $\Lambda_3\,(J \times M_3)$, $K_3\,(J \times G_3)$, $\alpha_3\,(M_3 \times 1)$, $B_3\,(M_3 \times M_3)$, $\Omega_3\,(M_3 \times R_3)$, and $\Gamma_3\,(M_3 \times G_3)$ are matrices of fixed effects (including intercepts and coefficients).

The within-level model is given by the following measurement model and structural model. It models the intraindividual differences occurring during the development process:

$$\begin{aligned}[Y_{1it} \mid S_{it} = s] = {} & v_{1s} + \Sigma_{l=0}^{L} \Lambda_{1ls} \eta_{1i(t-l)} \\ & + \Sigma_{l=0}^{L} R_{ls} Y_{1i(t-l)} + \Sigma_{l=0}^{L} K_{1ls} X_{1i(t-l)} \\ & + \epsilon_{1it} \end{aligned} \tag{8}$$

$$\begin{aligned}[\eta_{1it} \mid S_{it} = s] = {} & \alpha_{1s} + \Sigma_{l=0}^{L} B_{1ls} \eta_{1i(t-l)} \\ & + \Sigma_{l=0}^{L} \Sigma_{l'=0}^{L'} \Omega_{1ll's} h_{1ll'} \left(\eta_{1i(t-l)}, \eta_{1i(t-l')}\right) \\ & + \Sigma_{l=0}^{L} Q_{ls} Y_{1i(t-l)} + \Sigma_{l=0}^{L} \Gamma_{1ss} X_{1i(t-l)} + \zeta_{1it} \end{aligned} \tag{9}$$

$X_{1it}\,(G_1 \times 1)$ dimensional is a vector of individual- and time-dependent covariates (e.g., dosage of psychopharmaka). η_{1it} is a $(M_1 \times 1)$ dimensional vector of individualand time-dependent latent variables (such as a latent factor representing the relative deviance of the depression score compared to average person and time-specific score). Note that η_{1it} depends on a latent state s of individual i at time point t. $h_{1ll'}\left(\eta_{1i(t-l)}, \eta_{1i(t-l')}\right)$ is a $(R_1 \times 1)$ dimensional vector of (nonlinear) functions of latent variables at different lags (i.e., interactions and quadratic effects). $\epsilon_{1it}(J \times 1)$ and $\zeta_{1it}\,(M_1 \times 1)$ are multivariate residuals with zero expectations. $v_{1s}(J \times 1)$, $\Lambda_{1/s}\,(J \times M_1)$, $R_{ls}(J \times J)$, $K_{1/s}\,(J \times G_1)$, $\alpha_{1s}\,(M_1 \times 1)$, $B_{1/s}\,(M_1 \times M_1)$, $\Omega_{1/l_s}\,(M_1 \times R_1)$, and $\Gamma_{1/s}\,(M_1 \times G_1)$ are matrices with (fixed or random) coefficients. As can be seen, the measurement and structural model are time-series models which include latent classes.

The NDLC-SEM is capable of including observed categorical variables by a parameterization of threshold parameters. For a specific item j with the variable $\left[Y_{1jit} \mid S_{it} = s\right]$

with a variable-specific number of categories $k = 1, \ldots, m_j$, we assume a normally distributed latent variable $\left[Y^*_{1jit} \mid S_{it} = s\right]$ and threshold parameters $\tau_{j1s}, \ldots, \tau_{j(m_j-1)s}$ such that:

$$\left[Y_{1jit} = k \mid S_{it} = s\right] \Leftrightarrow \tau_{j(k-1)s} \leq \left[Y^*_{1jit} \mid S_{it} = s\right] < \tau_{jks} \tag{10}$$

with $\tau_{j0s} = -\infty$ and $\tau_{j(m_j)s} = \infty$ for all latent states $s = 1, \ldots, K$. For reasons of simplicity, we will assume continuous observed variables such that $\left[Y^*_{1jit} \mid S_{it} = s\right] = \left[Y_{1jit} \mid S_{it} = s\right]$ in Equations (8) and (9). Note that in Equation (6) and (7) $[Y_{1it} \mid S_{it} = s]$ refers to a $J \times 1$ vector of items and that for an item j, $\left[Y_{1jit} \mid S_{it} = s\right]$ is a specific element of this vector $[Y_{1it} \mid S_{it} = s]$.

The Markov switching model The latent state variable S_{it} is a person- and time-dependent variable which follows a Markov switching model with person- and time-specific transition probability:

$$P\left(S_{it} = d \mid S_{i(t-1)} = c\right) = \frac{\exp\left(\alpha_{itdc}\right)}{\sum_{k=1}^{K} \exp\left(\alpha_{itkc}\right)} \tag{11}$$

α_{itdc} are person- and time-specific random effects (see below) with $\alpha_{itKc} = 0$. For example, the latent state could represent the (time- and person-specific) adherence to the treatment regimen. Often, this variable is an unobserved (or unobservable) entity that nonetheless effects the development of the outcome variable (e.g., a decline in depression for persons that are adherent and a different growth pattern for those non-adherent)."

5.11.5 NDLC-SEM — ts-lvgvar — gVAR

R-Package: 'psychonetrics'

Epskamp (2020): "The framework emerges by combining a general factor model and a GVAR model and is referred to as the *lvgvar* model (An alternative perspective is that the lvgvar is a combination of a dynamic factor model (Molenaar, 1985, 2017) with a GGM on the autoregression's residuals). I will discuss estimation in two very distinct settings:

- Time-series data of a single subject. This setting will be termed *ts-lvgvar*.
- Panel data of many subjects measured on a few (at least three) measurement occasions. This setting will be termed *panel-lvgvar*.

These settings differ crucially from one another. *ts-lvgvar* concerns a fixed subject (fixed subject p), which involves variation over measurement occasions (random time T). Meanwhile, *panel-lvgvar* concerns variation over subjects (random subject P) at a few fixed time points $t, t+1, t+2, \ldots$. The *panel-lvgvar* is a multi-level model with random effects on the mean structure. [...] The *lvgvar* framework is implemented in the opensource software package https://psychonetrics.org/index.html. [...]

ts-lvgvar The *ts-lvgvar* concerns a fixed subject p and takes measurement T as random. Assuming multivariate normality for all variables, the following may be formulated:

$$\boldsymbol{y}_{p,T} \sim N\left(\boldsymbol{\mu}_p^{(y)}, \boldsymbol{\Sigma}_{p,0}^{(y)}\right) \quad \text{Within-subject distribution of observed variables.}$$

$$\boldsymbol{\eta}_{p,T} \sim N\left(\boldsymbol{\mu}_p^{(\eta)}, \boldsymbol{\Sigma}_{p,0}^{(\eta)}\right) \quad \text{Within-subject distribution of latent variables.}$$

More generally, the within-subject lag- k covariances can be defined as:

$$\operatorname{cov}\left(\boldsymbol{y}_{p,T+k}, \boldsymbol{y}_{p,T}\right) = \boldsymbol{\Sigma}_{p,k}^{(y)} \quad \text{Observed variables within-subject lag-k covariances.}$$
$$\operatorname{cov}\left(\boldsymbol{\eta}_{p,T+k}, \boldsymbol{\eta}_{p,T}\right) = \boldsymbol{\Sigma}_{p,k}^{(\eta)} \quad \text{Latent variables within-subject lag-k covariances,}$$

with the superscript indicating the variable of interest. As seen in Eq. (4), the expected mean vector $\boldsymbol{\mu}_p^{(y)}$ is a composite of the intercept, the latent means, and the systematic bias in subject p. Because $\boldsymbol{\mu}_p^{(\varepsilon)}$ has the same length as $\boldsymbol{y}_p^{(\varepsilon)}$, it is not possible to estimate this bias from time-series data of a single subject, or even from an analysis with a few subjects, even when equality constraints are placed on the intercepts and factor-loading structure (because the latent means may vary, which may be the topic of interest). Therefore, an identifying assumption $\boldsymbol{\mu}_p^{(\varepsilon)} = \boldsymbol{0}$ is required in the ts-lvgvar setting. Note, any within-subject variation is due to variations in deviations from the mean, which is denoted here with $\boldsymbol{\xi}$. As a consequence, the within-subject mean structure takes the following form:

$$\boldsymbol{\mu}_p^{(y)} = \boldsymbol{\tau} + \boldsymbol{\Lambda}\boldsymbol{\mu}_p^{(\eta)} \quad \text{Stationary within-subject means,}$$

and the variance-covariance structure takes the following form:

$$\boldsymbol{\Sigma}_{p,k}^{(y)} = \begin{cases} \boldsymbol{\Lambda}\boldsymbol{\Sigma}_{p,k}^{(\eta)}\boldsymbol{\Lambda}^\top + \boldsymbol{\Sigma}_p^{(\varepsilon)} & \text{if} k = 0, \\ \boldsymbol{\Lambda}\boldsymbol{\Sigma}_{p,k}^{(\eta)}\boldsymbol{\Lambda}^\top & \text{otherwise,} \end{cases} \quad \begin{array}{l}\text{Stationary within-subject} \\ \text{lag-k (co)variances.}\end{array} \tag{6}$$

This matrix captures within-subject variation, with k indicating the lag, which may differ person to person.

The relationships between the latent variables may be modeled as a lag-1 GVAR. Assuming stationarity, the structural model then becomes:

$$\operatorname{vec}\left(\boldsymbol{\Sigma}_{p,0}^{(\eta)}\right) = \left(\boldsymbol{I} \otimes \boldsymbol{I} - \boldsymbol{B}_p \otimes \boldsymbol{B}_p\right)^{-1} \operatorname{vec}\left(\boldsymbol{\Sigma}_{p,0}^{(\zeta)}\right) \quad \text{Stationary latent variation.}$$
$$\boldsymbol{\Sigma}_{p,k}^{(\eta)} = \boldsymbol{B}_p \boldsymbol{\Sigma}_{p,k-1}^{(\eta)} \quad \text{Stationary lag-k covariances between latent variables.}$$

The innovation variance-covariance matrix can further be modeled as a GGM to obtain a latent contemporaneous network:

$$\boldsymbol{\Sigma}_p^{(\zeta)} = \boldsymbol{\Delta}_p^{(\zeta)}\left(\boldsymbol{I} - \boldsymbol{\Omega}_p^{(\zeta)}\right)^{-1}\boldsymbol{\Delta}_p^{(\zeta)}.$$

This leads to two network structures: a temporal network modeled with $\boldsymbol{B}_p$, and a contemporaneous network modeled with $\boldsymbol{\Omega}_p^{(\zeta)}$. Typical SEM identifying constraints are needed to make the model discernible, such as having nonnegative degrees of freedom, placing $\boldsymbol{\mu}_p^{(\eta)} = \mathbf{0}$ in a singlesubject setting, and constraining either the first factor loadings in $\boldsymbol{\Lambda}$ or the diagonal elements of $\boldsymbol{\Delta}_p^{(\zeta)}$ to one. [...]

The presented modeling framework is closely related to several other frameworks:

1. The *ts*-lvgvar reduces to a general GVAR model (Epskamp, Waldorp, et al., 2018) when all variables in the network are observed-the factor-loading matrix then reduces to an identity matrix and the residual variances are set to zero. In this case, the model reduces further to a general vector-autoregression model if the contemporaneous level is modeled as a variance-covariance matrix rather than a GGM.

2. When latent variables are used in the ts-lvgvar, but the contemporaneous level is not modeled using a GGM, the model reduces to a type of dynamic factor model (Molenaar, 1985).

3. The panel-lvgvar can be considered a multi-level GVAR model (Epskamp, Waldorp, et al., 2018) if all variables in the networks are observed without measurement errorwith the exception that all network parameters are fixed (only intercepts are random). Extending the panel-lvgvar framework to include random effects on the network parameters will likely not be trivial, but may form an important direction of future research.

4. When the contemporaneous and between-subject effects of the panel-lvgvar are modeled as variance-covariance matrices and all variables in the networks are observed, the model reduces to a random intercept cross-lagged panel model (Hamaker et al., 2015), with the exception that stationarity is used to avoid estimating the variance-covariance structure of the first measurement.

5. When only cross-sectional data are used, the panel-lvgvar reduces to a latent network model (Epskamp, Rhemtulla, & Borsboom, 2017), after identifying all within-subject variance-covariance structures to be zero. Subsequently, when in cross-sectional data all variables in the network are assumed observed without measurement error, the model reduces to a standard GGM (Epskamp & Fried, 2018).

6. When the contemporaneous and between-subject levels of the panel-lvgvar are modeled using variance-covariance matrices, the model can be seen as a special case of general dynamic SEM models (Ciraki, 2007). When the data contain many persons and time series, it is in principle possible to obtain both fixed-effect and person-wise estimates (random effects), but this is beyond the scope of the current paper (e.g., Asparouhov, Hamaker, & Muthén, 2018; Epskamp, Waldorp, et al., 2018)."

NDLC-SEM **Andriamiarana et al. (2023)**: "[...] One might be interested in studying the intention of N students to quit their studies and the impact of such unobserved intention in, for instance, their daily/weekly affective states. First, we can assume that the change into a state of intention to quit is explained by some more or less stable vulnerability factors on the between-level, namely the lack of conscientiousness ($\boldsymbol{\eta}_{22}$) and the neuroticism ($\boldsymbol{\eta}_{23}$). A latent interaction effect ($\boldsymbol{\eta}_{22} \times \boldsymbol{\eta}_{23}$) can also be added to describe a synergistic dropout related consequence of these two latent vulnerabilities. Second, a discrete latent class variable is used to refer to the two potential classes of intentions (intention vs. no-intention to quit). This discrete latent intention variable can be assumed to follow a HMM with individual-specific transition probabilities between the states, explained by a latent factor $\boldsymbol{\eta}_{21}$. In that case, $\boldsymbol{\eta}_{21}$ is determined by $\boldsymbol{\eta}_{22}$, $\boldsymbol{\eta}_{23}$ and their interaction effect on the between level. Third, a latent within factor, for instance describing affective states, is measured on the intra-individual level across T time points so that the trajectory of such a latent factor conditionally depends on the latent class (intention to quit or not) for each individual and on the previous time-dependent affective states. An empirical data set that follows a very similar design was presented by Kelava et al. (2022). In that specific example, $N = 122$ students were measured $T = 50$ times with regard to their time-dependent affective states.

Within the NDLC-SEM framework this translates to a given 3-dimensional vector of observed variables $\boldsymbol{Y}_{\text{it}}$, varying between individuals $i = 1, \ldots, N$ and time points $t = 1, \ldots, T$. We can represent $\boldsymbol{Y}_{it}$ as

$$\boldsymbol{Y}_{it} = \boldsymbol{Y}_{2i} + \boldsymbol{Y}_{1it} \tag{1}$$

where $\boldsymbol{Y}_{2i}$ collects two 3-dimensional vectors of observed time-invariant variables on the between level, as described in Figure 1. More precisely, this is a set of variables which represents (stable) inter-individual differences in observed outcomes. $\boldsymbol{Y}_{1\text{it}}$ represents the within level 3-dimensional vector of observed variables (intra-individual changes). [...]

The inter-individual differences (at baseline) are defined on the between level. In our model, this is represented by a random effect explained by a combination of two latent factors. These factors are directly derived from the measurement model based on $\boldsymbol{Y}_{2i}$'s components. From a practical point of view, we assume that $\boldsymbol{Y}_{2i}$ is a 6-dimensional vector involving indicators that serve as a basis for factors of vulnerability. Formally, the between level can be expressed as follows:

$$\boldsymbol{Y}_{2i} = \Lambda_2 \boldsymbol{\eta}_{2i} + \epsilon_{2i} \tag{2}$$

$$\eta_{21i} = \beta_0 + \beta_1 \eta_{22i} + \beta_2 \eta_{23i} + \beta_3 \eta_{22i} \eta_{23i} \tag{3}$$

where Equation (3) represents the random effect η_{21i} that is based on the 2 -dimensional vector of latent variables $\boldsymbol{\eta}_{2i} = (\eta_{22i}, \eta_{23i})'$. This random effect η_{21i} influences the HMM's transition probabilities. In the measurement model in Equation (2), the 6 -dimensional vector ϵ_{2i} represents the error terms, $\boldsymbol{\eta}_{2i}$ the 2 -dimensional vector of latent variables, and $\boldsymbol{\Lambda}_2$ the corresponding (6×2)-dimensional loading matrix. For Bayesian modeling, we can rewrite Equation (2) as

$$\boldsymbol{Y}_{2i} \sim \mathcal{N}\left(\mu_{\boldsymbol{Y}_{2i}}, \sigma^2_{\epsilon_{2i}} \mathrm{I}_6\right) \tag{4}$$

with $\mu_{\mathrm{Y}_{2i}} = \Lambda_2 \boldsymbol{\eta}_{2i}$. For the 2 -dimensional vector of latent variables, we assume $\boldsymbol{\eta}_{2i} \sim \mathcal{N}(0, \Phi_2)$ with $\Phi_2 = (\phi_{kl})_{1 \leq k,l \leq 2}$. [...]"

5.12 VAR — DLM — HMM

Hidden Markov Model (HHM) Murphy (2012): "Figure 10.3(a) illustrates a first-order Markov chain as a DAG. Of course, the assumption that the immediate past, x_{t-1}, captures everything we need to know about the entire history, $\boldsymbol{x}_{1:t-2}$, is a bit strong. We can relax it a little by adding a dependence from x_{t-2} to x_t as well; this is called a second order Markov chain, and is illustrated in Figure 10.3(b). The corresponding joint has the following form:

$$\begin{aligned} p(\boldsymbol{x}_{1:T}) &= p(x_1, x_2)\, p(x_3 \mid x_1, x_2)\, p(x_4 \mid x_2, x_3) \dots \\ &= p(x_1, x_2) \prod_{t=3}^{T} p(x_t \mid x_{t-1}, x_{t-2}) \end{aligned} \tag{10.9}$$

We can create higher-order Markov models in a similar way. See Section 17.2 for a more detailed discussion of Markov models.

Unfortunately, even the second-order Markov assumption may be inadequate if there are lon-grange correlations amongst the observations. We can't keep building ever higher order models, since the number of parameters will blow up. An alternative approach is to assume that there is an underlying hidden process, that can be modeled by a first-order Markov chain, but that the data is a noisy observation of this process. The result is known as a hidden Markov model or HMM, and is illustrated in Figure 10.4. Here z_t is known as a hidden variable at "time" t, and x_t is the observed variable. (We put "time" in quotation marks, since these models can be applied to any kind of sequence data, such as genomics or language, where t represents location rather than time.) The Conditional Probability Distribution (CPD) $p(z_t \mid z_{t-1})$ is the transition model, and the CPD $p(\boldsymbol{x}_t \mid z_t)$ is the observation model.

The hidden variables often represent quantities of interest, such as the identity of the word that someone is currently speaking. The observed variables are what we measure, such as the acoustic waveform. What we would like to do is estimate the hidden state given the data, i.e., to compute $p(z_t \mid \boldsymbol{x}_{1:t}, \boldsymbol{\theta})$. This is called state estimation, and is just another form of probabilistic inference. See Chapter 17 for further details on HMMs."

Dynamic Linear Model (DLM) Laine (2020): "Dynamic linear models (DLM) offer a very generic framework to analyse time series data. Many classical time series models can be formulated as DLMs, including ARMA models and standard multiple linear regression models. The models can be seen as general regression models where the coefficients can vary in time. In addition, they allow for a state space representation and a formulation as hierarchical statistical models, which in turn is the key for efficient estimation by Kalman formulas and by Markov chain Monte Carlo (MCMC) methods. A dynamic linear model can handle non-stationary processes, missing values and non-uniform sampling as well as observations with varying accuracies. [...]

The state space description offers a unified formulation for the analysis of dynamic regression models. The same formulation is used extensively in signal processing and geophysical data assimilation studies, for example. A general dynamic linear model with an observation equation and a model equation is

$$y_t = \mathrm{H}_t x_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \mathrm{R}_t), \tag{3.1}$$

$$x_t = \mathrm{M}_t x_{t-1} + E_t, \quad E_t \sim N(0, \mathrm{Q}_t). \tag{3.2}$$

Above y_t is a vector of length k of observations at time t, with $t = 1, \ldots, n$. Vector x_t of length m contains the unobserved states of the system that evolve in time according to a linear system operator M_t (a $m \times m$ matrix). In time series settings x_t will have elements corresponding to various components of the time series process, like trend, seasonality, etc. We observe a linear combination of the states with noise ε_t, and matrix $\mathrm{H}_t(m \times k)$ is the observation operator that transforms the model states into observations. Both observations and the system states can have additive Gaussian errors with covariance matrices $\mathrm{R}_t(k \times k)$ and $\mathrm{Q}_t(m \times m)$, respectively. In univariate time series analysis we will have $k = 1$. With multivariate data, the system matrices $\mathrm{M}_t, \mathrm{H}_t, \mathrm{R}_t$ and Q_t can be used to define correlations between the observed components. [...]"

6 IM — Kinetic IM — VDAR — VAR

Campajola et al. (2021): "The dynamics of a large variety of systems, from Physics to Economics and Finance, can be represented as time series of binary variables. The most telling examples are the spin systems in Statistical Physics, where magnetic moments of the particles in a lattice are described as two-state variables, or binary time series in Quantitative Finance, capturing for instance the occurrence of extreme events of prices [1, 2], or buy and sell orders in the order book of financial markets [3]. Different models have been introduced to capture the multivariate interaction structure of such binary systems, in particular the Kinetic Ising Model (KIM) [4] in Physics and the discrete autoregressive processes [5, 6] in time series analysis, together with the (Markovian) multivariate generalization recently introduced by [7], namely the Vector Discrete Auto Regressive Process VDAR(1). In this paper we prove analytically that, under some condition, the KIM is equivalent to the VDAR(1) model. Furthermore it is well known that the Ising model, in both static [8] and kinetic [9] version, is a maximum entropy model, given mean magnetizations and pairwise correlations (at lag one in the kinetic case), see also [10, 11, 12], and, among other aspects, maximum entropy arguments can be used to define without ambiguity the temperature in such non-equilibrium spin systems [13]. Here, by exploiting the equivalence between the two models, we prove also that the Markov chain associated with the vector discrete autoregressive process VDAR(1) can be interpreted as the maximum entropy distribution of binary random variables with given means, auto-correlations, and lagged cross-correlations (of order one). [...]

The Kinetic Ising Model [4, 14, 15, 16], was originally proposed as the out of equilibrium version of the classical Ising spin glass [17, 18, 19] to describe a Markovian dynamics of *spins* σ_t^i, i.e. binary random variables taking values -1 and $+1$, interacting with each other according to some generic matrix of couplings. The KIM has found countless applications in many contexts [...]

In mathematical terms, the KIM is a logistic regression model specified by the following transition probability for a set of N spins $\boldsymbol{\sigma}_t \equiv \left\{\sigma_t^i\right\}_{i=1,\ldots,N}$,

$$p_{KIM}\left(\boldsymbol{\sigma}_t \mid \boldsymbol{\sigma}_{t-1}, \boldsymbol{J}, \boldsymbol{h}\right) = Z_{t-1}^{-1} \exp\left(\sum_{i,j=1}^{N} \sigma_t^i J_{ij} \sigma_{t-1}^j + \sum_{i=1}^{N} \sigma_t^i h_i\right) \tag{1}$$

where $\boldsymbol{J} \equiv \left\{J_{ij}\right\}_{i,j=1,\ldots,N}$ is a matrix of real-valued parameters giving the multivariate auto-regressive structure of the model or, equivalently, representing the couplings between spins, $\boldsymbol{h} \equiv \{h_i\}_{i=1,\ldots,N}$ is a set of variable-specific parameters representing the external magnetic fields associated with each spin, and Z_{t-1} is the partition function $Z_{t-1} = \prod_{i=1}^{N} 2\cosh\left(\sum_j J_{ij}\sigma_{t-1}^j + h_i\right)$ guaranteeing that the probability distribution is properly normalized.

The KIM of Eq. (1) is a *maximum entropy* [29, 30] model for a set of binary random variables which display on average given means, and both auto- and (lagged) cross-correlations. For the sake of clarity and in preparation to the section below, let us move from the spin variables $\sigma_t^i \in \{-1, 1\}$ to the binary variables $X_t^i \in \{0, 1\}$.

Given a set of N binary variables $\boldsymbol{X}_t \equiv \left\{X_t^i\right\}_{i=1,\ldots,N}^{t=1,\ldots,T}$, let us consider the following metrics,

$$2\sum_t X_t^i, \forall i = 1, \ldots, N, \tag{2}$$

$$2\sum_t X_t^i X_{t-1}^j + \left(1 - X_t^i\right)\left(1 - X_{t-1}^j\right), \forall i, j = 1, \ldots, N, \tag{3}$$

related (under stationarity conditions) to the mean of the binary random variables and the correlation between them, respectively.

The metric (3) for $i = j$ is known as *stability* [31], which is connected with the sample auto-correlation of a binary sequence, i.e. $\sum_t X_t^i X_{t-1}^i$. Similarly, when $i \neq j$ the metric is related to lagged cross-correlations. The maximum entropy probability distribution of $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_T$, i.e. the one maximizing the entropy $-\sum_{\boldsymbol{X}_1,\ldots,\boldsymbol{X}_T} p\left(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_T\right) \log p\left(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_T\right)$ while preserving on average some given values for the metrics (2) and (3), has transition probability (by assuming a given initial condition $\boldsymbol{X}_0$ and exploiting the Markov property)

$$p\left(\boldsymbol{X}_t \mid \boldsymbol{X}_{t-1}; \boldsymbol{J}, \boldsymbol{h}\right) = \prod_{i=1}^{N} \frac{\exp\left[2X_t^i\left(h_i + \sum_{j=1}^{N} J_{ij}\left(2X_{t-1}^j - 1\right)\right)\right]}{1 + \exp\left[2\left(h_i + \sum_{j=1}^{N} J_{ij}\left(2X_{t-1}^j - 1\right)\right)\right]} \tag{4}$$

where $\boldsymbol{J} = \left\{J_{ij}\right\}_{i,j=1,\ldots,N}$ and $\boldsymbol{h} = \{h_i\}_{i=1,\ldots,N}$ are $N^2 + N$ Lagrange multipliers solving the maximum entropy problem [29,32]. It is trivial to show that the transition probability of the Kinetic Ising Model (1) can be stated equivalently as the maximum entropy probability (4) in terms of the binary variables $\boldsymbol{X}_t \forall i, t$, through the relation $X_t^i = \frac{1+\sigma_t^i}{2}$ (with the same parameters $\boldsymbol{J}$ and $\boldsymbol{h}$).

The VDAR(1) model describes the dependence structure of a set of binary random variables which has Markov property and Bernoulli marginal distribution. It has been proposed originally in its univariate version [5] and followed by several extensions such as the Discrete AutoRegressive Moving Average (DARMA) model [6] and recently proposed in its multivariate formulation [7], the VDAR model. Models from this family have seen applications in [...].

In terms of the N binary variables $\boldsymbol{X}_t \equiv \left\{X_t^i\right\}_{i=1,\ldots,N}$ with $X_t^i \in \{0,1\}$ (and initial condition $\boldsymbol{X}_0$), the VDAR(1) process describes the evolution of X_t^i as

$$X_t^i = V_t^i X_{t-1}^{A_t^i} + \left(1 - V_t^i\right) Z_t^i \tag{5}$$

with $V_t^i \sim \mathcal{B}(\nu_i)$ a Bernoulli random variable with parameter $\nu_i \in [0,1]$, $A_t^i \sim \mathcal{M}(\lambda_{i1},\ldots\lambda_{iN})$ a multinomial random variable taking integer value in $\{1,\ldots,\mathrm{N}\}$, with parameters $\lambda_{i1},\ldots\lambda_{iN}$ such that $\sum_{j=1}^N \lambda_{ij} = 1$, and $Z_t^i \sim \mathcal{B}(\chi_i)$ with $\chi_i \in [0,1]$. In other words, the VDAR(1) process captures the (multivariate) mechanism of copying from the past: with probability ν_i, X_t^i is copied from the past and, in this case, λ_{ij} is the probability that X_t^i is equal to X_{t-1}^j (including also the past itself with probability λ_{ii}); otherwise, with probability $1-\nu_i, X_t^i$ is not copied and is instead sampled according to a Bernoulli marginal with probability χ_i. Hence, the VDAR(1) model describes N binary random variables with both Markov property and some autoregressive dependency structure, similarly to the KIM. The model is formalized by the transition probability

$$p_{VDAR}\left(\boldsymbol{X}_t \mid \boldsymbol{X}_{t-1};\boldsymbol{\pi}\right) = \prod_{i=1}^N \left[\nu_i\left(\sum_{j=1}^N \lambda_{ij}\delta_{X_i^i,X_{t-1}^j}\right) + (1-\nu_i)\,(\chi_i)^{X_t^i}\,(1-\chi_i)^{1-X_t^i}\right] \tag{6}$$

where $\delta_{X_i^i,X_{i-1}^j}$ is the Kronecker delta, $\boldsymbol{\pi} = \left\{\{\nu_i\},\left\{\lambda_{ij}\right\},\{\chi_i\}\right\}_{i,j=1,\ldots,N}$.

Notice that the model has N^2+N parameters, exactly as the Kinetic Ising Model. It is thus immediate to ask the question whether a mapping between the two models exists, as well as finding under which conditions the two models can be considered equivalent. In the following, we indicate the KIM model as $\{\{\boldsymbol{X}_t\}, p_{KIM}, \boldsymbol{\theta}\}$ with set of parameters $\boldsymbol{\theta} \equiv (\boldsymbol{J},\boldsymbol{h})$, while the VDAR(1) model is summarized as $\{\{\boldsymbol{X}_t\}, p_{VDAR}, \boldsymbol{\pi}\}$. Calling $\Theta = \mathbb{R}^{N\times N}\times\mathbb{R}^N$ the space of all possible KIM parameters $\boldsymbol{\theta}$ and Π the space of all possible VDAR(1) parameters π,

Definition 1 The KIM and the VDAR(1) models are said to be equivalent on $(\hat{\Theta},\hat{\Pi})$ if there exist an unique invertible map $f:\hat{\Pi}\subseteq\Pi\to\hat{\Theta}\subseteq\Theta$ such that

$$p_{KIM}\left(\boldsymbol{X}_t \mid \boldsymbol{X}_{t-1}; f(\boldsymbol{\pi})\right) = p_{VDAR}\left(\boldsymbol{X}_t \mid \boldsymbol{X}_{t-1};\boldsymbol{\pi}\right)$$

for any $\boldsymbol{X}_t$ and $\boldsymbol{X}_{t-1}$."

7 IM — time-dependent IM — HMM

Failenschmid and Marsman (2023): "The proposed time-dependent Ising model is based on Fortuin and Kasteleyn's (1972) theory of the random-cluster model, which introduces heterogeneity to the Ising model and allows for individualized network topologies. To introduce temporal dynamics, we apply a hidden Markov process that allows the idiographic topologies to transition into different states over time. This allows the model to capture individual progressions through time, in a group level change process within the hidden Markov chain. We present a candidate probability distribution for the transitions of the idiographic topologies and derive the resulting distributions for the complete Markov chain as well as the time-dependent Ising model."

8 (Markov) Random Fields

- Markov Random Field (MRF) → Time Dynamic MRF (TD-MRF)
- Hybrid MRF (HRF) → Dynamic (D-HRF)
- Bayesian Network (BN) → Dynamic BN (DBN)
- Mixed Graphical Model (MGM)
- Moderated Network Model (MNM; Haslbeck et al., 2021)
- Ordinal MRF (OMRF; Marsman et al., 2025)

R-Package: 'MRFcov', 'mrf2d'

Hernández-Lemus (2021): "The theory and applications of random fields born out of the fortunate marriage of two simple but deep lines of reasoning. On the one hand, physical intuition, strongly founded in the works of Boltzmann and the Ehrenfests, but also in other originators of the kinetic theory of matter, was that large scale, long range phenomena may originate from (a multitude of) local interactions. On the other hand, probabilistic reasoning induced us to think that such multitude of local interactions would be stochastic in nature. These two ideas, paramount to statistical mechanics, have been extensively explored and develop into a full theoretical subdiscipline, the theory of random fields. Perhaps the archetypal instance of a random field was laid out in the doctoral thesis of Ernst Ising, the Ising model of ferromagnetism [...] The formal establishment of the theory of Markov-Gibbs random fields, however, is often attributed to the works of Dobruschin, Lanford and Ruelle [8, 9], in particular to their DLR equations for the probability measures. Also remarkable is the contribution of Hammersley and Clifford, who developed a proof of the equivalence of Gibbs random fields and Markov random fields, provided positive definite probabilities [10]."

8.1 MRF — HRF — (D)BN — HMM

R-Package: 'HMM', 'statespacer', 'bnlearn'

Hybrid Random Field (HRF; Freno et al., 2009)

Freno and Trentin (2011): "Probabilistic graphical models [154, 174], where statistical correlations are jointly modeled by means of graphical structures and conditional probability distributions involving sets of random variables. [...]

The two most popular families of graphical models are reviewed in detail in the book, namely Bayesian networks and Markov random fields (Chapter 2 and Chapter 3, respectively). Since the novel hybrid random field we introduce in the following chapters builds on these traditional models, their thorough description and understanding will be necessary. Several extensions to the models were proposed in the literature. Some of the most significant will be reviewed concisely as well, including dynamic Bayesian networks and hidden Markov models (which extend the basic BN model to time series), conditional random fields (which are a discriminative version of Markov random fields, also suitable for sequence modeling), and some families of neural networks (Hopfield networks and Boltzmann machines, thought of as particular cases of MRFs). Dependency networks [128] and graphical chain models (also known as chain graphs) [191, 314] are also importantly related to hybrid random fields. [...]

Dynamic Bayesian networks (DBN) are a major instance of Bayesian networks extended to sequences [110]. From a general standpoint, DBNs are simply BNs where time-indexed variables $X_0, ..., X_T$ are treated (and, represented within the graphical structure) as separate variables (i.e., as individual vertexes in the graph), and the edges—being of causal nature—follow the natural time order (e.g., a directed edge exists between X_t and X_{t+1}). The model may also take into account hidden state variables, referring to an underlying, non-observable random process which evolves over time and that is responsible for the dynamics of the observable state variables (either discrete, or continuous). Although DBNs are a very general, unifying framework for the probabilistic representation of time series, some problems arise at the algorithmic level when it comes down to inference and learning. [...] It is a fact of the utmost interest that several, popular paradigms can be interpreted as particular (yet, simple) instances of DBNs, namely Kalman filters [158, 264], hidden Markov models [251] [...]

Hidden Markov Model (HMM) — DBN A hidden Markov model is a pair of stochastic processes: a *hidden* Markov chain and an *observable* process which is a probabilistic function of the states of the former. This means that observable events in the real world (e.g., the amino-acids along a protein primary structure) are modeled with (possibly continuous) probability distributions, that are the observable part of the model, associated with individual states of a discrete-time, first-order Markov process. The semantics of the model (conceptual correspondencewith physical phenomena) is usually encapsulated in the hidden part. [...] More precisely, a hidden Markov model is defined by:

1. A set S of Q states, $S = \{S_1, \ldots, S_Q\}$, which are the distinct values that the discrete, hidden stochastic process can take;

2. An *initial state* probability distribution, i.e. $\pi = \{P(S_i \mid t = 0), S_i \in S\}$, where t is a discrete time index;

3. A probability distribution that characterizes the allowed transitions between states, that is $\boldsymbol{a} = \left\{P\left(S_j \text{ at time } t \mid S_i \text{ at time } t-1\right), S_i \in S, S_j \in S\right\}$ where the *transition probabilities* a_{ij} are assumed to be independent of time t;

4. An *observation* or *feature* space F, which is a discrete or continuous universe of all possible observable events (usually a subset of $\mathbb{R}^d$, where d is the dimensionality of the observations);

5. A set of probability distributions (referred to as emission or output probabilities) that describes the statistical properties of the observations for each state of the model: $\boldsymbol{b_x} = \{b_i(\boldsymbol{x}) = P(\boldsymbol{x} \mid S_i), S_i \in S, \boldsymbol{x} \in F\}$.

[...] The interpretation of HMMs as dynamic Bayesian networks goes, roughly, as follows. Nodes are inserted in the graph for each hidden state q_t at time t, and for the emission probabilities associated with these states. Causal, directed edges are inserted between each pair of adjacent states (meaning that the state at time $t+1$ is independent of all non-descendant states given the state at time t, due to the Markov assumption), and between each state and the corresponding emission (the latter being a function of the former, independently from the specific time t as well as from any other state in the model). The distribution of initial and transition probabilities, and the emission distributions as well, provide us with the quantities needed in order to fill the CPTs (depending only on the parents of a given node in the graph). [...]

Markov Random Field (MRF) Markov random fields [121, 167, 282] are used to represent joint probability distributions underlying sets of random variables. A Markov random field is composed of an undirected graph and a set of potential functions. Each node in the graph represents a random variable, and for each maximal clique C in the graph we have a corresponding potential function φ_C. Each potential function φ_C takes as argument the state of clique C in the graph and returns as value a non-negative real number. The state of a clique C in a Markov random field is nothing but a specific realization of the variables in C, i.e. an event described by a specific configuration of the values of those variables. [...] Consider a set $\boldsymbol{X}$ of random variables $X_1, \dots, X_n$, a graph $\mathcal{G}$ whose nodes are the variables in $\boldsymbol{X}$, and the set γ of all maximal cliques in $\mathcal{G}$. Given the potential functions for the cliques in γ, the joint probability distribution of $\boldsymbol{X}$ is computed as follows:

$$P(\boldsymbol{x}) = \frac{1}{Z} \prod_{C \in \gamma} \varphi_C(\boldsymbol{x}_C) \tag{3.3}$$

where Z is a normalization factor called 'partition function' and $\boldsymbol{x}_C$ is the state of clique C as this state is determined by $\boldsymbol{x}$. If $\mathfrak{X}$ is the set of all possible configurations of the values of the variables in $\boldsymbol{X}$, the partition function is given by:

$$Z = \sum_{\boldsymbol{x} \in \mathfrak{X}} \prod_{C \in \gamma} \varphi_C(\boldsymbol{x}_C) \tag{3.4}$$

Since the cardinality of $\mathfrak{X}$ is given by $|\mathfrak{X}| = \prod_{i=1}^{n} |\mathcal{D}_i|$, where $\mathcal{D}_i$ denotes the domain of variable X_i, it is clear that computing Z requires to sum over a number of states that grows exponentially with the number of variables in $\boldsymbol{X}$. Therefore, if we stick to the probability function specified in Equation 3.3, estimating the likelihood of a state $\boldsymbol{x}$ for a given Markov random field remains intractable. A solution to this problem, first suggested in [25], will be detailed in Section 3.2.2.

The potential functions are represented as exponential functions. This choice involves exploiting the notion of *feature function*. For each state $\boldsymbol{x}_C$ of each maximal clique C, we introduce a feature function $f_{\boldsymbol{x}_C}$ such that:

$$f_{\boldsymbol{x}_C}(\boldsymbol{X}_C) = \begin{cases} 1 \text{ if } \boldsymbol{X}_C = \boldsymbol{x}_C \\ 0 \text{ otherwise} \end{cases} \tag{3.5}$$

In other words, a feature function returns 1 if the respective clique is in the state corresponding to that feature function, while it returns 0 if the clique is in any other state. To each feature function f_j we attach a real-valued weight w_j. Given the feature functions and their weights, each potential function φ_e takes the following form:

$$\varphi_C(\boldsymbol{x}_C) = \exp\left(\sum_{j=1}^{m_C} w_j f_j(\boldsymbol{x}_C)\right) \tag{3.6}$$

where m_C is the number of features defined for clique C. This leads to the following representation of the joint probability distribution:

$$\begin{aligned} P(\boldsymbol{x}) &= \frac{1}{Z}\prod_{C\in\gamma}\exp\left(\sum_{j=1}^{m_C} w_j f_j(\boldsymbol{x}_C)\right) \\ &= \frac{1}{Z}\exp\left(\sum_{C\in\gamma}\sum_{j=1}^{m_C} w_j f_j(\boldsymbol{x}_C)\right) \end{aligned} \tag{3.7}$$

Concerning the weights, they can be learned from data so as to maximize the probability of the data given the Markov random field. This problem will be addressed in section 3.3.

Hammersley-Clifford Theorem The kind of probability distribution specified in Equation 3.7 is usually referred to as a *Gibbs* (or *Boltzmann*) distribution. At this point, the reader may well wonder why we should choose to model joint probabilities by means of Gibbs distributions, and in particular why we should choose to factorize our Gibbs distribution based on the maximal cliques of a certain (undirected) graph. The key to understanding the connection between the Gibbs distribution and the graphical structure of Markov random fields is provided by the Hammersley-Clifford theorem [24].

Before stating the theorem, let us first define the notion of *(undirected) separation*:

Definition 3.1. Consider an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. If $\mathcal{A}, \mathcal{B}$, and $\mathcal{S}$ are three disjoint subsets of $\mathcal{V}$, then $\mathcal{S}$ separates $\mathcal{A}$ from $\mathcal{B}$ if, for any node X_i in $\mathcal{A}$ and any node X_j in $\mathfrak{B}$, every possible path connecting X_i to X_j contains at least a node X_k such that $X_k \in \mathcal{S}$.

Given the concept of separation, the Hammersley-Clifford theorem can be stated as follows:

Theorem 3.1. Given a random vector $\boldsymbol{X} = (X_1, \ldots, X_n)$, if $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is an undirected graph such that $\mathcal{V} = \{X_1, \ldots, X_n\}$ and P is a probability distribution over $\boldsymbol{X}$ such that $P(\boldsymbol{x}) > 0$ for any realization $\boldsymbol{x}$ of $\boldsymbol{X}$, then the following conditions are equivalent:

1. $P(\boldsymbol{X})$ is a Gibbs distribution which factorizes according to the maximal cliques in $\mathcal{G}$;

2. If $\mathcal{N}(X_i)$ is the set of neighbors of X_i in $\mathcal{G}$, then $P(X_i \mid \boldsymbol{X} \setminus \{X_i\}) = P(X_i \mid \mathcal{N}(X_i))$ (local Markov property);

3. If $\mathcal{A}, \mathcal{B}$, and $\mathcal{S}$ are three disjoint subsets of $\boldsymbol{X}$, and $\mathcal{S}$ separates $\mathcal{A}$ from $\mathcal{B}$, then $P(\mathcal{A} \mid \mathcal{B} \cup \mathcal{S}) = P(\mathcal{A} \mid \mathcal{S})$ (global Markov property).

Proof. See e.g. [24].

In other words, given a certain domain of random variables, the Hammersley-Clifford theorem entails that assuming either that the domain can be modeled by a Gibbs distribution or that the domain satisfies the local or the global Markov property means assuming one and the same thing. Therefore, using a Markov random field to model a certain probability distribution is justified exactly when any one of the three conditions specified in the theorem is satisfied by the distribution at hand. That is to say, when we use Markov random fields we are in fact making any one of the three assumptions proved to be equivalent by Hammersley and Clifford.

The positivity condition on the probability distribution, i.e. the requirement that $P(\boldsymbol{x}) > 0$, is strictly necessary in order for the theorem to hold. This condition played a singular role in the history of the Hammersley-Clifford theorem. The theorem was first proved by John Hammersley and Peter Clifford in 1971, assuming the positivity condition. However, the authors of the proof felt that this assumption was an undesired feature of their result. The drawback of the positivity condition is that it does not allow to apply Markov random fields to any domain featuring forbidden states, i.e. to any system such that some constraints prevent it from assuming certain states. This limitation ruled out the possibility of modeling some problems in statistical mechanics through Markov random fields. For this reason, Hammersley and Clifford postponed the publication of the result, in the hope of proving the theorem without assuming the positivity condition. In the following years, Julian Besag worked out a more elegant proof of the theorem than Hammersley and Clifford's original proof, which was revealed to be unduly complex, and Besag's result was published [24]. Of course, the positivity condition was assumed by Besag too in his proof of the theorem. Finally, the last chapter in the history of the Hammersley-Clifford theorem was written by one of Hammersley's graduate students at Oxford, namely John Moussouris, who showed (by means of a counter-example) that the positivity of the probability distribution was a strictly necessary condition of the theorem [221]. [...]

Hybrid Markov Random Field (HRF) [...] the model we are going to present is aimed at overcoming the limitations of both Bayesian networks and Markov random fields in terms of the computational cost of estimating the model from data. To this aim, the basic strategy behind hybrid random field estimation is to exploit on the one hand the ease of learning Bayesian networks for relatively small domains, and on the other hand the factorization of joint distributions employed in Markov random fields in the form of the pseudo-likelihood measure. In particular, hybrid random fields model the conditional distribution of each variable X_i given its Markov blanket in $\boldsymbol{X}$ through a *local* Bayesian network, i.e. a Bayesian network containing only a suitable subset of variables from $\boldsymbol{X}$. [...]

Just like Bayesian networks and Markov random fields, hybrid random fields are aimed at representing joint probability distributions underlying sets of random variables. In order to define the concept of hybrid random field, we need to introduce a few preliminary notions concerning directed and undirected graphs. First, let us define the concept of *(directed) union* of directed graphs:

Definition 4.1. If $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1)$ and $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2)$ are directed graphs, then the (directed) union of $\mathcal{G}_1$ and $\mathcal{G}_2$ (denoted by $\mathcal{G}_1 \cup \mathcal{G}_2$) is the directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ such that $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ and $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2$.

Clearly, for a set of directed graphs $\mathcal{G}_1, \ldots, \mathcal{G}_n$, the directed union $\mathcal{G} = \bigcup_{i=1}^{n} \mathcal{G}_i$ simply results from iterated application of the binary union operator. Similarly, we define the notion of *(undirected) union* for undirected graphs:

Definition 4.2. If $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1)$ and $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2)$ are undirected graphs, then the (undirected) union of $\mathcal{G}_1$ and $\mathcal{G}_2$ (denoted by $\mathcal{G}_1 \cup \mathcal{G}_2$) is the undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ such that $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ and $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2$.

Now, if $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a directed graph, then we say that $\mathcal{G}^* = (\mathcal{V}^*, \mathcal{E}^*)$ is the undirected version of $\mathcal{G}$ if $\mathcal{V}^* = \mathcal{V}$ and $\mathcal{E}^* = \left\{\left\{X_i, X_j\right\} : \left(X_i, X_j\right) \in \mathcal{E}\right\}$. Thus, we also define the undirected union for a pair of directed graphs:

Definition 4.3. If $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1)$ and $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2)$ are directed graphs, then the undirected union of $\mathcal{G}_1$ and $\mathcal{G}_2$ (denoted by $\mathcal{G}_1 \Cup \mathcal{G}_2$) is the (undirected) graph $\mathcal{G}^* = \mathcal{G}_1^* \cup \mathcal{G}_2^*$ such that $\mathcal{G}_1^*$ and $\mathcal{G}_2^*$ are the undirected versions of $\mathcal{G}_1$ and $\mathcal{G}_2$ respectively.

Given the concepts introduced above, a hybrid random field can be defined as follows:

Definition 4.4. Let $\boldsymbol{X}$ be a set of random variables $X_1, \ldots, X_n$. A *hybrid random field* for $X_1, \ldots, X_n$ is a set of Bayesian networks $BN_1, \ldots, BN_n$ (with DAGs $\mathcal{G}_1, \ldots, \mathcal{G}_n$) such that:

1. Each BN_i contains X_i plus a subset $\mathcal{R}(X_i)$ of $\boldsymbol{X} \setminus \{X_i\}$;

2. If $\mathcal{MB}_i(X_i)$ denotes the Markov blanket of X_i in BN_i (i.e. the set containing the parents, the children, and the parents of the children of X_i in $\mathcal{G}_i$) and $P(X_i \mid \mathcal{MB}_i(X_i))$ is the conditional distribution of X_i given $\mathcal{MB}_i(X_i)$, as derivable from BN_i, then at least one of the following conditions is satisfied:

   a. The directed graph $\mathcal{G} = \bigcup_{i=1}^{n} \mathcal{G}_i$ is acyclic and there is a Bayesian network $h_{\mathcal{G}}$ with DAG $\mathcal{G}$ such that, for each X_i in $\boldsymbol{X}$, $P(X_i \mid \mathcal{MB}_i(X_i)) = P(X_i \mid \mathcal{MB}_{\mathcal{G}}(X_i))$, where $\mathcal{MB}_{\mathcal{G}}(X_i)$ is the Markov blanket of X_i in $h_{\mathcal{G}}$ and $P(X_i \mid \mathcal{MB}_{\mathcal{G}}(X_i))$ is the conditional distribution of X_i given $\mathcal{MB}_{\mathcal{G}}(X_i)$, as entailed by $h_{\mathcal{G}}$;

   b. There is a Markov random field $h_{\mathcal{G}^*}$ with graph $\mathcal{G}^*$ such that $\mathcal{G}^* = \mathcal{G}_1 \Cup \ldots \Cup \mathcal{G}_n$ and, for each X_i in $\boldsymbol{X}$, $P(X_i \mid \mathcal{MB}_i(X_i)) = P(X_i \mid \mathcal{MB}_{\mathcal{G}^*}(X_i))$, where $\mathcal{MB}_{\mathcal{G}^*}(X_i)$ is the Markov blanket of X_i in $h_{\mathcal{G}^*}$ and $P(X_i \mid \mathcal{MB}_{\mathcal{G}^*}(X_i))$ is the conditional distribution of X_i given $\mathcal{MB}_{\mathcal{G}^*}(X_i)$, as derived from $h_{\mathcal{G}^*}$.

The elements of $\mathcal{R}(X_i)$ are called 'relatives of X_i '. That is, the relatives of a node X_i in a hybrid random field are the nodes appearing in graph $\mathcal{G}_i$ (except for X_i itself). An illustration of a hybrid random field is provided in Figure 4.1.

The crucial issue concerning Definition 4.4 can be stated by means of two (tightly correlated) questions:

1. Is it possible to extract a joint probability distribution from a hybrid random field?

2. Is it possible to characterize a set of conditional independence statements entailed by a hybrid random field?

Both questions are given a positive answer by the following theorem:

Theorem 4.1. Suppose that h is a hybrid random field for the random vector $\boldsymbol{X} = (X_1, \ldots, X_n)$, and let h be made up by Bayesian networks $BN_1, \ldots, BN_n$ (with DAGs $\mathcal{G}_1, \ldots, \mathcal{G}_n$). Then, if each conditional distribution $P(X_i \mid \mathcal{MB}_i(X_i))$ is strictly positive (where $1 \leq i \leq n$), h has the following properties:

1. An ordered Gibbs sampler applied to h, i.e. an ordered Gibbs sampler which is applied to the conditional distributions $P(X_1 \mid \mathcal{MB}_1(X_1)), \ldots, P(X_n \mid \mathcal{MB}_n(X_n))$, defines a joint probability distribution over $\boldsymbol{X}$ via its (unique) stationary distribution $P(\boldsymbol{X})$;

2. For each variable X_i in $\boldsymbol{X}, P(\boldsymbol{X})$ is such that

   $P(X_i \mid \boldsymbol{X} \setminus \{X_i\}) = P(X_i \mid \mathcal{MB}_i(X_i))$.

[...]"

8.1.1 HRF — D-HRF

Bongini et al. (2018): "The paper introduces a dynamic extension of the hybrid random field (HRF), called dynamic HRF (D-HRF). The D-HRF is aimed at the probabilistic graphical modeling of arbitrary-length sequences of sets of (time-dependent) discrete random variables under Markov assumptions. Suitable maximum likelihood algorithms for learning the parameters and the structure of the D-HRF are presented. [...]

Definition a dynamic HRF $\mathcal{DH}$ is a tuple $\mathcal{DH} = (\mathcal{X}, S, \pi, \mathcal{F}, \mathcal{A}, \mathscr{H})$ where

1. $\mathcal{X}$ is a set of n observable random variables, $\mathcal{X} = \{X_1, \ldots, X_n\}$, having discrete definition domains. We will write $x_{i,t}$ to represent the specific outcome of the variable X_i at time t, for $t = 1, 2, \ldots$, and $i = 1, \ldots, n$.

2. S is a set of Q discrete, latent states of the system, namely $S = \{S_1, \ldots, S_Q\}$. It is assumed that sequences of such states are responsible for the generation of sequences of outcomes of the observable variables, and that the elements of S can be thought of as the states of a discrete-time, first-order Markov chain (latent Markov assumption). We write q_t to denote the state of the system at time t.

3. π represents the initial probability distribution of the latent states, i.e. $\pi = \{P(S_i \mid t = 1) : S_i \in S\}$, where t is the discrete time index. The notation π_i is used for representing the quantity $P(S_i \mid t = 1)$ according to π, for $i = 1, \ldots, Q$. For instance, if a certain state S_j can never occur at time $t = 1$, then $\pi_j = 0$.

Contrariwise, if the Markov chain over S may equally-likely start with any state, then π is uniform over S.

4. $\mathcal{F} \subseteq S$ is the set of final states, i.e. the states which can legitimately generate sets of outcomes of the observable variables at the end of the sequences.

5. $\mathcal{A} = \left[a_{ij}\right]$ is a transition matrix, expressing the probability distribution that characterizes the (allowed) transitions between states, that is $a_{ij} = P\big(S_j$ at time $t+1 \mid S_i$ at time $t)$ for all $S_i, S_j \in S$. The transition probabilities a_{ij} are assumed to be independent of time t. Note that the definition is meaningful due to the latent Markov assumption.

6. $\mathscr{H}$ is a set of HRFs over $\mathcal{X}$, $\mathscr{H} = \{\mathcal{H}_1, \ldots, \mathcal{H}_Q\}$, where $\mathcal{H}_q$ is uniquely associated to q-th state S_q such that the joint emission probability $b_q(\mathcal{X}) = P\left(X_1, \ldots, X_n \mid S_q\right)$ is modeled via HRF $\mathcal{H}_q$ over $\mathcal{X}$, independently of time t. We assume that, given the state S_q, the outcomes of the variables in $\mathcal{X}$ at time t are independent of the outcomes of the variables in $\mathcal{X}$ at time t' for all $t' \neq t$ (emission Markov assumption). Bearing in mind the definition of HRF and the modularity property of HRFs (see Appendix A), it is seen that $\mathcal{H}_q$ involves a set of Bayesian networks $BN_{q,1}, \ldots, BN_{q,n}$ (with directed acyclic graphs $\mathcal{G}_{q,1}, \ldots, \mathcal{G}_{q,n}$) such that:

   (a) each $BN_{q,i}$ contains X_i plus a subset $\mathcal{R}_q(X_i)$ of $\mathcal{X} \setminus \{X_i\}$, namely the set of relatives of X_i in $BN_{q,i}$;

   (b) for each X_i, $P\left(X_i \mid S_q, \mathcal{X} \setminus \{X_i\}\right) = P\left(X_i \mid \mathcal{MB}_{q,i}(X_i)\right)$, where $\mathcal{MB}_{q,i}(X_i)$ is the set containing the parents, the children, and the parents of the children of X_i in $\mathcal{G}_{q,i}$ (namely, the Markov blanket [36] of X_i in $BN_{q,i}$).

   Note that since $\mathcal{H}_q$ is time-independent (for $q = 1, \ldots, Q$) then the corresponding BNs $BN_{q,1}, \ldots, BN_{q,n}$ do not change over time The Markov assumption holding for $\mathcal{H}_1, \ldots, \mathcal{H}_Q$ is referred to as the *observable Markov assumption* in the present framework.

Note that the overall D-HRF could be thought of as a probabilistic graphical model over the set of random variables $S \cup \mathcal{X}$, once the elements of S were thought of as time-dependent Boolean random variables (where $S_i = 1$ if the system is currently in the corresponding state, otherwise $S_i = 0$). Nonetheless, this definition allows for separate sets of BNs (i.e. different HRFs) for each state, meaning that it does not extend regular HRFs to sequences by defining them as sets of dynamic Bayesian networks. The definition given is flexible, and it allows us to exploit efficient algorithmic tools (e.g. the usual recursive calculation scheme on the "trellis" [37]) found in traditional HMMs. In fact, from the definition it is immediately seen that HMMs are a special case of D-HRFs (the general modeling capabilities of D-HRFs are surveyed in the next Section). In particular, the Viterbi algorithm [37] can be applied to D-HRFs to perform sequence recognition and segmentation tasks right away, as long as all the quantities in ($\mathcal{X}, S, \pi, \mathcal{F}, \mathcal{A}, \mathscr{H}$) are known. Therefore, learning algorithms capable of estimating these quantities from data samples of training sequences are required. [...]"

8.2 MRF — TD-MRF

Cabello (2022): "Time-dynamic (or time-variying) Markov Random Fields (TD-MRFs) are the plain extension of MRFs for dynamic scenarios (see [19]). These are a special sort of dynamic graphical models (DGMs) which explicitly model the correlations in space and in time as dependencies amongst the random variables such that the corresponding joint distribution function can be written by means of functions of the corresponding time-varying random variable over the cliques $C \in \mathcal{C}$:

$$\mathbb{P}[X] = \frac{1}{Z} \exp\left[-\sum_{C \in \mathcal{C}} f\left(X_C^t\right)\right] \text{where t represents the time step."} \tag{1}$$

8.3 IM — MRF — GGM

Murphy (2012): "The Ising model is an example of an MRF that arose from statistical physics. It was originally used for modeling the behavior of magnets. In particular, let $y_s \in \{-1, +1\}$ represent the spin of an atom, which can either be spin down or up. In some magnets, called ferro-magnets, neighboring spins tend to line up in the same direction, whereas in other kinds of magnets, called anti-ferromagnets, the spins "want" to be different from their neighbors.

We can model this as an MRF as follows. We create a graph in the form of a 2D or 3D lattice, and connect neighboring variables, as in Figure 19.1(b). We then define the following pairwise clique potential:

$$\psi_{st}(y_s, y_t) = \begin{pmatrix} e^{w_{st}} & e^{-w_{st}} \\ e^{-w_{st}} & e^{w_{st}} \end{pmatrix} \tag{19.17}$$

Here w_{st} is the coupling strength between nodes s and t. If two nodes are not connected in the graph, we set $w_{st} = 0$. We assume that the weight matrix $\boldsymbol{W}$ is symmetric, so $w_{st} = w_{ts}$. Often we assume all edges have the same strength, so $w_{st} = J$ (assuming $w_{st} \neq 0$).

If all the weights are positive, $J > 0$, then neighboring spins are likely to be in the same state; this can be used to model ferromagnets, and is an example of an associative Markov network. If the weights are sufficiently strong, the corresponding probability distribution will have two modes, corresponding to the all +1 's state and the all -l's state. These are called the ground states of the system.

If all of the weights are negative, $J < 0$, then the spins want to be different from their neighbors; this can be used to model an anti-ferromagnet, and results in a frustrated system, in which not all the constraints can be satisfied at the same time. The corresponding probability distribution will have multiple modes. [...]

There is an interesting analogy between Ising models and Gaussian graphical models. First, assuming $y_t \in \{-1, +1\}$, we can write the unnormalized log probability of an Ising model as follows:

$$\log \tilde{p}(\boldsymbol{y}) = -\sum_{s \sim t} y_s w_{st} y_t = -\frac{1}{2} \boldsymbol{y}^T \boldsymbol{W} \boldsymbol{y} \tag{19.18}$$

(The factor of $\frac{1}{2}$ arises because we sum each edge twice.) If $w_{st} = J > 0$, we get a low energy (and hence high probability) if neighboring states agree.

Sometimes there is an external field, which is an energy term which is added to each spin. This can be modelled using a local energy term of the form $-\boldsymbol{b}^T \boldsymbol{y}$, where $\boldsymbol{b}$ is sometimes called a bias term. The modified distribution is given by

$$\log \tilde{p}(\boldsymbol{y}) = \sum_{s \sim t} w_{st} y_s y_t + \sum_s b_s y_s = \frac{1}{2} \boldsymbol{y}^T \boldsymbol{W} \boldsymbol{y} + \boldsymbol{b}^T \boldsymbol{y} \tag{19.19}$$

where $\boldsymbol{\theta} = (\boldsymbol{W}, \boldsymbol{b})$. If we define $\boldsymbol{\mu} \triangleq -\frac{1}{2}\boldsymbol{\Sigma}^{-1}\boldsymbol{b}, \boldsymbol{\Sigma}^{-1} = -\boldsymbol{W}$, and $c \triangleq \frac{1}{2}\boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, we can rewrite this in a form that looks similar to a Gaussian:

$$\tilde{p}(\boldsymbol{y}) \propto \exp\left(-\frac{1}{2}(\boldsymbol{y} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{y} - \boldsymbol{\mu}) + c\right) \tag{19.20}$$

One very important difference is that, in the case of Gaussians, the normalization constant, $Z = |2\pi\boldsymbol{\Sigma}|$, requires the computation of a matrix determinant, which can be computed in $O\left(D^3\right)$ time, whereas in the case of the Ising model, the normalization constant requires summing over all 2^D bit vectors; this is equivalent to computing the matrix permanent, which is NP-hard in general (Jerrum et al. 2004).

[...] A Hopfield network (Hopfield 1982) is a fully connected Ising model with a symmetric weight matrix, $\boldsymbol{W} = \boldsymbol{W}^T$."

Masuda et al. (2025): "Gaussian graphical models assume normality. One approach to relax this assumption is to use a so-called nonparanormal distribution [231]. The idea is to assume that the transformed random variables $(f_1(X_1), \ldots, f_N(X_N))$ obey a multivariate normal distribution, where $f_1, \ldots, f_N$ are monotone and differentiable functions. In this case, we say that the original variables $(X_1, \ldots, X_N)$ have a nonparanormal distribution or a Gaussian copula (given by $f_1, \ldots, f_N$, and the mean vector and the covariance matrix of the normal distribution after the transformation). The nonparanormal distribution is nonidentifiable. For example, if we replace $f_1(X_1)$ by a new function $\tilde{f}_1(X_1) \equiv 2f_1(X_1)$ and appropriately scale the first row and column of the covariance matrix and the first entry of mean vector used for the multivariate normal distribution that the transformed N variables obey, then one gets the same distribution of $(X_1, \ldots, X_N)$. Therefore, we further impose that the transformations $f_1, \ldots, f_N$ conserve the mean and variance of the original $X_1, \ldots, X_N$. Then, as in the case of Gaussian graphical models, $\Omega_{ij} = 0$ implies that X_i and X_j are conditionally independent of each other given all the other $N - 2$ variables, in addition to that $f(X_i)$ and $f\left(X_j\right)$ are conditionally independent of each other. There are methods to estimate from data the functions $f_1, \ldots, f_N$ as well as a sparse covariance matrix Ω, assuming a lasso penalty. Naturally, the nonparanormal distribution outperforms the graphical lasso when the true distribution of the data is not multivaraite normal [231]. Later work proposed algorithms for estimating the nonparanormal distribution with optimal convergence rate [232,233]; the idea behind the improved algorithms is to avoid explicitly calculating

$f_1, \ldots, f_N$ and to exploit statistics of rank correlation coefficient estimators. There are also other methods for estimating non-normal graphical models without relying on copula [234, 235].

Extreme cases of non-Gaussian distribution of the random variables are discrete multivariate distributions, in particular when each X_i is binary (i.e., $\in \{-1, 1\}$.). In this case, the form of the distribution of $(X_1, \ldots, X_N)$ proportional to $e^{\boldsymbol{x}\Omega\boldsymbol{x}^\top}$, where $\boldsymbol{x} = (X_1 - \mu_1, \ldots, X_N - \mu_N)$, defines the Ising model. Note that the same form of distribution for continuous variables is the multivariate normal distribution, which we discussed above for graphical lasso. There are many methods for inferring the Ising model from observed multivariate binary data, which is the task referred to as the inverse Ising model and Boltzmann machine learning [236-240]. Although exact likelihood maximization is available, it is practical only for small N. Therefore, various methods aim to overcome this limitation by allowing some approximation. Similar to graphical models, inference algorithms for the Ising model respecting the sparsity of the underlying network have also been investigated. For example, ℓ_1-regularized methods based on pseudo-likelihood maximization were developed for estimating a sparse Ising Markov random field, or a graph in which the absence of the edge signifies the conditional independence between two nodes [241-243]. Decimation, or to recursively set the small edge weights to zero, combined with a stopping criterion and other heuristics, improves the performance of pseudo-likelihood maximization [244]."

8.4 MRF — MGM — MNM — GGM

R-Package: 'mgm'

Haslbeck et al. (2021): "Pairwise network models such as the Gaussian Graphical Model (GGM) are a powerful and intuitive way to analyze dependencies in multivariate data. A key assumption of the GGM is that each pairwise interaction is independent of the values of all other variables. However, in psychological research, this is often implausible. In this article, we extend the GGM by allowing each pairwise interaction between two variables to be moderated by (a subset of) all other variables in the model, and thereby introduce a Moderated Network Model (MNM). [...]

A graphical (or network) model is a statistical model for which an undirected graph/network encodes the conditional dependence structure between random variables (Lauritzen, 1996). A popular network model for continuous data is the multivariate Gaussian distribution. We introduce this distribution here, because we will use it as a basis for constructing MNMs for continuous data in the following section:

$$P(X = x) = \frac{1}{\sqrt{(2\pi)^p|\Sigma|}} \exp\left\{-\frac{1}{2}(x-\mu)^\top\Sigma^{-1}(x-\mu)\right\}, \tag{6}$$

where x is p-dimensional vector of random variables, μ is a p-dimensional vector of means, Σ is a $p \times p$ variance-covariance matrix, and $|\Sigma|$ denotes the determinant of Σ.

In the case of the multivariate Gaussian distribution, it is easy to obtain the graph/network that encodes the conditional dependence structure between random variables: if an entry in the inverse variancecovariance matrix Σ^{-1} is nonzero, the two corresponding variables are conditionally dependent (present edge in the network); if the entry is zero, the two corresponding variables are conditionally independent (no edge in the network). The resulting conditional (in-)dependence network is also called Gaussian Graphical Model (GGM). [...]

Moderated Network Model (MNM) The central goal of this article is to construct a joint distribution over p continuous variables which allows each pairwise interaction between variables X_i and X_j to be a linear function of all other variables. Another way of saying this is that each pairwise interaction between X_i and X_j is linearly moderated by all other variables. Here we construct such a joint distribution by adding moderation effects to the multivariate Gaussian distribution, which models (unmoderated) linear pairwise interactions between variables.

The density of the multivariate Gaussian distribution (6) shown in the previous section can be rewritten in its exponential family form as

$$P(X) = \exp\left\{\sum_{i}^{p} \alpha_i \frac{X_i}{\sigma_i} + \sum_{\substack{i,j \in V \\ 1 \neq j}} \beta_{i,j} \frac{X_i}{\sigma_i}\frac{X_j}{\sigma_j} + \sum_{i}^{p} \frac{X_i^2}{\sigma_i^2} - \Phi(\alpha, \beta)\right\} \tag{7}$$

where $X \in \mathbb{R}^p$, $V = \{1, 2, \ldots, p\}$ is the index set for the p variables, α is a p-vector of intercepts, β is a $p \times p$ matrix of $\binom{p}{2}$ partial correlations, σ_i^2 is the variance of X_i, and $\Phi(\alpha, \beta)$ is the log normalizing constant which ensures that the probability distribution integrates to 1 .

To see how the common form of the Gaussian density in Equation (6) can be written as Equation (7) above, momentarily assume that all means are equal to zero $\mu = 0$. Then the term in the exponential simplifies to $-\frac{1}{2}x^\top \Sigma^{-1} x$. This inner product is the same as the sum $\sum_{i \neq j^{j,j \in V}} \beta_{i,j} \frac{X_i}{\sigma_i}\frac{X_j}{\sigma_j}$ in (7), except that we summed over each (i, j) combination twice, which is why we multiply with $\frac{1}{2}$. With means unequal zero, one can expand the expression in the exponential and get a similar expression for the interactions, plus an expression for the intercepts α, which are a function of the means and the interaction parameters.

Now, to extend the Gaussian distribution in (7) with all possible moderation effects, we add all 3-way interactions to the model:

$$\begin{aligned} P(X) = \exp\Bigg\{ & \sum_{i}^{p} \alpha_i \frac{X_i}{\sigma_i} + \sum_{\substack{i,j \in V \\ i \neq j}} \beta_{i,j} \frac{X_i}{\sigma_i}\frac{X_j}{\sigma_j} \\ & + \sum_{\substack{i,j,q \in V \\ i \neq j \neq q}} \omega_{i,j,q} \frac{X_i}{\sigma_i}\frac{X_j}{\sigma_j}\frac{X_q}{\sigma_q} + \sum_{i}^{p} \frac{X_i^2}{\sigma_i^2} - \Phi(\alpha, \beta, \omega)\Bigg\}, \end{aligned} \tag{8}$$

where ω is a $p \times p \times p$ array of $\binom{p}{3}$ 3-way interactions.

How many parameters are introduced by adding 3-way interactions? For $p = 10$ variables, adding all 3-way interactions means that the number of interaction parameters increases from $\binom{10}{2} = 45$ to $\binom{10}{2} + \binom{10}{3} = 45 + 120 = 165$. Instead of adding all moderation effects (3-way interactions) one can also add single moderation effects, or all moderation effects of a subset of variables. For instance, adding all moderation effects of $M \in \{1, 2, \ldots, p\}$ moderators would result in $\sum_{m=1}^{M} \sum_{i=1}^{\min\{0,p-1-m\}} i$ additional moderation parameters.

The distribution of X_s conditioned on all remaining variables $X_{\setminus s}$ is given by

$$P(X_s \mid X_{\setminus s}) = \exp\left\{ \alpha_s \frac{X_s}{\sigma_s} + \sum_{\substack{i \in \mathbb{V} \\ i \neq s}} \beta_{i,j} \frac{x_i}{\sigma_i} \frac{X_s}{\sigma_s} + \sum_{\substack{i,j \in \mathbb{V} \\ i \neq j \neq s}} \omega_{i,j,s} \frac{x_i}{\sigma_i} \frac{x_j}{\sigma_j} \frac{X_s}{\sigma_s} + \frac{X_s^2}{\sigma_s^2} - \Phi^*(\alpha, \beta, \omega) \right\}, \tag{9}$$

where $\Phi^*(\alpha, \beta, \omega)$ is the log-normalizing constant with respect to the conditional distribution $P(X_s \mid X_{\setminus s})$, and we use lower case letters x_i to indicate fixed values opposed to random variables X_i. We now show that (9) is a conditional Gaussian distribution. To make the presentation more clear, we set $\sigma_s = 1$ without loss of generality. If we let

$$\mu_s = \alpha_s + \sum_{\substack{i \in V \\ i \neq s}} \beta_{i,s} x_i + \sum_{\substack{i,j \in V \\ i \neq j \neq s}} \omega_{i,j,s} x_i x_j \tag{10}$$

and

$$\Phi^*(\alpha, \beta, \omega) = \frac{\mu_s}{2} - \log\left(\frac{1}{\sqrt{2\pi}}\right)$$

we can rewrite (9) into

$$\begin{aligned} P(X_s \mid X_{\setminus s}) &= \frac{1}{\sqrt{2\pi}} \exp\left\{ \mu_s X_s - \frac{X_s}{2} - \frac{\mu_s}{2} \right\} \\ &= \frac{1}{\sqrt{2\pi}} \exp\left\{ \frac{(X_s - \mu_s)^2}{2} \right\}, \end{aligned} \tag{11}$$

which is the well-known form of the conditional Gaussian distribution. In Appendix B, we provide more intuition for the MNM by deriving the joint distribution in (8) from the conditionals as in (11) for the case of $p = 3$ variables. [...]

To fit the MNM, we use the R-package mgm, which implements functions to estimate k-order Mixed Graphical Models (MGMs), of which GGMs and MNMs are special cases. [...]"

Haslbeck and Waldorp (2020): "Undirected graphical models are families of probability distributions that respect a set of conditional independence statements represented in an undirected graph G (Lauritzen 1996). [...]

Definition 1 (Global Markov property). If $X_A \perp\!\!\!\perp X_B \mid X_U$ whenever U is a node cutset that breaks the graph into disjoint subsets A and B, then the random vector $X := \left(X_1, \ldots, X_p\right)$ is Markov with respect to the graph G.

[...] In the remainder of this paper we focus on exponential family distributions, which are strictly positive distributions. For these distributions the global Markov property is equivalent to the Markov factorization property by the Hammersley-Clifford Theorem (Lauritzen 1996). Consider for each clique C in the set of all clique sets $\mathcal{C}$ a clique-compatibility function $\psi_C\left(X_C\right)$ that maps configurations $x_C = \{x_s, s \in C\}$ to $\mathbb{R}^+$such that ψ_C only depends on the variables X_C corresponding to the clique C.

Definition 2 (Markov factorization property). The distribution of X factorizes according to G if it can be represented as a product of clique functions

$$P(X) \propto \prod_{C \in \mathcal{C}} \psi_C\left(X_C\right). \tag{1}$$

This equivalence implies that if we have distributions that are represented as a product of clique functions, then we can represent the conditional dependence statements in this distribution in a graph G. This is the case for the exponential family distributions we use in the present paper

$$P(X) = \exp\left\{\sum_{C \in \mathcal{C}} \theta_C \phi_C\left(X_C\right) - \Phi(\theta)\right\}, \tag{2}$$

where the functions $\phi_C\left(X_C\right) = \log \psi_C\left(X_C\right)$ are sufficient statistic functions specified by the exponential family member at hand (e.g., Gaussian, Exponential, Poisson, etc.), θ_C are parameters associated with the clique functions and $\Phi(\theta)$ is the log-normalizing constant

$$\Phi(\theta) = \log \int_{\mathcal{X}} \sum_{C \in \mathcal{C}} \theta_C \phi_C\left(X_C\right) \nu(dx),$$

where depending on the distribution of X, the measure ν is a counting measure or Lesbesgue measure (for details see Wainwright and Jordan 2008).

The graph G represents a *family* of distributions because its edges do not indicate the strength of the dependency and the nodes can represent different conditional distributions. Hence there is a one to one mapping from the density to the graph, but a one to many mapping from graph to densities.

Mixed Graphical Model (MGM) In this section, we introduce the class of mixed graphical models (MGMs), which are a special case of the distribution in Equation 2 that allow one to combine an arbitrary set of conditional univariate members of the exponential family in a joint distribution (Yang, Baker, Ravikumar, Allen, and Liu 2014; Chen, Witten, and Shojaie 2015).

Consider a p-dimensional random vector $X = \left(X_1, \ldots, X_p\right)$ with each variable X_s taking values in a potentially different set $\mathcal{X}_s$, and let $G = (V, E)$ be an undirected graph over p nodes corresponding to the p variables. Now suppose the node-conditional distribution of node X_s conditioned on all other variables $X_{\setminus s}$ is given by an arbitrary univariate exponential family distribution

$$P\left(X_s \mid X_{\setminus s}\right) = \exp\left\{E_s\left(X_{\setminus s}\right)\phi_s\left(X_s\right) + B_s\left(X_s\right) - \Phi\left(X_{\setminus s}\right)\right\}, \tag{3}$$

where the functions of the sufficient statistic $\phi_s(\cdot)$ and the base measure $B_s(\cdot)$ are specified by the choice of exponential family and the canonical parameter $E_s\left(X_{\setminus s}\right)$ is a function of all variables except X_s. Wainwright and Jordan (2008) make these functions explicit for a number of exponential family distributions.

These node-conditional distributions are consistent with a joint distribution over the random vector X as in (1), that is Markov with respect to graph $G = (V, E)$ with the set of cliques C_k of size at most k, if and only if the canonical parameters $\{E_s(\cdot)\}_{s\in V}$ are a linear combination of products of univariate sufficient statistic functions $\{\phi\left(X_r\right)\}_{r\in N(s)}$ of order up to k

$$\theta_s + \sum_{r\in N(s)} \theta_{s,r}\phi_r\left(X_r\right) + \ldots + \sum_{r_1,\ldots,r_{k-1}\in N(s)} \theta_{r_1,\ldots,r_{k-1}} \prod_{j=1}^{k-1} \phi_{r_j}\left(X_{r_j}\right), \tag{4}$$

where $\theta_{s:} := \{\theta_s, \theta_{s,r}, \ldots, \theta_{sr_2\ldots r_k}\}$ is a set of parameters and $N(s)$ is the set of neighbors of node s according to graph G (Yang et al. 2014). Factorizing p conditional distributions as in Equation 3 gives the joint distribution

$$\begin{aligned} P(X) = \exp\Bigg\{ &\sum_{s\in V} \theta_s\phi_s\left(X_s\right) + \sum_{s\in V}\sum_{r\in N(s)} \theta_{s,r}\phi_s\left(X_s\right)\phi_r\left(X_r\right) + \\ &\cdots + \sum_{r_1,\ldots,r_k\in C} \theta_{r_1,\ldots,r_k} \prod_{j=1}^{k} \phi_{r_j}\left(X_{r_j}\right) + \sum_{s\in V} B_s\left(X_s\right) - \Phi(\theta)\Bigg\}, \end{aligned} \tag{5}$$

where $\Phi(\theta)$ is the log-normalization constant.

The dimensionality of the parameter vector θ depends both on the type of modeled variables and the order of interactions. If one only models continuous variables with pairwise interactions ($k = 2$), the MGM simplifies to the multivariate Gaussian distribution which is parameterized by a $1 \times p$ vector of intercepts and a $p \times p$ matrix of $\binom{p}{2}$ partial correlations. Including all 3 -way interactions would lead to an additional $\binom{p}{3}$ parameters, etc. At the end of Section 2.2, we discuss the dimensionality of the parameter vector in the presence of categorical variables. [...]

We take the Ising-Gaussian model as a specific example of the joint distribution in Equation 5. [...]"

8.5 MRF — OMRF

R-Package: `bgms`

Marsman et al. (2025): "We present an MRF for ordinal data that has been proposed in both the psychometric and machine learning literature but has so far not been considered in network psychometrics. In psychometrics, Anderson and Vermunt

(2000) derived a joint graphical model to manifest ordinal and unobserved continuous variables. By imposing a conditional Gaussian distribution (Lauritzen & Wermuth, 1989) on the latent variables, given the ordinal variables, they were able to derive a log-multiplicative association model for the distribution of the ordinal variables. We will show below that this log-multiplicative association model is an MRF for ordinal variables. Independently of the work of Anderson and Vermunt, in the machine learning literature, Suggala et al. (2017) derived an MRF for ordinal variables via a Hammersley–Clifford style analysis (Besag, 1974), introducing a model for the full conditional distribution for each variable in the network, given the remaining variables, and showing that this leads to a unique joint distribution. We will introduce this ordinal MRF and connect the work of Anderson and Vermunt (2000) and Suggala et al. (2017). [...]

Anderson and Vermunt (2000) [...] Let X_i denote an ordinal variable with $m + 1$ response categories and realizations $x_i = 0, 1, \dots, m$, for $i = 1, 2, \dots, p$ variables. We assume that the joint distribution for the full vector of p response variables $\boldsymbol{X}$ can be modeled using an item response theory (IRT) model, $P(\boldsymbol{X} \mid \boldsymbol{\theta})$, in which a vector of continuous latent variables $\boldsymbol{\theta}$ of at most dimension $p - 1$ fully accounts for the dependencies between the p response variables. In this context, Muraki (1990) proposed the generalized partial credit model (GPCM) for ordinal responses. The multidimensional version of this GPCM is characterized by the following probability of observing category h of the ordinal response variable:

$$P\left(X_i = h \mid \boldsymbol{\theta}\right) = \frac{\exp\left(h\boldsymbol{\alpha}_i^{\mathrm{T}}\boldsymbol{\theta} + \beta_{ih}\right)}{1 + \sum_{u=1}^{m} \exp\left(u\boldsymbol{\alpha}_i^{\mathrm{T}}\boldsymbol{\theta} + \beta_{iu}\right)},$$

where β_{ih} denotes a threshold parameter for category h of variable or item i, with β_{i0} set to zero for identification, and $\boldsymbol{\alpha}_i$ denotes a vector of factor-loadings that relate the latent variables to the response probabilities for item i. If we let $f(\boldsymbol{\theta})$ denote the distribution for the latent variables, the full statistical model has the form

$$P(\boldsymbol{X}) = P\left(X_1, \dots, X_\rho\right) = \int_{\mathbb{R}^{p-1}} \prod_{i=1}^{p} P\left(X_i \mid \boldsymbol{\theta}\right) f(\boldsymbol{\theta}) \boldsymbol{d\theta}$$

We aim to find an expression for $P(\boldsymbol{X})$, the marginal distribution of the response variables.

Marsman et al. (2019) showed that if an IRT model is in an exponential family form, we can define a latent variable distribution that allows us to analytically express the marginal distribution $P(\boldsymbol{X})$. We define the exponential family form of the IRT model as follows:

$$\prod_{i=1}^{p} P\left(X_i \mid \boldsymbol{\theta}\right) = \prod_{i=1}^{p} \frac{1}{\mathrm{z}_i(\boldsymbol{\theta})} \mathrm{b}_i\left(X_i\right) \exp\left(\boldsymbol{S}_i\left(X_i\right)^{\mathrm{T}} \boldsymbol{\theta}\right),$$

where $\boldsymbol{S}_i\left(X_i\right)$ is a (possibly vector-valued) statistic that is sufficient for the (possibly vector-valued) latent variable $\boldsymbol{\theta}, \mathrm{b}_i\left(X_i\right)$ is a base measure that does not depend on the latent variable, and $\mathrm{z}_i(\boldsymbol{\theta})$ is a normalizing constant that does not depend on the observed response variable and ensures that the probabilities add up to one. The normalizing constant is equal to

$$\mathrm{z}_i(\boldsymbol{\theta}) = \sum_{X_i=0}^{m} \mathrm{b}_i(X_i) \exp\left(\boldsymbol{S}_i(X_i)^{\mathrm{T}} \boldsymbol{\theta}\right)$$

We can express the multidimensional GPCM in this form. That is, we define the sufficient statistic as $\boldsymbol{S}_i(X_i) = X_i \boldsymbol{\alpha}_i$, the logarithm of the base measure as $\ln(\mathrm{b}_i(X_i)) = \beta_{iX_i}$, and the normalizing constant as

$$z_i(\boldsymbol{\theta}) = 1 + \sum_{u=1}^{m} \exp\left(u \boldsymbol{\alpha}_i^{\mathrm{T}} \boldsymbol{\theta} + \beta_{iu}\right)$$

We can now define a latent variable distribution as follows:

$$f(\boldsymbol{\theta}) = \frac{1}{\mathrm{Z}} \prod_{i=1}^{p} \mathrm{z}_i(\boldsymbol{\theta}) k(\boldsymbol{\theta}),$$

where $k(\boldsymbol{\theta})$ is a density function and Z is a normalizing constant. The normalizing constant is equal to

$$\mathrm{Z} = \int_{\mathbb{R}^{p-1}} \prod_{i=1}^{p} \mathrm{z}_i(\boldsymbol{\theta}) k(\boldsymbol{\theta}) \boldsymbol{d\theta} \tag{1}$$

The proposed distribution $f(\boldsymbol{\theta})$ is valid for any density function $k(\boldsymbol{\theta})$ for which $0 < \mathrm{Z} < \infty$. Corollary 1 in Marsman et al. (2019) shows that if we assume that the density $k(\boldsymbol{\theta})$ is a multivariate normal distribution with a mean vector $\boldsymbol{m} = 0$ and covariance matrix $\boldsymbol{V} = \boldsymbol{I}$, the identity matrix, we find the following expression for the marginal distribution:

$$P(\boldsymbol{X} \mid \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{1}{\mathrm{Z}} \exp\left(\sum_{i=1}^{p} \beta_{iX_i} + \sum_{i=1}^{p} X_i^2 \boldsymbol{\alpha}_i^{\mathrm{T}} \boldsymbol{\alpha}_i + 2 \sum_{i=1}^{p-1} \sum_{j=i+1}^{p} X_i X_j \boldsymbol{\alpha}_i^{\mathrm{T}} \boldsymbol{\alpha}_j\right)$$

From here, we recognize the quadratic form

$$P(\boldsymbol{X} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{Z} \exp\left(\sum_{i=1}^{p} \sum_{h=1}^{m} \mathbb{I}(X_i = h) \mu_{ih} + 2 \sum_{i=1}^{p-1} \sum_{j=i+1}^{p} X_i X_j \sigma_{ij}\right), \tag{2}$$

where $\mu_{ih} = \beta_{ih} + h^2 \boldsymbol{\alpha}_i^{\mathrm{T}} \boldsymbol{\alpha}_i$ and $\boldsymbol{\Sigma} = \left[\sigma_{ij}\right] = \boldsymbol{\alpha}_i^{\mathrm{T}} \boldsymbol{\alpha}_j$, and $\sigma_{ii} = 0$, for $i = 1, \ldots, p$. The marginal distribution $P(\boldsymbol{X} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the desired MRF and equal to Eq. (20) in Anderson and Vermunt (2000; see also Theorem 2 in Hessen, 2020). The normalizing constant that we defined in Eq. (1) can now be restated as

$$\mathrm{Z} = \sum_{\boldsymbol{X} \in \Omega_X} \exp\left(\sum_{i=1}^{p} \sum_{h=1}^{m} \mathbb{I}(X_i = h) \mu_{ih} + 2 \sum_{i=1}^{p-1} \sum_{j=i+1}^{p} X_i X_j \sigma_{ij}\right), \tag{3}$$

where $\Omega_X = \{0, \ldots, m\}^p$ denotes the space of possible response patterns. Observe that when $m = 1$, the MRF in Eq. (2) simplifies to the Ising model (Cox, 1972; Ising, 1925).

Suggala et al. (2017) [...] The Hammersley-Clifford theorem states that if a multivariate distribution $P\left(X_1, \ldots, X_p\right)$ is consistent with certain full conditional distributions $P\left(X_i \mid X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_p\right) = P\left(X_i \mid \boldsymbol{X}^{(i)}\right)$, then it is the only multivariate distribution consistent with these conditional distributions. Besag (1974) suggested using this idea to determine whether there is a multivariate distribution that is consistent with certain univariate full conditional distributions. Suggala et al. (2017) use this idea to analyze three popular models for ordinal variables: the cumulative ratio, the continuation ratio, and the consecutive ratio (or adjacent category) logits (Agresti, 2010, 2018). Theorem 3 in Suggala et al. (2017) shows that there is a multivariate distribution consistent with the consecutive ratio or adjacent categories logit,

$$\log\left(\frac{P(X=h)}{P(X=h+1)}\right) = \eta_h - \beta, \text{ for } h = 0, \ldots, m-1.$$

We use this logit to define the full conditional of a node i given the remaining variables $\boldsymbol{X}^{(i)}$, by setting $\eta_h = \eta_{ih}$ and $\beta = \beta_i\left(\boldsymbol{X}^{(i)}\right)$, such that the full conditional is of the form

$$P\left(X_i = h \mid \boldsymbol{X}^{(i)}\right) = \frac{\exp\left(\sum_{q=0}^{h}\left(\eta_{iq} - \beta_i\left(\boldsymbol{X}^{(i)}\right)\right)\mathbb{I}\left(x_i \le q\right)\right)}{1 + \sum_{u=0}^{m-1}\exp\left(\sum_{q=0}^{m-1}\left(\eta_{iq} - \beta_i\left(\boldsymbol{X}^{(i)}\right)\right)\mathbb{I}(u \le q)\right)}, \text{ for } h = 0, \ldots, m-1.$$

Theorem 3 in Suggala et al. (2017) shows that the multivariate distribution that is consistent with this full conditional is equal to

$$\begin{aligned}
P(\boldsymbol{X} \mid \boldsymbol{\eta}, \boldsymbol{\Sigma}) &= \frac{1}{Z}\exp\left(\sum_{i=1}^{p}\sum_{q=0}^{m-1}\eta_{iq}\mathbb{I}\left(X_i \le q\right) + \sum_{i=1}^{p-1}\sum_{j=i+1}^{p}\sum_{q=0}^{m-1}\sum_{u=0}^{m-1}\sigma_{ij}\mathbb{I}\left(X_i \le q\right)\mathbb{I}\left(X_j \le u\right)\right) \\
&= \frac{1}{Z}\exp\left(\sum_{i=1}^{p}\sum_{q=0}^{m-1}\eta_{iq}\mathbb{I}\left(X_i \le q\right) + \sum_{i=1}^{p-1}\sum_{j=i+1}^{p}\sigma_{ij}\left(m - X_i\right)\left(m - X_j\right)\right).
\end{aligned}$$

When we redefine μ_{ih} as $\sum_{q=h}^{m-1}\eta_{iq}$ and X_i^* as $m - X_i$, this multivariate distribution is equal to the MRF in Eq. (2). Thus, the proposed MRF in Eq. (2) is a multivariate extension of the adjacent category model.

Suggala et al. (2017) showed that no multivariate distribution is consistent with either of the other two models for ordinal variables. The cumulative ratio or proportional odds is defined as

$$\log\left(\frac{p(X \le h)}{p(X > h)}\right) = \eta_h - \beta, \text{ for } h = 0, \ldots, m-1$$

This logit model is similar in nature to the probit model (Guo et al., 2015), in which the realizations of the ordinal variable correspond to adjacent intervals of an underlying latent variable. Theorem 1 in Suggala et al. (2017) shows that no multivariate distribution is consistent with having this logit model as full-conditional distribution. Furthermore, their Theorem 2 shows that there is also no multivariate distribution with full conditionals based on the

$$\log\left(\frac{p(X = h)}{p(X > h)}\right) = \eta_h - \beta, \text{ for } h = 0, \ldots, m-1$$

This implies that the proposed MRF is a rather unique multivariate model for ordinal variables."

8.6 IM — Exponential Trace Model – GGM

Zhuang et al. (2016): "We introduce a general framework for undirected graphical models. It generalizes Gaussian graphical models to a wide range of continuous, discrete, and combinations of different types of data. The models in the framework, called exponential trace models, are amenable to estimation based on maximum likelihood.

[...] For a brief overview, consider a random vector $X \in \mathbb{R}^p$ that follows a centered normal distribution with density

$$f_\Sigma(\boldsymbol{x}) = \frac{1}{(2\pi)^{p/2}\sqrt{|\Sigma|}} e^{-\boldsymbol{x}^\top \Sigma^{-1} \boldsymbol{x}/2} \tag{1}$$

with respect to Lebesgue measure, where the population covariance $\Sigma \in \mathbb{R}^{p \times p}$ is a symmetric and positive definite matrix. Gaussian graphical models associate these densities with a graph (V, E) that has vertex set $V := \{1, \ldots, p\}$ and edge set $E := \left\{(i, j) : i, j \in \{1, \ldots, p\}, i \neq j, \Sigma^{-1}_{ij} \neq 0\right\}$. The graph encodes the dependence structure of X in the sense that any two entries X_i, X_j, $i \neq j$, are conditionally independent given all other entries if and only if $(i, j) \notin E$. A natural and straightforward estimator of Σ^{-1}, and thus of E, is the inverse of the empirical covariance, which is also the maximum likelihood estimator. Inference can then be approached via the quantiles of the normal distribution. [...]

Let us have a glance at the framework. For this, we start with the Gaussian densities (1). Our first observation is that $-\boldsymbol{x}^\top \Sigma^{-1} \boldsymbol{x}/2 = -\left\langle \Sigma^{-1}, \boldsymbol{x}\boldsymbol{x}^\top/2 \right\rangle_{\mathrm{tr}}$, where $\langle \cdot, \cdot \rangle_{\mathrm{tr}}$ is the trace inner product. This formulation looks "somewhat less revealing" (Eaton 2007, Page 125) on first sight, but it has two conceptual advantages. First, we argue that writing the parameters and the data as algebraic duals of each other makes their relationship more symmetric. Second, we argue that it is a good starting point for generalizations. For this, we take the viewpoint that the exponents in the densities are linear functions of the matrix $\boldsymbol{x}\boldsymbol{x}^\top/2$, and then replace this matrix by a general matrix-valued function T of $\boldsymbol{x}$. Our second observation is that the fundamental quantity in the family of Gaussian graphical models is the inverse covariance matrix Σ^{-1} rather than the covariance matrix Σ itself. This suggests a reparametrization of the model using the matrix Σ^{-1}, which is then replaced by a general matrix M . This subtlety is important: as we will see later, the matrix M contains all information about the dependence structure of X, while the equality of M^{-1} and the covariance matrix is a mere coincidence in the Gaussian case. With these two observations in mind, and denoting the log-normalization by $\gamma(\mathrm{M})$, with $\gamma(\mathrm{M}) = \log\left((2\pi)^p \left|\mathrm{M}^{-1}\right|\right)/2$ in the Gaussian case, we can then generalize the densities (1) to

$$f_{\mathrm{M}}(\boldsymbol{x}) = e^{-\langle \mathrm{M}, T(\boldsymbol{x})\rangle_{\mathrm{tr}} - \gamma(\mathrm{M})}$$

with respect to an arbitrary σ-finite measure ν, and with $\mathrm{M}, T \in \mathbb{R}^{q \times q}$ a matrix-valued parameter and data function, respectively. While additional data terms can be absorbed in the measure ν, it is sometimes illustrative to write them explicitly. We thus consider distributions with densities of the form

$$f_{\mathrm{M}}(\boldsymbol{x}) = e^{-\langle \mathrm{M}, T(\boldsymbol{x}) \rangle_{\mathrm{tr}} + \xi(\boldsymbol{x}) - \gamma(\mathrm{M})}$$

where $\xi(\boldsymbol{x})$ depends on $\boldsymbol{x}$ only. These densities form an exponential family indexed by M and are called exponential trace models in the following for convenience.

[...] exponential trace models generalize pairwise interaction models in the sense that all generic examples of pairwise interaction models are encompassed. [...] we argue that the exponential trace framework is a practical starting point for general studies of graphical models, because it comprises a very large variety of examples and still ensures a firm theoretical structure. [...]

[...] standard graphical models, such as Ising models with binary and multinomial responses and Gaussian and non-paranormal graphical models, fit the exponential trace framework. [...]"

9 Recurrence Plot (RP) / Recurrence Network (RN)

R-Package: 'casnet', 'crqa', 'nonlinearTseries'
Python: 'pyunicorn'

Donner et al. (2010): "Since the early stages of quantitative nonlinear sciences, numerous conceptual approaches have been introduced for studying the characteristic features of dynamical systems based on observational time series [1]–[4]. [...]

As a particular concept, the basic ideas of which originated in the pioneering work of Poincaré in the late 19th century [6], the quantification of recurrence properties in phase space has recently attracted considerable interest [7]. One particular reason for this is that these recurrences can be easily visualized (and subsequently quantified in a natural way) by means of the so-called recurrence plots obtained from a single trajectory of the dynamical system under study [8, 9]. When observing this trajectory as a scalar time series $x(t)(t = 1, \ldots, N)$, one may use a suitable m-dimensional time delay embedding of $x(t)$ with delay τ [10], $\boldsymbol{x}^{(m)}(t) = (x(t), x(t+\tau), \ldots, x(t+(m-1)\tau))$, for obtaining a recurrence plot as a graphical representation of the binary recurrence matrix

$$R_{i,j}(\epsilon) = \Theta\left(\epsilon - \left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|\right), \tag{1}$$

where $\Theta(\cdot)$ is the Heaviside function, $\|\cdot\|$ denotes a suitable norm in the considered phase space and ϵ is a threshold distance that should be reasonably small (in particular, much smaller than the attractor diameter [9, 11, 12]). To simplify our notation, we have used the abbreviation $\boldsymbol{x}_i = \boldsymbol{x}^{(m)}\,(t = t_i)$ (with t_i being the point in time associated with the i th observation recorded in the time series) wherever appropriate.

Experimental time series often yield a recurrence plot displaying complex structures, which are particularly visible in sets of recurrence points (i.e. $R_{i,j}(\epsilon) = 1$) forming diagonal or vertical 'line' structures. A variety of statistical characteristics of the length distributions of these lines (such as maximum, mean, or Shannon entropy) can be used for defining additional quantitative measures that characterize different aspects of dynamic complexity of the studied system in more detail. This conceptual framework is known as recurrence quantification analysis (RQA) [13]-[15] and is nowadays frequently applied to a variety of real-world applications of time series analysis in diverse fields of research [16]. However, most of these RQA measures are sensitive to the choice of embedding parameters, which are found to sometimes induce spurious correlations in a recurrence plot [17].

Recent studies have revealed that the fundamental invariant properties of a dynamical system (i.e. its correlation dimension D_2 and correlation entropy K_2) are conserved in the recurrence matrix [18, 19], the estimation of which is independent of the particular embedding parameters. What is more, recurrence plots preserve all the topologically relevant phase space information of the system, such that we can completely reconstruct a time series from its recurrence matrix (modulo some rescaling of its probability distribution function) [20,21].

A further appealing paradigm for analysing the structural features of complex systems is based on their representation as complex networks of passive or active (i.e. mutually interacting) subsystems. For this purpose, classical graph theory has been systematically extended by a large variety of different statistical descriptors of the topological features of such networks on local, intermediate and global scales [22]-[24]. [...]

As an alternative that may provide a unifying conceptual and practical framework for nonlinear time series analysis using complex networks, we reconsider the concept of recurrences in phase space for defining complex network structures directly based on time series. For this purpose, it is straightforward to interpret the recurrence matrix $\boldsymbol{R}(\epsilon)$ as the adjacency matrix $\boldsymbol{A}(\epsilon)$ of an unweighted and undirected complex network, which we suggest to call the recurrence network associated with a given time series. To be more specific, the associated adjacency matrix is given by

$$A_{i,j}(\epsilon) = R_{i,j}(\epsilon) - \delta_{i,j}, \tag{2}$$

where $\delta_{i,j}$ is the Kronecker delta introduced here in order to avoid artificial self-loops, corresponding to a specific choice of the Theiler window in RQA [9]. A related conceptual idea has recently been independently suggested by different authors [35, 36, 39, 40, 42], but not yet systematically studied. In this work, however, we aim to give a rigorous and detailed interpretation of a variety of quantitative characteristics of recurrence networks, yielding novel concepts for statistically evaluating distinct phase space structures captured in recurrence plots. In this sense, our presented work continues and complements the developments in the field of RQA within the last two decades [9, 43]. It shall be noted that a generalization to weighted networks (as partially studied in [33,34]) is straightforward if the recurrence matrix is replaced by the associated distance matrix between pairs of states. In any case, recurrence networks based on the mutual phase space distances of observational points on a single trajectory are spatial networks, i.e. fully embedded in an m-dimensional space, which has important implications for their specific topological features. We will raise this point in detail within this paper.

The consideration of recurrence plots as graphical representations of complex networks allows a reinterpretation of many network-theoretic measures in terms of characteristic phase space properties of a dynamical system. According to the ergodicity hypothesis, we suppose that we can gain full information about these properties by either ensembles of trajectories, or sufficiently long observations of a single trajectory. Following this line of ideas, we approximate the (usually unknown) invariant density $p(\boldsymbol{x})$ (which is related to the associated invariant measure μ by $d\mu = p(\boldsymbol{x})d\boldsymbol{x}$) of the studied system by some empirical estimate $\hat{p}^{(\epsilon)}(\boldsymbol{x})$ obtained from a time series, where ϵ defines the level of coarse-graining of phase space involved in this procedure. Transforming the time series into a recurrence network then allows us to quantitatively characterize the higher-order statistical properties (i.e. properties that are based on joint features of different states) of the invariant density $p(\boldsymbol{x})$ by means of network-theoretic measures. [...]

In particular, correlation networks [37] (see section 2.3) can be regarded as a special case of recurrence networks where the usual metric distance has been replaced by the correlation distance [76]

$$d_C\left(\boldsymbol{x}_i, \boldsymbol{x}_j\right) = 1 - r_{i,j}, \tag{8}$$

where $r_{i,j}$ is the correlation coefficient between two embedded state vectors defined in equation (5). [...]"

10 Dynamic Time Warping (DTW)

R-Package: 'dtw'

van der Does et al. (2025): "Psychopathological disorders are increasingly conceptualized as complex dynamic systems, which can be represented as networks of interconnected symptoms. These dynamic networks are often constructed using Multilevel Vector AutoRegression (mlVAR) models. However, psychological processes frequently violate these assumptions. An alternative approach for examining temporal relationships between variables is Dynamic Time Warping (DTW). This paper evaluates the potential applications, advantages, and disadvantages of DTW and mlVAR. As part of the Netherlands Study of Depression and Anxiety, an Ecological Momentary Assessment module was administered five times daily for 2 weeks, to 376 participants [...]

The dynamic interplay between symptoms or mood states can be studied and mapped in a network, using time-series data. Ecological Momentary Assessment (EMA), wherein participants complete brief questionnaires multiple times a day using a mobile device, may be an especially valuable approach. EMA provides an opportunity to capture detailed real-time information on mood states and their interactions, offering insights into subtle nuances and connections that may not be captured when relying on less frequent and retrospective assessments [9]. When constructing EMA networks, researchers often employ the Multi-Level Vector Autoregression (mlVAR) model. This model enables the fitting of an autoregressive model across multiple individuals, where the value of a symptom or mood state is predicted by its own value and the value of other items in time-lag analyses [10]. When researching psychopathology, it is common to use a lag-1 model, which regresses the values of the item scores over values from the previous time point [11,12]. Findings are explained in terms of Granger causality, meaning that at least the temporal sequence requirements of causal effects are met, and fluctuation in a 'causal' item precedes fluctuation in the 'effect' [13].

The mlVAR model, however, faces several challenges that need to be addressed [14]. Firstly, some assumptions connected to the model are frequently violated. For instance, the assumption of stationarity, assuming that the mean, variance, and autocorrelation of variables remain constant over time [15], is violated in developmental and treatment processes [16]. In previous research using mlVAR in the analysis of EMA data, stationarity has been assumed due to the relatively short time span in which EMA data are captured. [13] However, this is not always realistic, as developments and changes can occur in the individual in short time periods. Furthermore, it precludes the study of networks where change is expected, for instance during an intervention [17]. Secondly, the undirected and between-subjects mlVAR approaches rely on partial correlations, and the directed mlVAR approach relies on multivariate regression, meaning that associations between two mood items are dependent on their relationships with all other mood items. This means that when all relevant variables are included and statistical assumptions are met, causality can be inferred from the connections in the directed network. However, in psychological systems a common occurrence is the presence of colliders-items that are influenced by two or more statistically unrelated nodes. Conditioning on the relationship with these colliders in the partial correlations network introduces a negative statistical dependency between the items influencing the collider, which can show up as a negative connection in the network when no such relationship exists in the data. Therefore, in psychological systems where colliders are present, the mlVAR approach might introduce spurious connections or distort the strength of positive connections in the network, leading to misinterpretations regarding the importance of certain nodes and edges [18]. Previous research has indeed demonstrated instances where mlVAR models produced such spurious connections in the network, which might be the consequence of such colliders. [18,19] Additionally, items in the network may have construct overlap and can be highly intercorrelated, such that relationships among items can change noticeably if such an item is added or removed from the multivariate mlVAR network.

Dynamic Time Warping (DTW) offers a promising alternative for examining the relationships between items in a dynamic network. DTW is a statistical method that identifies patterns in time-series data by optimizing alignment, even when the covariation between symptoms is not simultaneous or linear [20]. This technique quantifies the shape-based similarity between variable trajectories over time, with the possibility of incorporating time lags, resulting in a DTW distance measure. Where mlVAR aims to identify causal connections, the DTW technique can be applied to study co-occurrence and synchrony between variables, with the possibility of identifying lagged and non-linear relationships. DTW has been applied to studying time-series data across various fields [21–23]. In the field of psychiatry, DTW has recently shown success in estimating symptom networks of depression [8], and examining changes in depression symptom networks after electroconvulsive therapy [24]. It has also recently been used to map symptoms of post-traumatic stress [25], bipolar disorder [26,27], and eating disorder psychopathology y[28].

Unlike mlVAR, DTW is a non-linear technique utilized for comparing and aligning time-series data, making it less susceptible to the effects of non-stationarity. Moreover, in DTW the calculation of the relationships between items is independent of their relationships with other items in the network. This means that the choice of variables included in the network does not influence the DTW distance between other variables.

Additionally, using the DTW algorithm makes it easier to incorporate time-varying lags in the model. This allows for the inclusion of lag-0, lag-1, and lag-2 co-variation within the same model [29], unlike mlVAR, which requires advanced script extensions or custom equations [30]. However, DTW also has its potential downsides, as it cannot capture multivariate associations or autocorrelation. Therefore, under ideal conditions where assumptions are met, Granger causality may be more reliably inferred from mlVAR networks. However, the mlVAR network is more susceptible to unmeasured confounding, as this can distort the mlVAR results but not the DTW distance calculations. [...]"

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